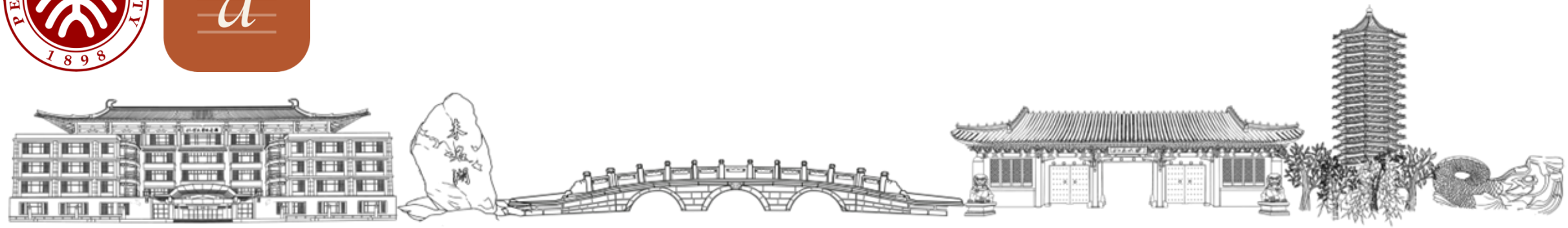
Cognitive Reasoning
Fall 2026



Lecture 06
Intuitive Physics

Instructor: Dr. Chi Zhang
School of Intelligence Science and Technology



Logistics

- Proposals and Related Work Check
- DDL: 11.03
- Topics
  - Any topic related to current AI research (Not Chemistry, Physics, Math, etc)
  - Best if related to contents we touch about in this course (AI and Cognition, Reasoning, etc)
  - For students not into research, any AI tools you'd like to create, deemed practical and useful
- Submit a short 2-page PDF with introduction and related work, in the style of NeurIPS 2026 latex template
- Late penalty: 5 points each day late

Logistics

- Milestone Presentation
- DDL: 11.04
- ATTENDENCE REQUIRED
- 5 min each group

Contents

1. Introduction
2. Cognitive basis
3. Computational models
    a. Observation
    b. Prediction
    c. Interaction
4. Modern AI systems on physics

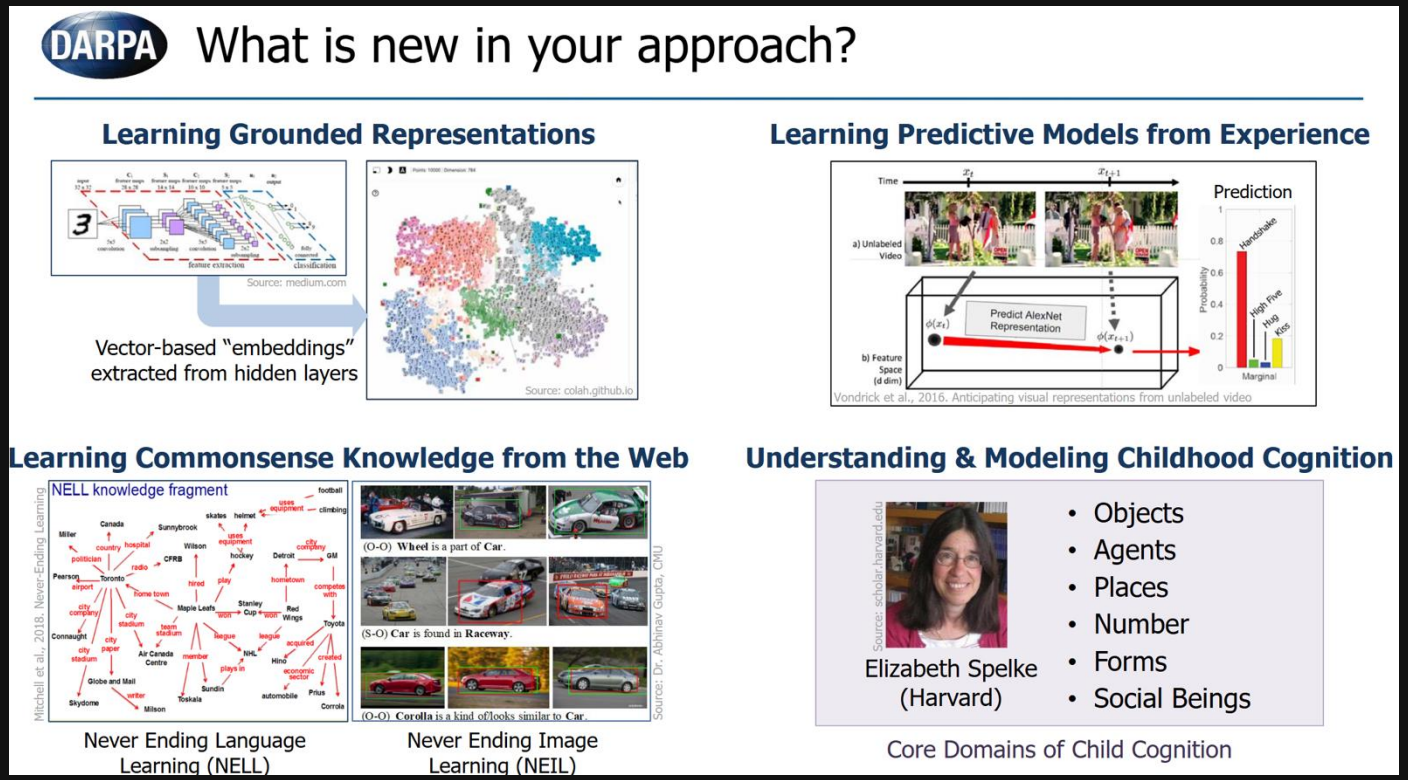
Part 1
Introduction

Introduction

DARPA: Machine Common Sense

Supervised learning (ImageNet)

Multi-models (GPT-4o)

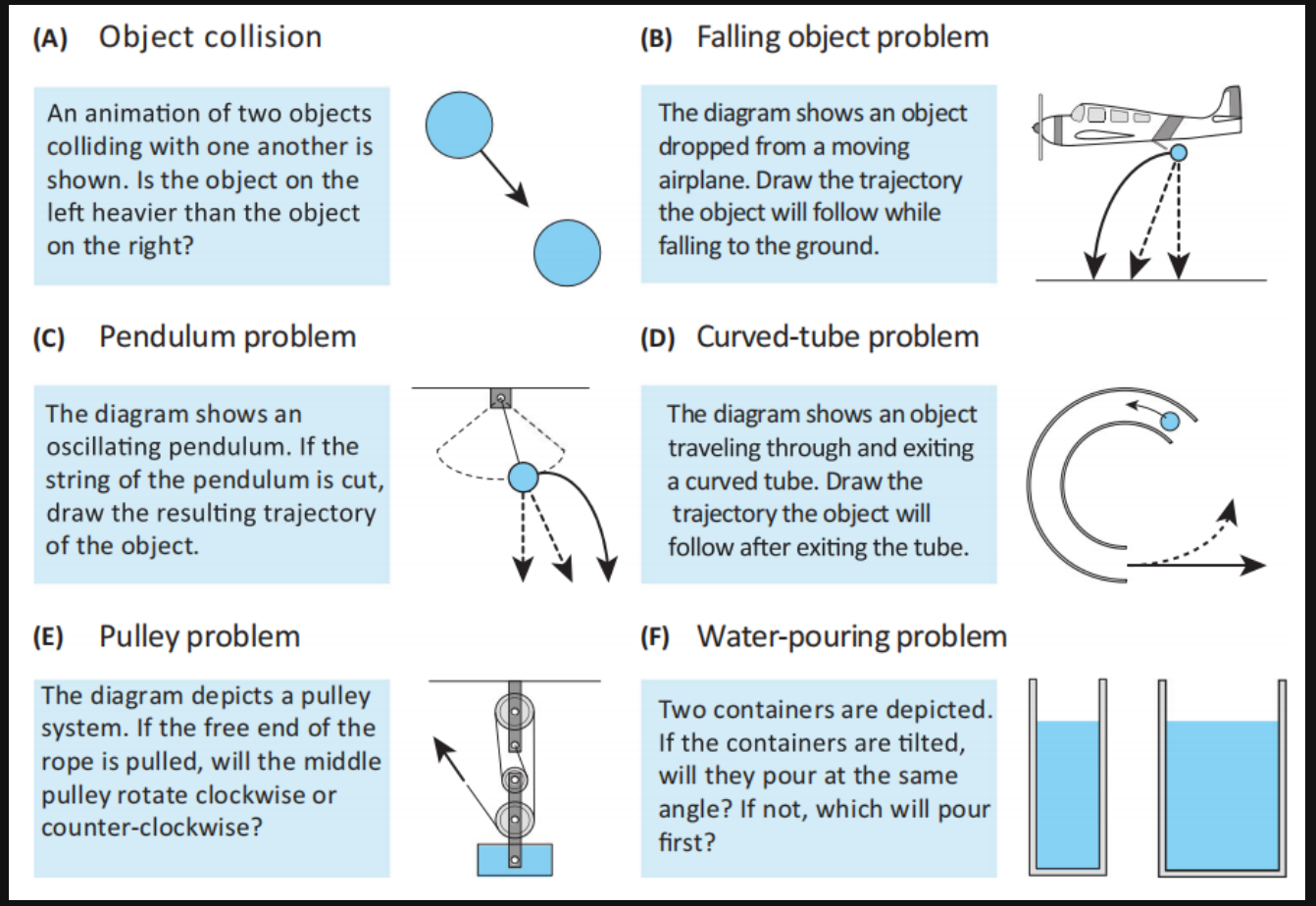


Self-supervised learning (Sora)

Cognitive architecture

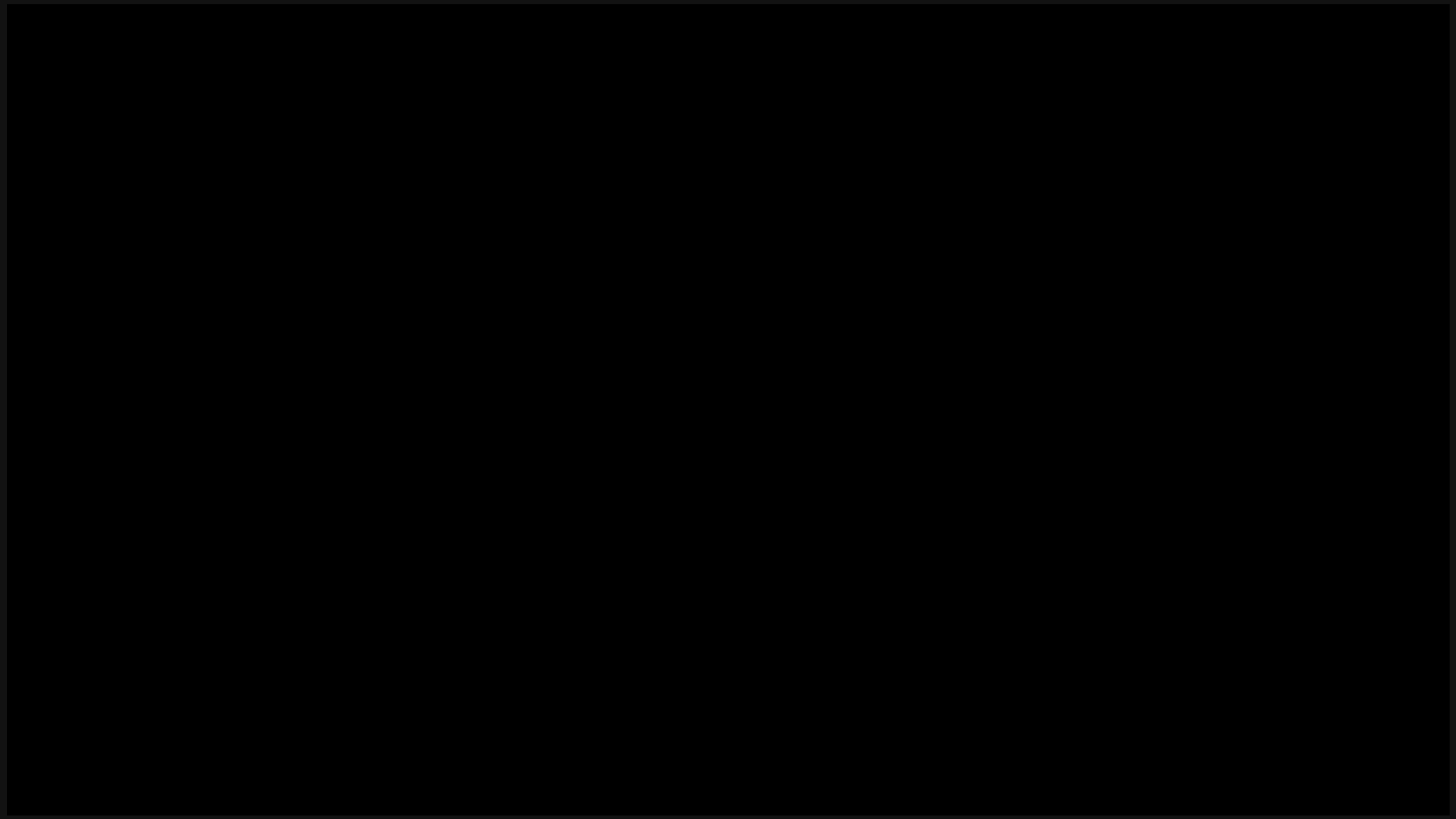
Introduction

- Knowledge to understand their physical environment and interact with objects and substances that undergo dynamic state changes, making at least approximate predictions about how observed events will unfold;
- Research primarily focused on misconceptions that people demonstrate when reasoning about the attributes and movements of objects and substances in the world.



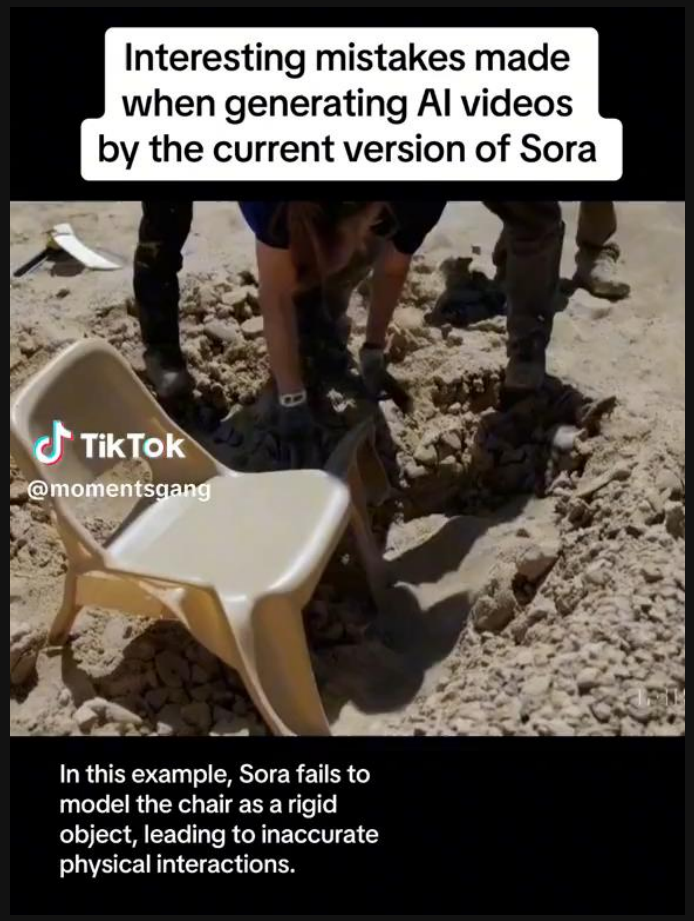

FRI HD
Shaun Murphy
Vs O'Sullivan
WC SNOOKER
MURPHY
0
1 (25) 3
1
O'SULLIVAN
WC SNOOKER!






100x
MICKEY MOUSE

SOTA models still fall short in understanding physics



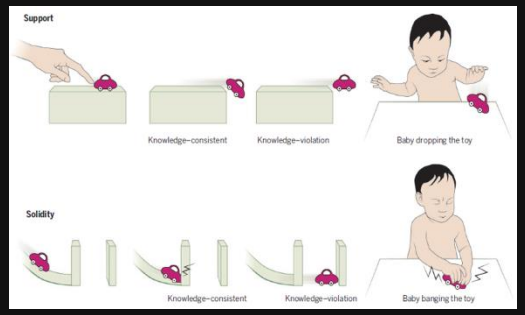
Part 2
Cognitive basis

Developmental psychology, neural basis, VoE paradigm

History of intuitive physics on cognitive basis



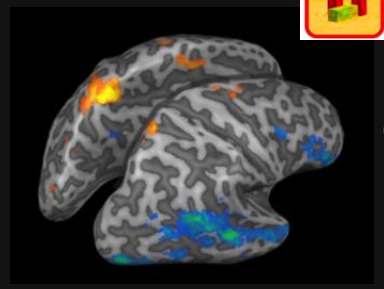
Infant studies
1980s



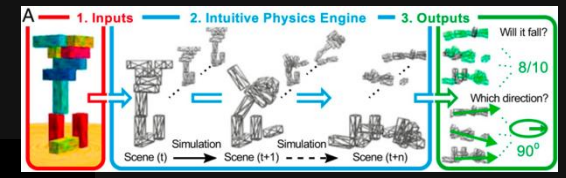
Deeper exploration
1990s



Animal cognition
2000s



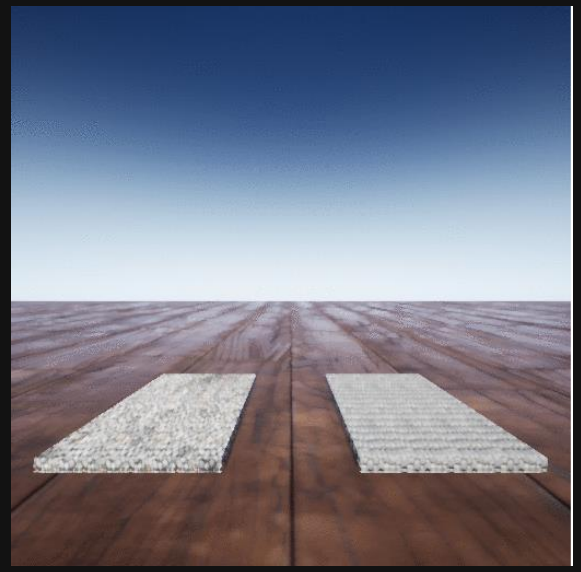
Neural basis
2010s

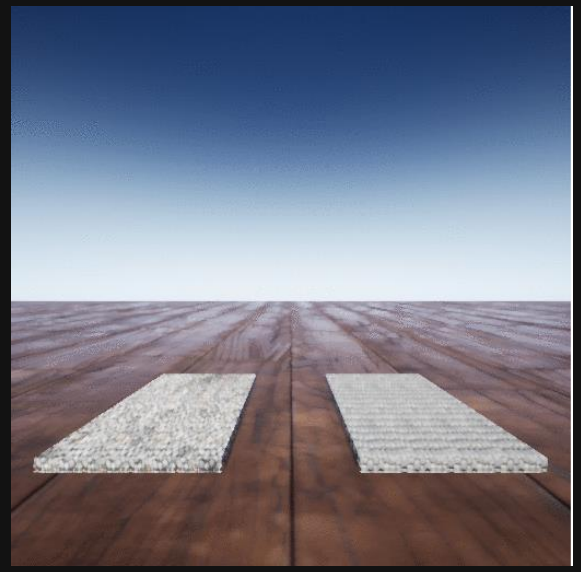


Computational
mechanisms
2010-present

Violation of Expectation (VoE)

Which one is abnormal?





This is abnormal!

Violation of Expectation (VoE)

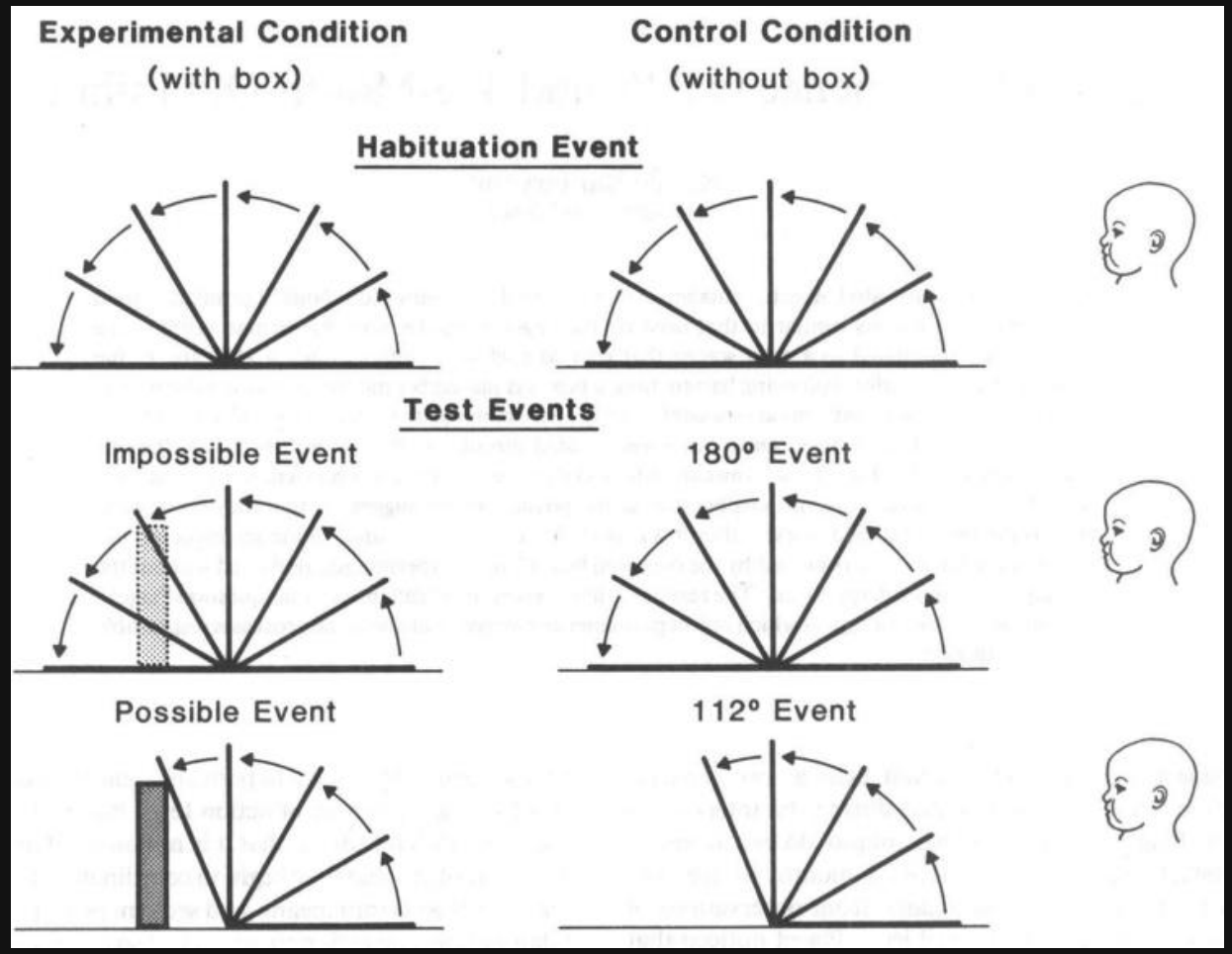


Violation of Expectation (VoE)



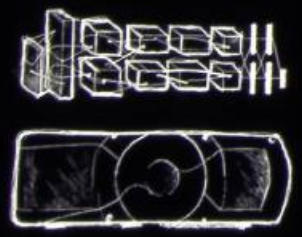
Cognitive basis--Infant studies

- Infants develop the ability to recognize the persistence of hidden objects earlier than nine months
- The infants in the experimental group looked at the "impossible event" for a significantly longer time



Baillargeon R. Object Permanence in 3 1/2-and 4 1/2-Month-Old Infants[J]. An Introductory Reader in Developmental Psychology, 1995, 23(5): 103.

2012 ALEXNET



PERCEPTION AI
SPEECH RECOGNITION
DEEP RECSYS
MEDICAL IMAGING

GENERATIVE AI
DIGITAL MARKETING
CONTENT CREATION

AGENTIC AI
CODING ASSISTANT
CUSTOMER SERVICE
PATIENT CARE

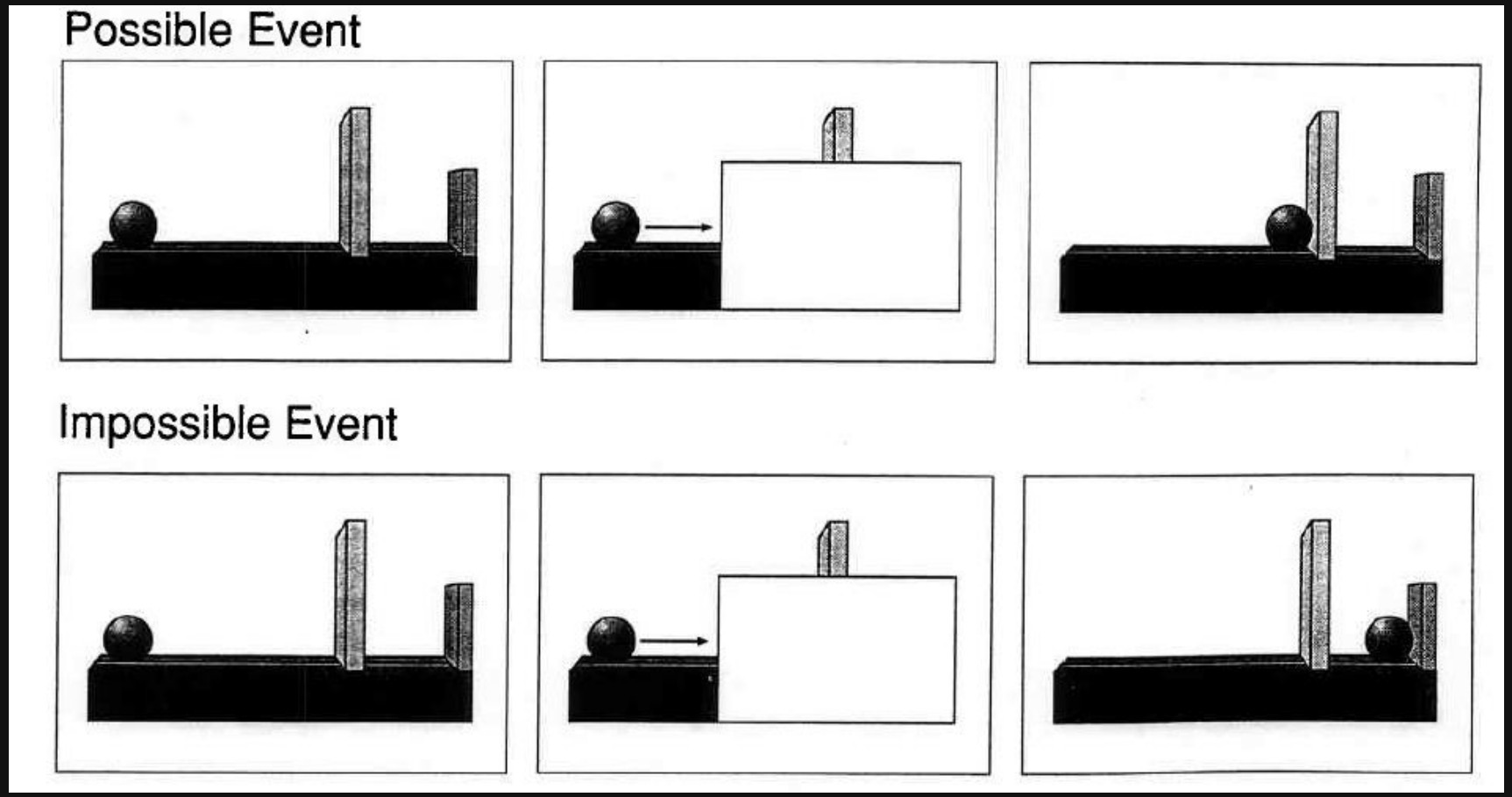
PHYSICAL AI
AUTONOMOUS VEHICLES
GENERAL ROBOTICS



Developmental psychology: infants gradually acquire distinct physical concepts over time

Cognitive basis--Infant studies

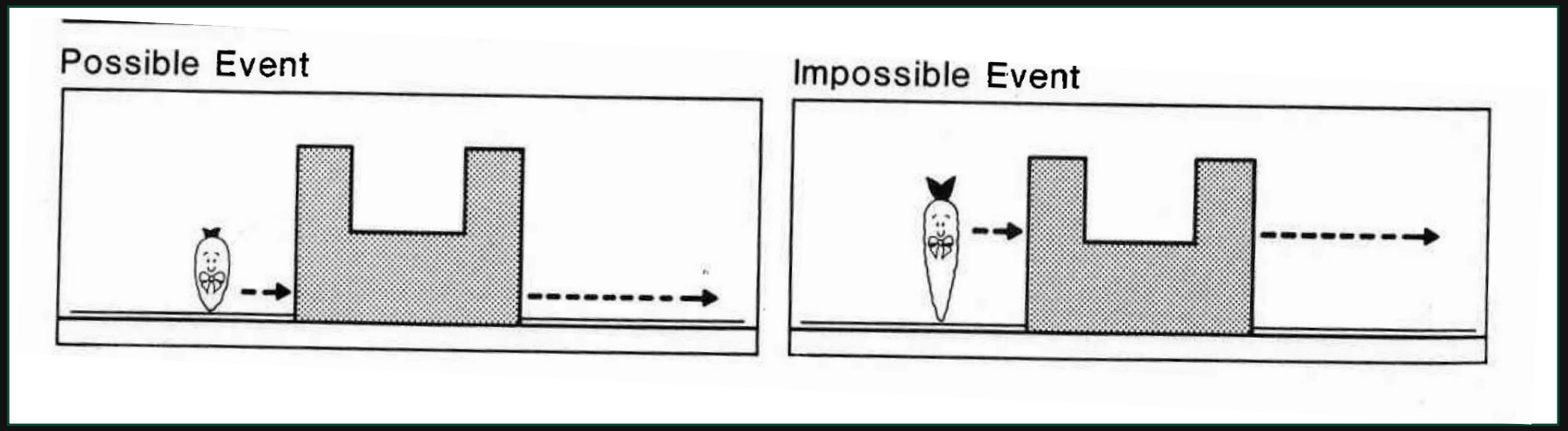
- 2.5 months



Object solidity

Cognitive basis--Infant studies

- 3.5 months

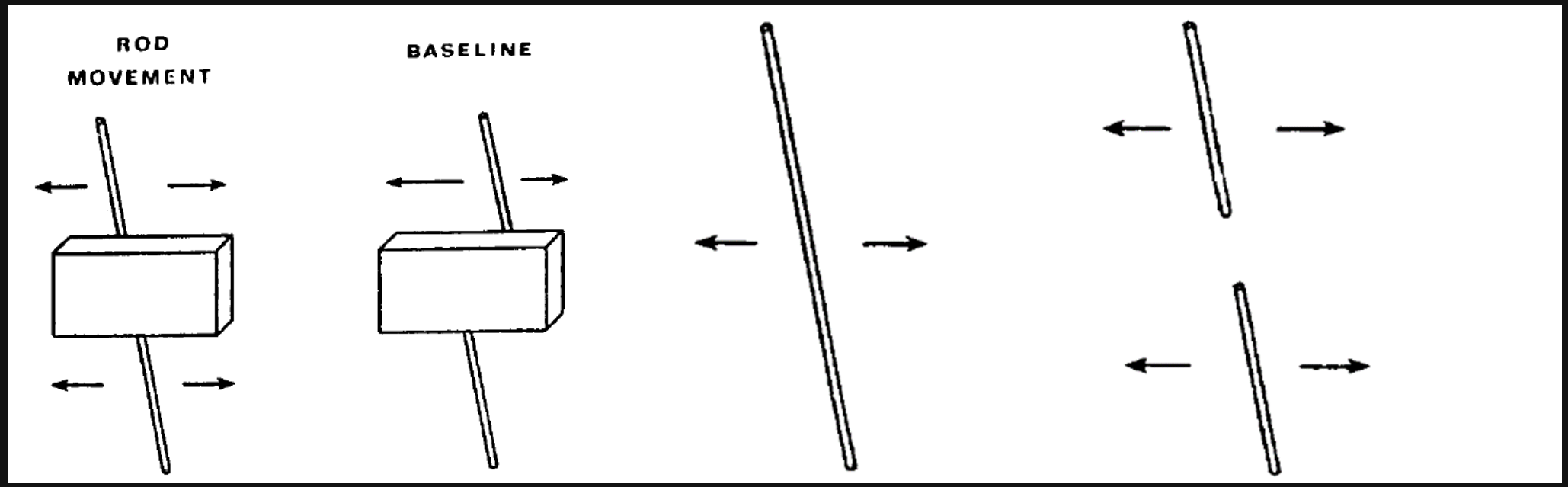


Integrating height, occlusion and continuity

Cognitive basis--Infant studies

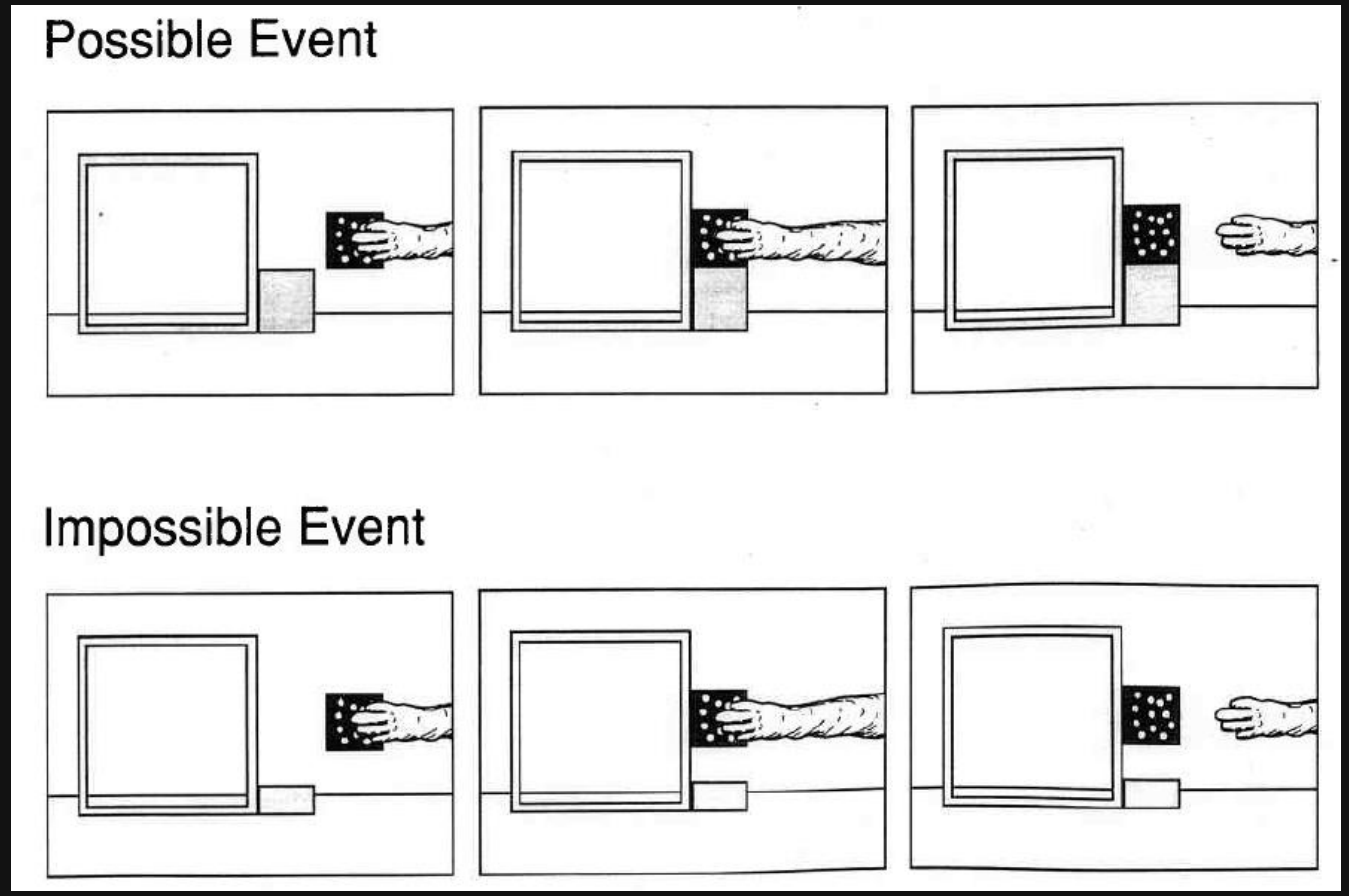
- 4.0 months

Partly occluded



Cognitive basis--Infant studies

- 4.5 months

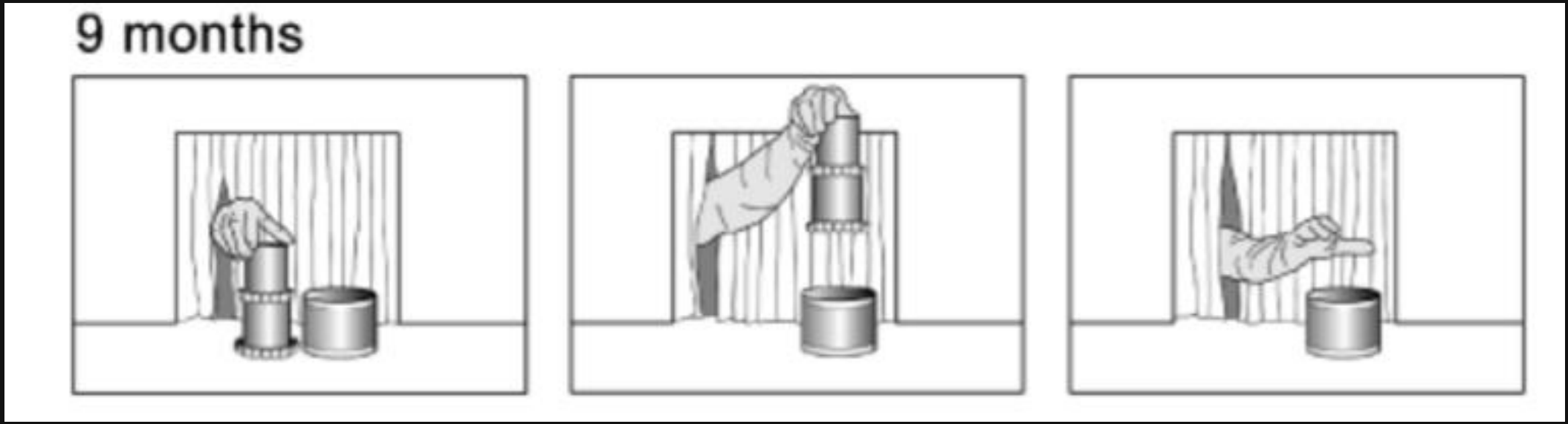


Support

Cognitive basis--Infant studies

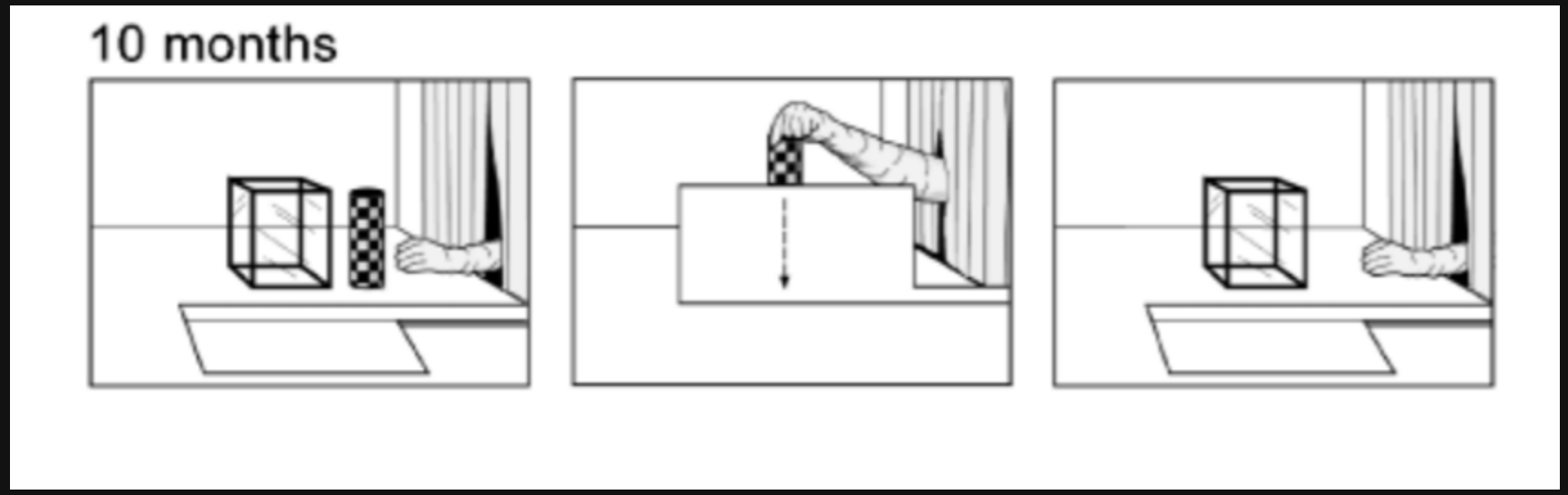
- 9.0 months

Containment



Cognitive basis--Infant studies

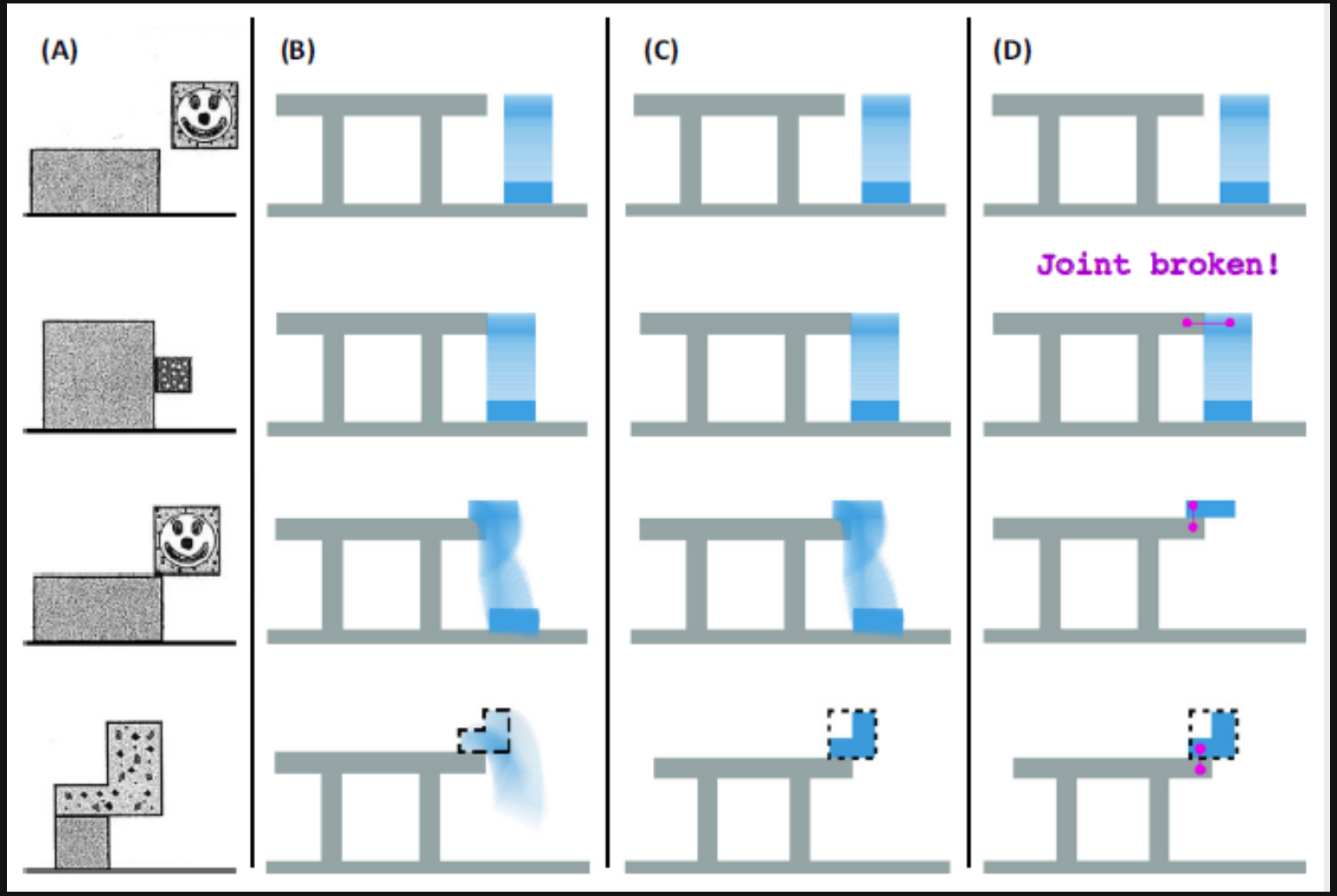
- 10.0 months



Transparency and containment

Cognitive basis--Infant studies

- Different ages



> 12.5 months 5 - 12.5 months <5 months

Cognitive basis--Deeper exploration

- Seeking explanations for impossible events

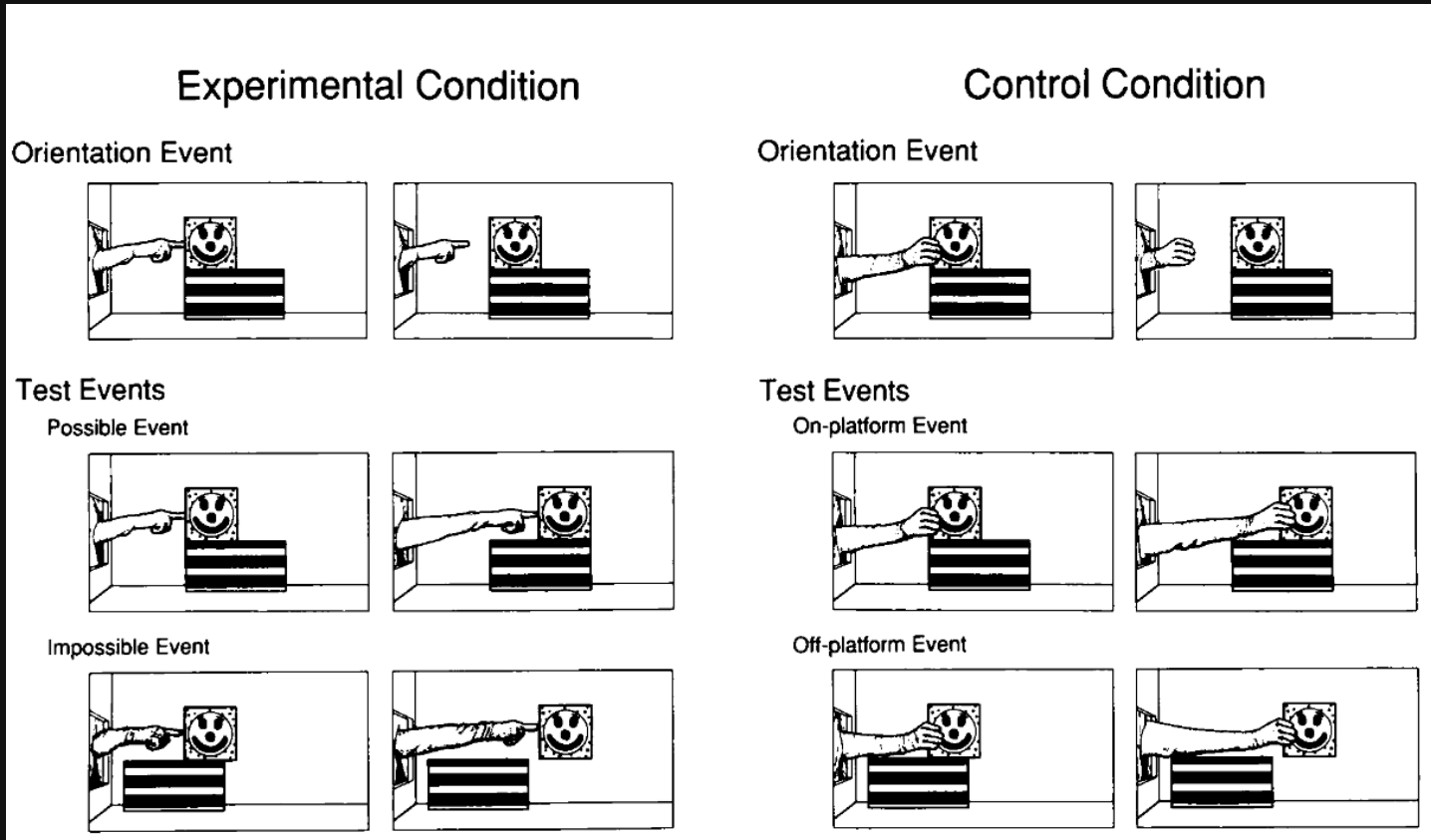


Figure 7. Schematic drawing of the orientation and test events used in Needham & Baillargeon (1994, Expt 1).

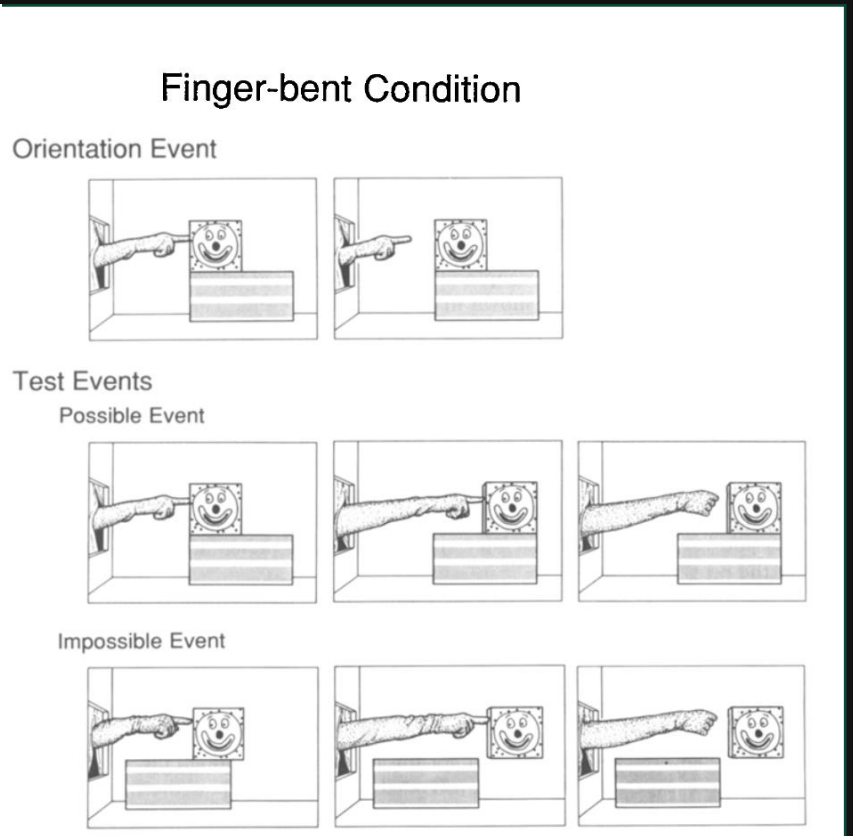
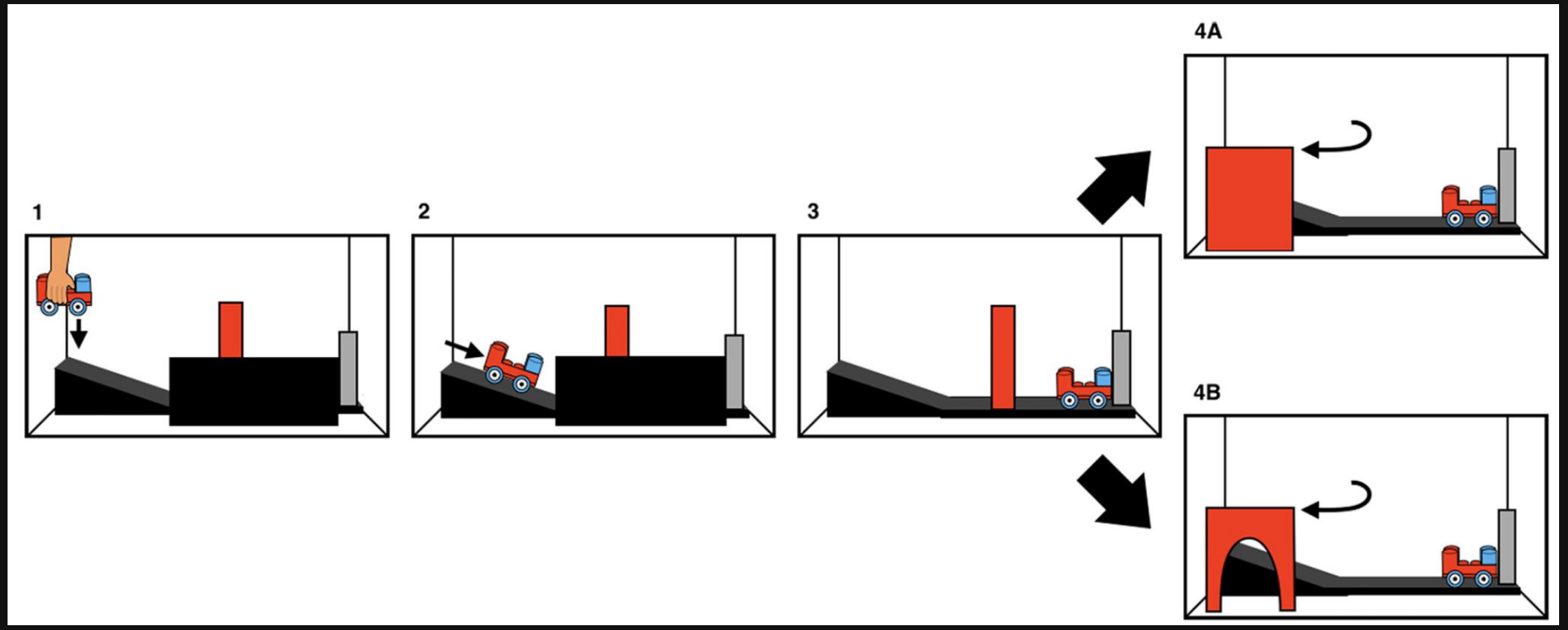


Figure 8. Schematic drawing of the orientation and test events used in Needham & Baillargeon (1994, Expt 2).

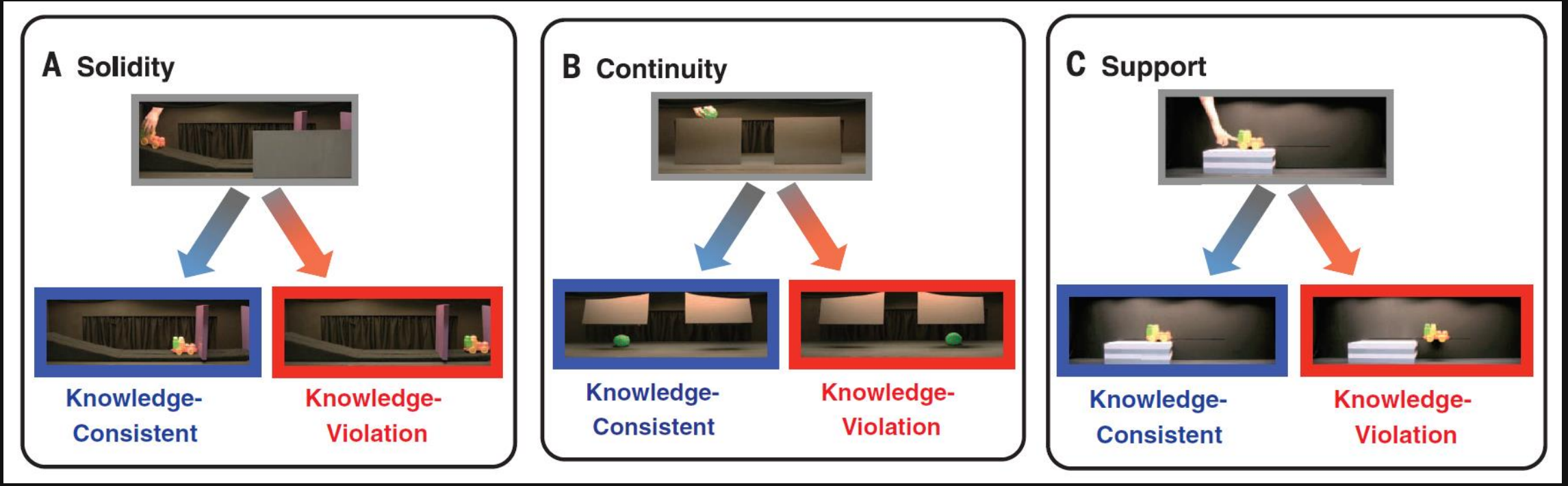
Cognitive basis--Deeper exploration

- Seeking explanations for impossible events

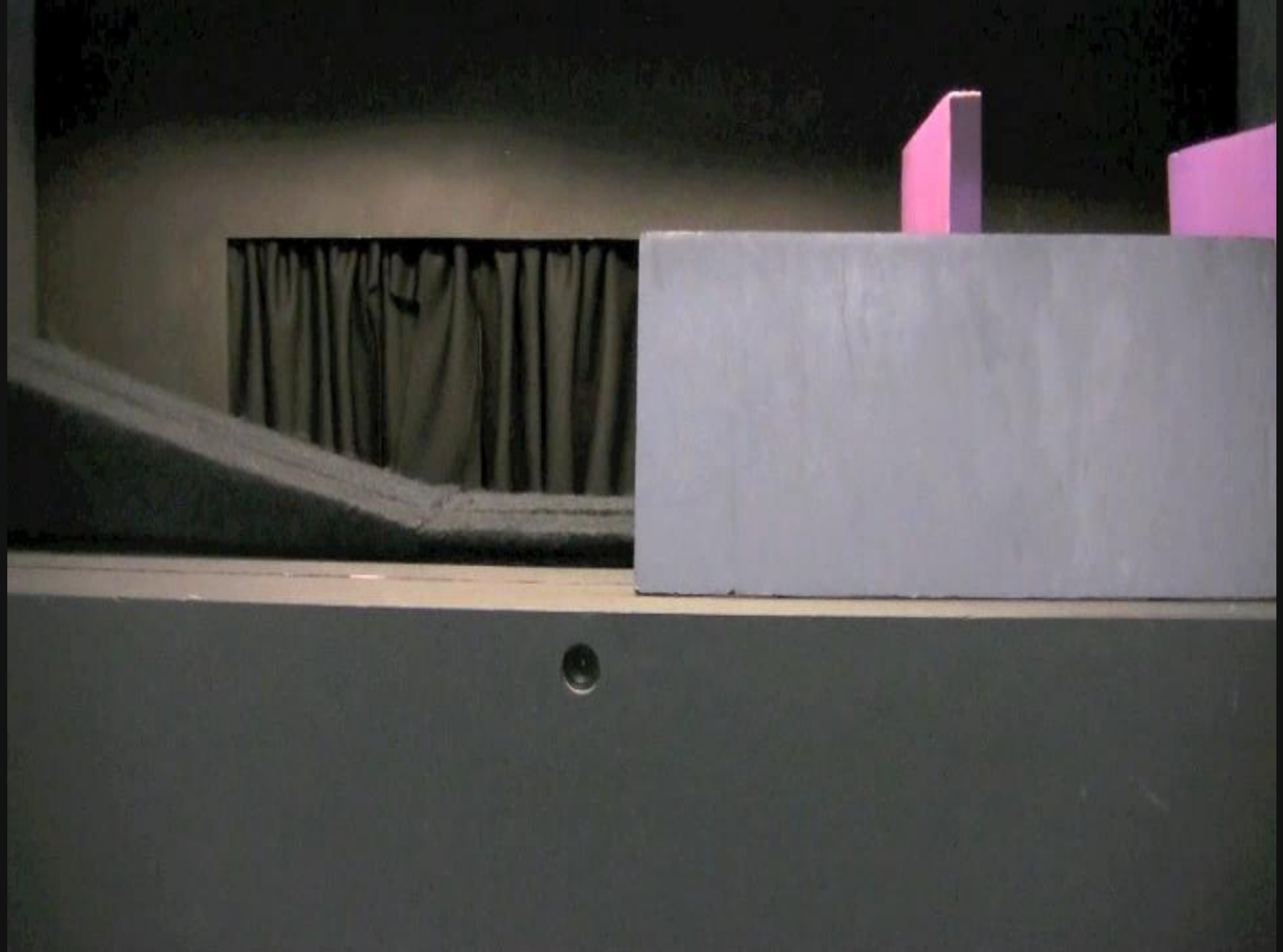


Cognitive basis--Deeper exploration

- Exploring for impossible events

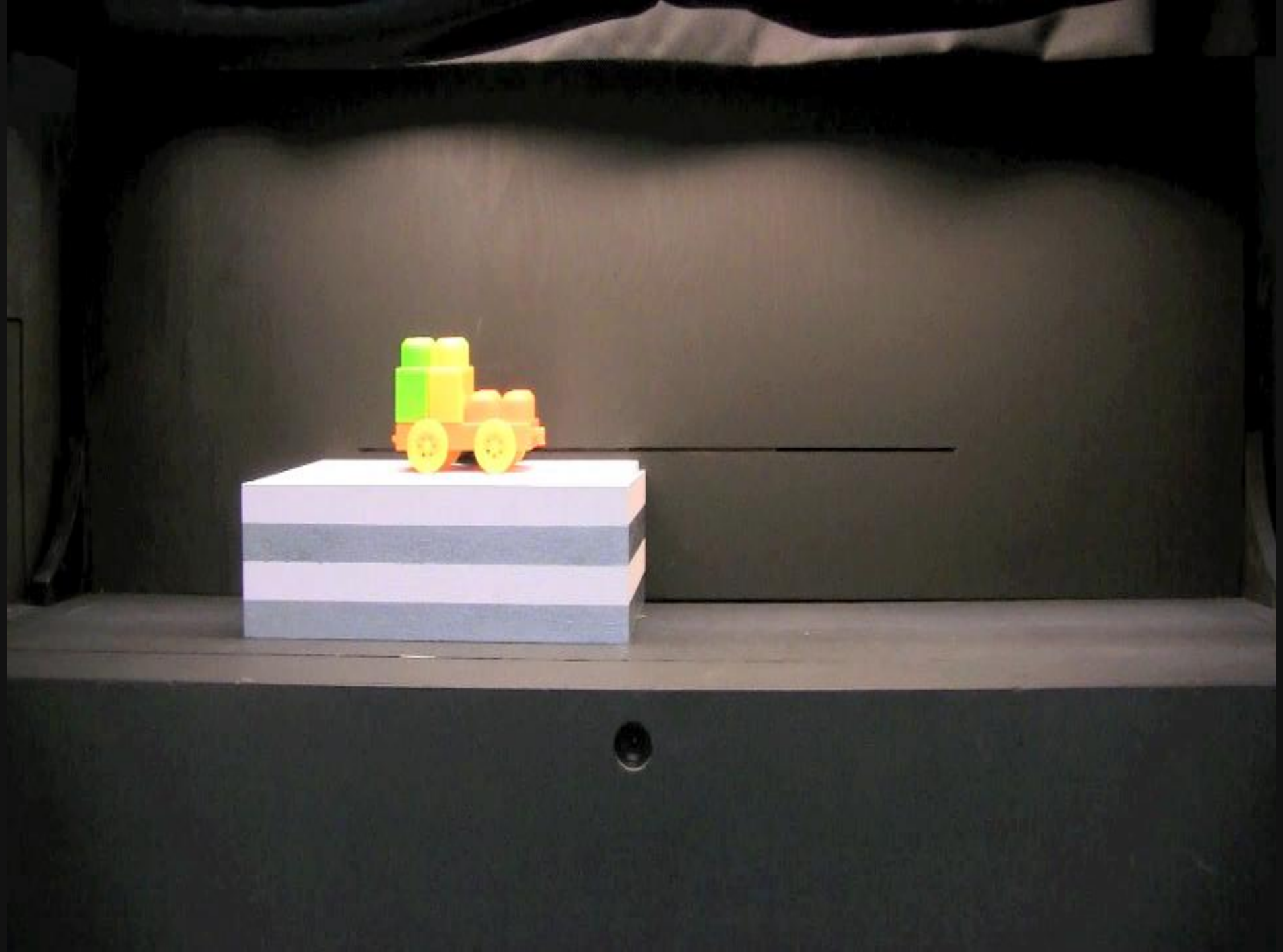






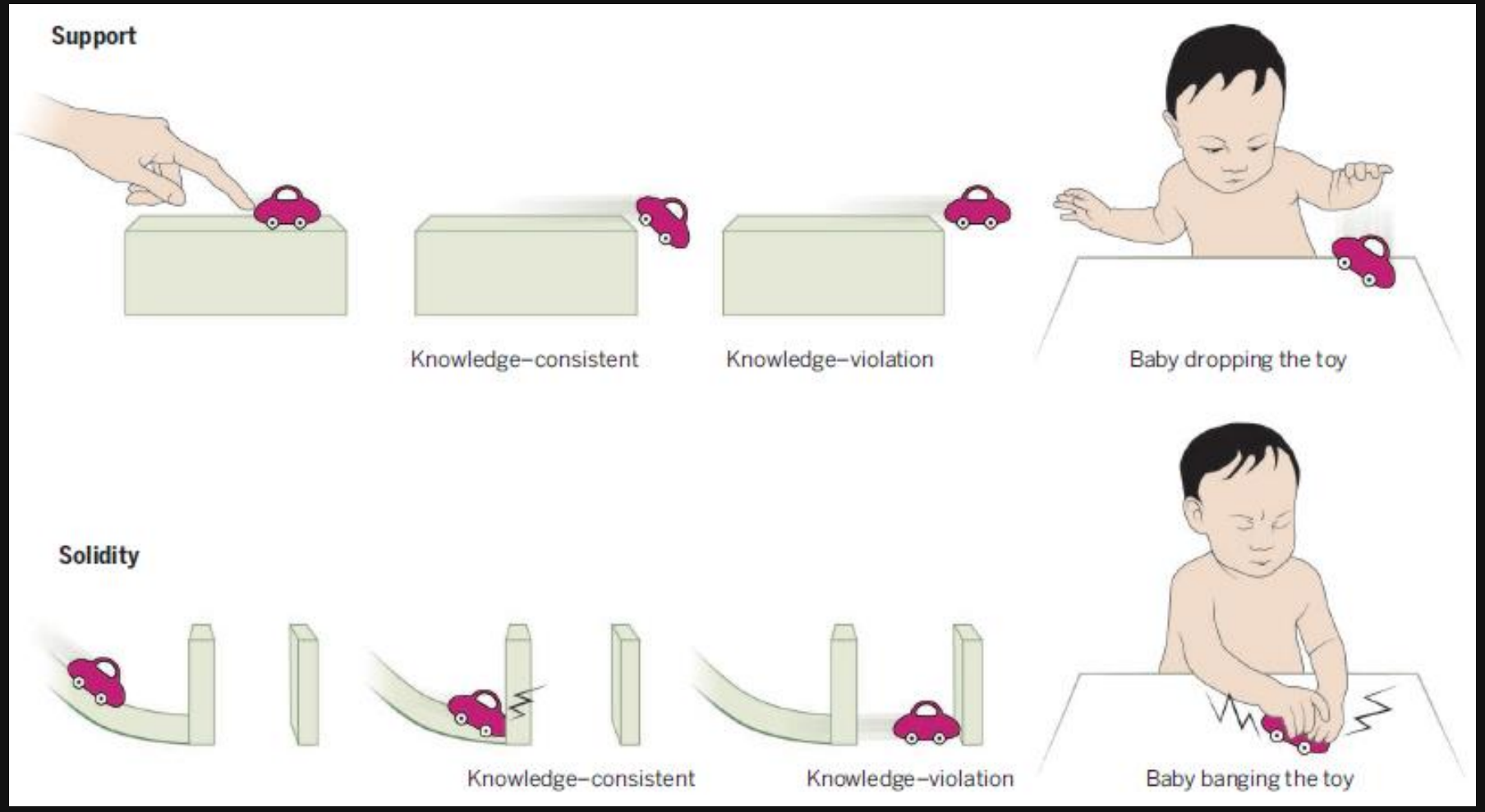






Cognitive basis--Deeper exploration

- Exploring for impossible events



Cognitive basis--Animal cognition



https://www.youtube.com/watch?v=B-HF-wBwQsc

Cognitive basis--Animal cognition

New Caledonian Crows Behave Optimistically after Using Tools

Dakota E. McCoy,[1,4,*] Martina Schiestl,[2,3] Patrick Neilands,[2] Rebecca Hassall,[2] Russell D. Gray,[2,3] and Alex H. Taylor[2]
[1]Department of Organismic and Evolutionary Biology, Harvard University, 26 Oxford Street, Cambridge, MA 02138, USA
[2]School of Psychology, University of Auckland, 23 Symonds Street, Auckland 1010, New Zealand
[3]Max Planck Institute for the Science of Human History, Jena 07745, Germany
[4]Lead Contact
*Correspondence: dakotamccoy@g.harvard.edu
https://doi.org/10.1016/j.cub.2019.06.080



Figure 1. Spatial Cognitive Bias Test
(A) Experimental aviary setup.
(B–D) Box is placed in one of three locations on the table, along with a wooden marker that does not change location. Birds learned that the box on one side (B) had a large reward (3 meat cubes); the box on the other side (C) had a small reward (¼ of a meat cube). The box in the middle (D) was ambiguous and used to test optimism.
(E–G) Crows (E) fly or walk to table, (F) remove box lid, and (G) retrieve meat from inside.

Quicker approaches imply a relatively more positive affective state, and slower approaches imply a relatively more negative affective state.

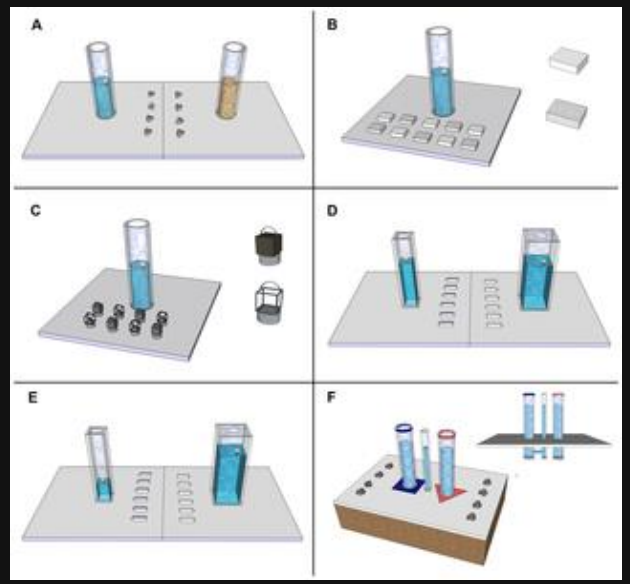
We compared latency to approach the ambiguous box after crows encountered one of four conditions, each of which resulted in the same reward amount (Figure 2). We compared (1) tool use, using a stick tool to extract one piece of food from an apparatus (Figure 2A), with (2) no tool, extracting one piece of food with their beak from the front of the same apparatus (Figure 2B). This tested whether crows were more optimistic after using a tool rather than simply after gaining food. It also controlled for the presence of the apparatus itself. However, differences between these two conditions might also

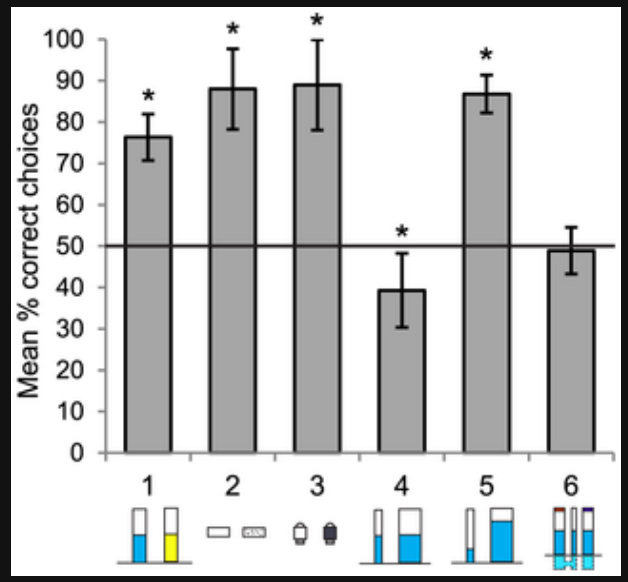
https://www.youtube.com/watch?v=B-HF-wBwQsc

Cognitive basis--Animal cognition

Crows preferentially dropped stones into a water-filled tube instead of a sand-filled tube; they dropped sinking objects rather than floating objects; solid objects rather than hollow objects, and they dropped objects into a tube with a high water level rather than a low one. New Caledonian crows possess a sophisticated, but incomplete, understanding of the causal properties of displacement, rivalling that of 5–7 year old children.



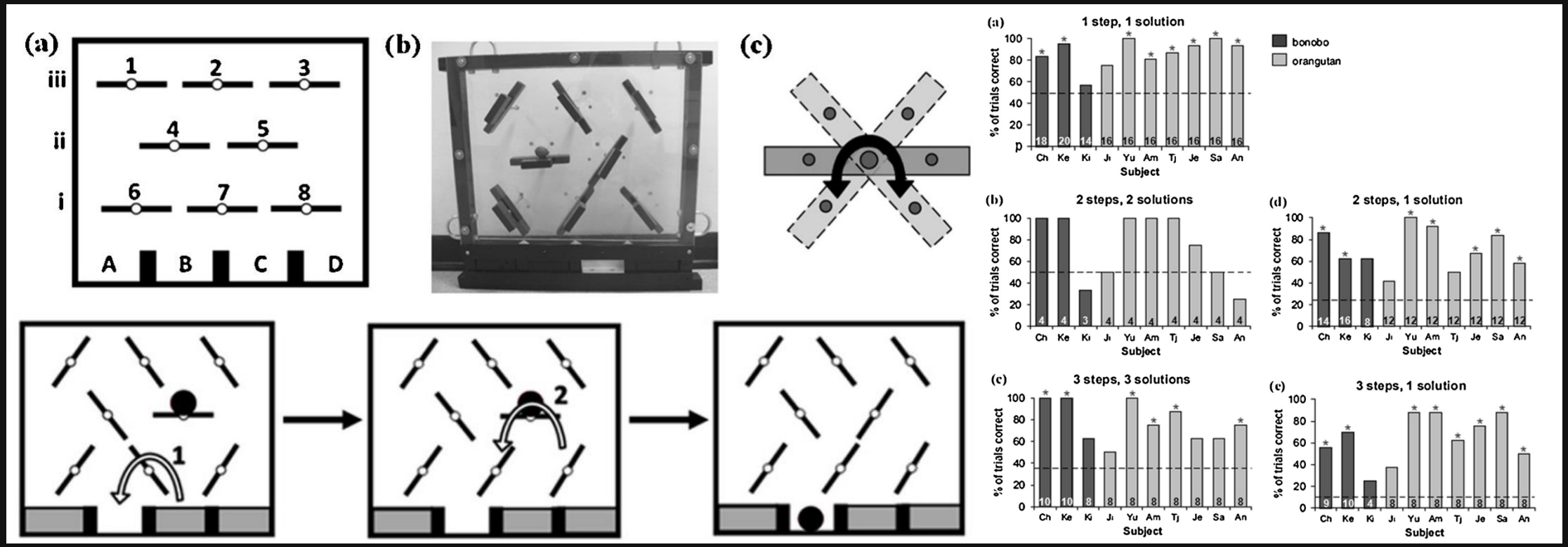




Cognitive basis--Animal cognition

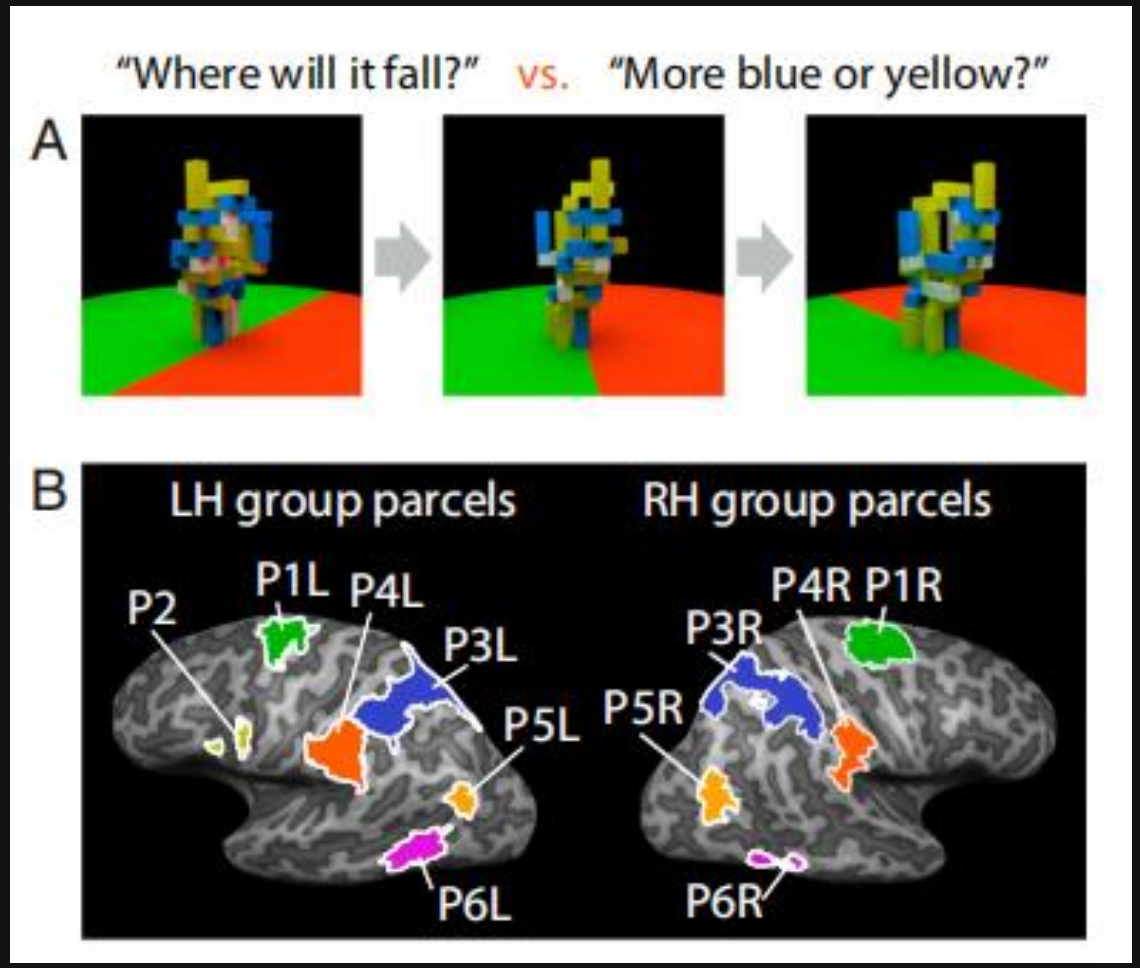
Task: Rotate the paddles to make the ball fall into a specific hole.

Conclusion: Great apes can effectively plan actions step-by-step to solve complex physical reasoning tasks.



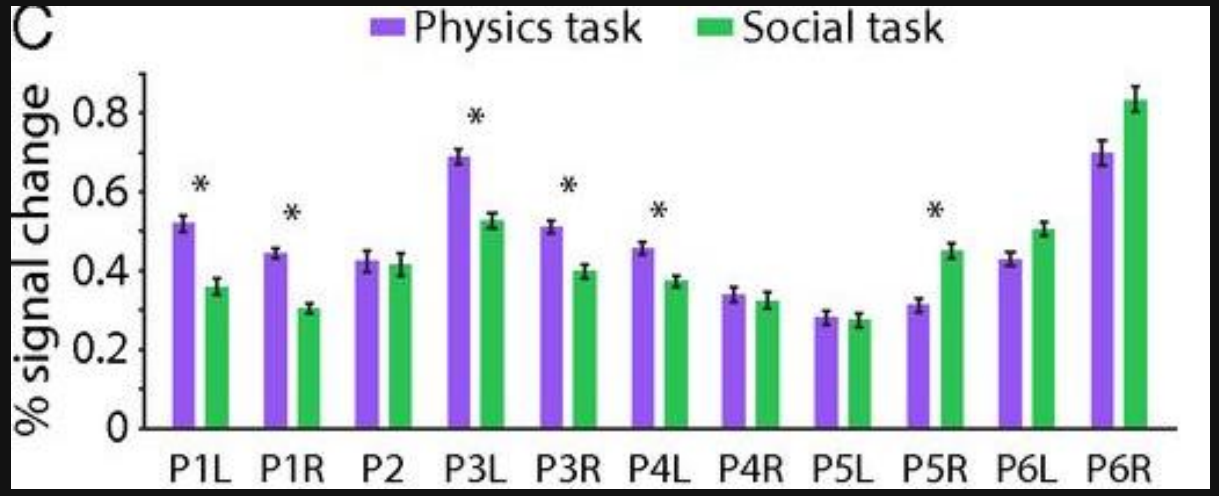
Cognitive basis--Neural basis

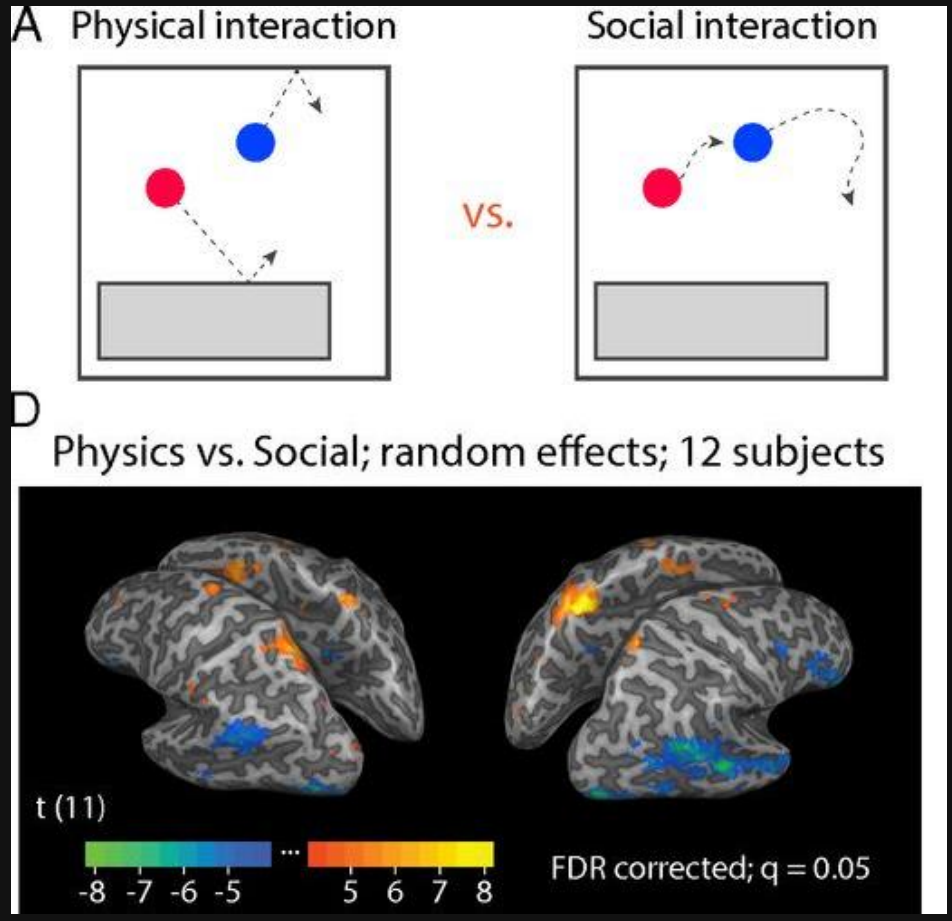
- There is a group of brain areas related to physical reasoning: including the bilateral dorsal premotor cortex（双侧背侧前运动皮层）, supplementary motor area（辅助运动区）, parietal cortex （顶叶皮层）and left supramarginal gyrus（左侧缘上回）



Cognitive basis--Neural basis

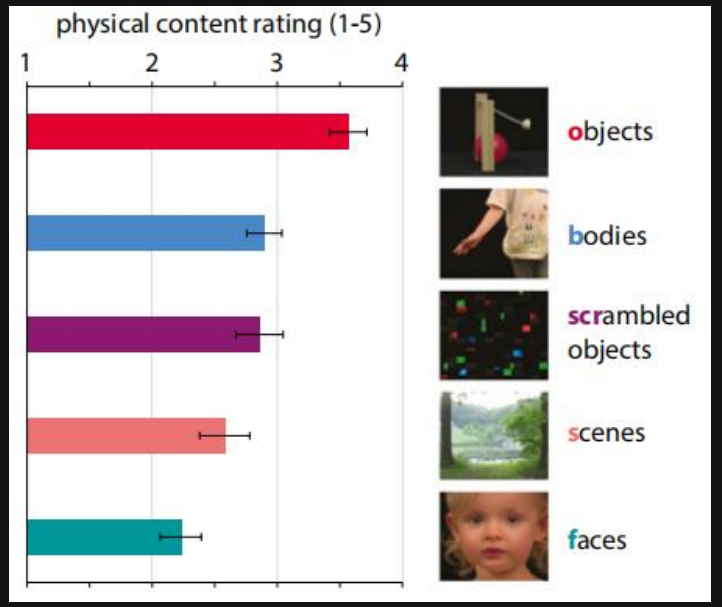
- Also, specific brain areas fire in physical interaction tasks but not in social interaction tasks.

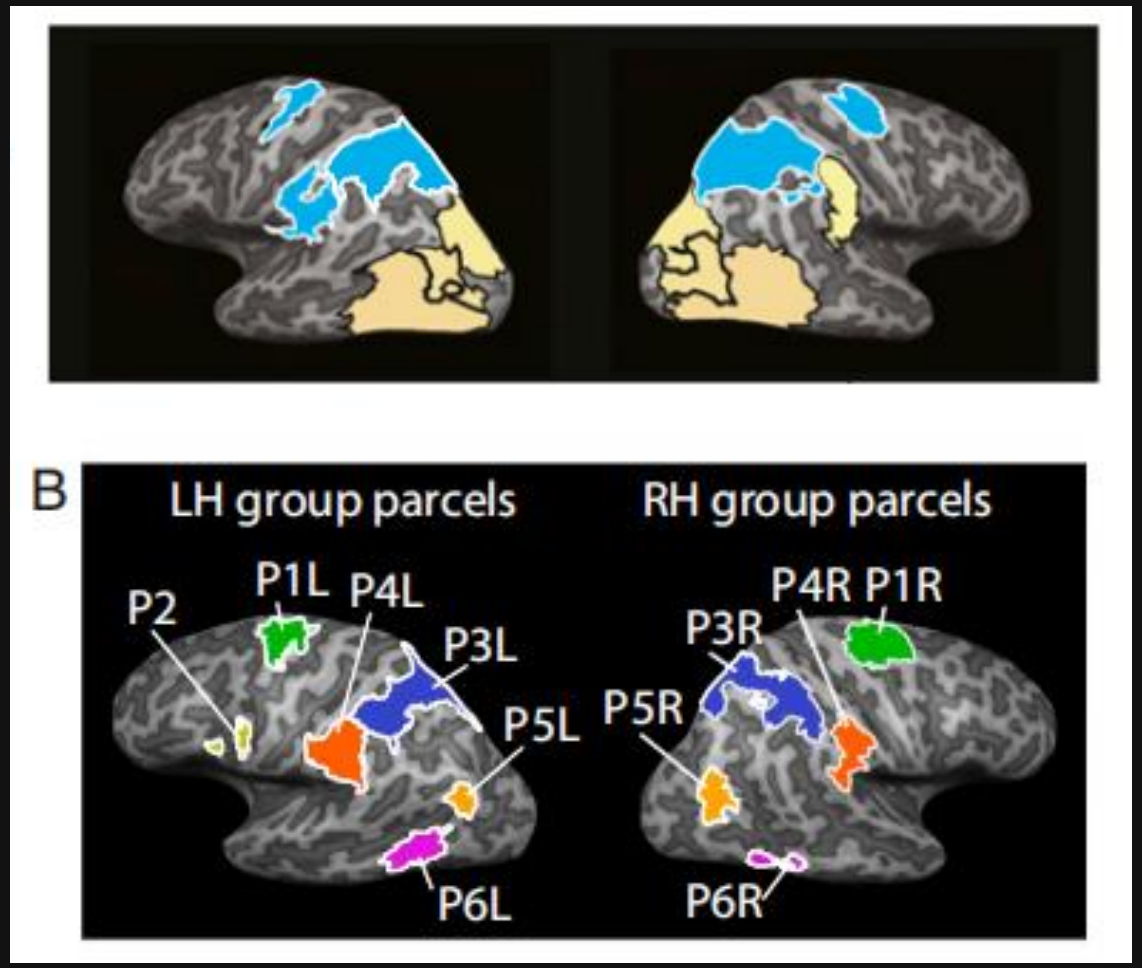




Cognitive basis--Neural basis

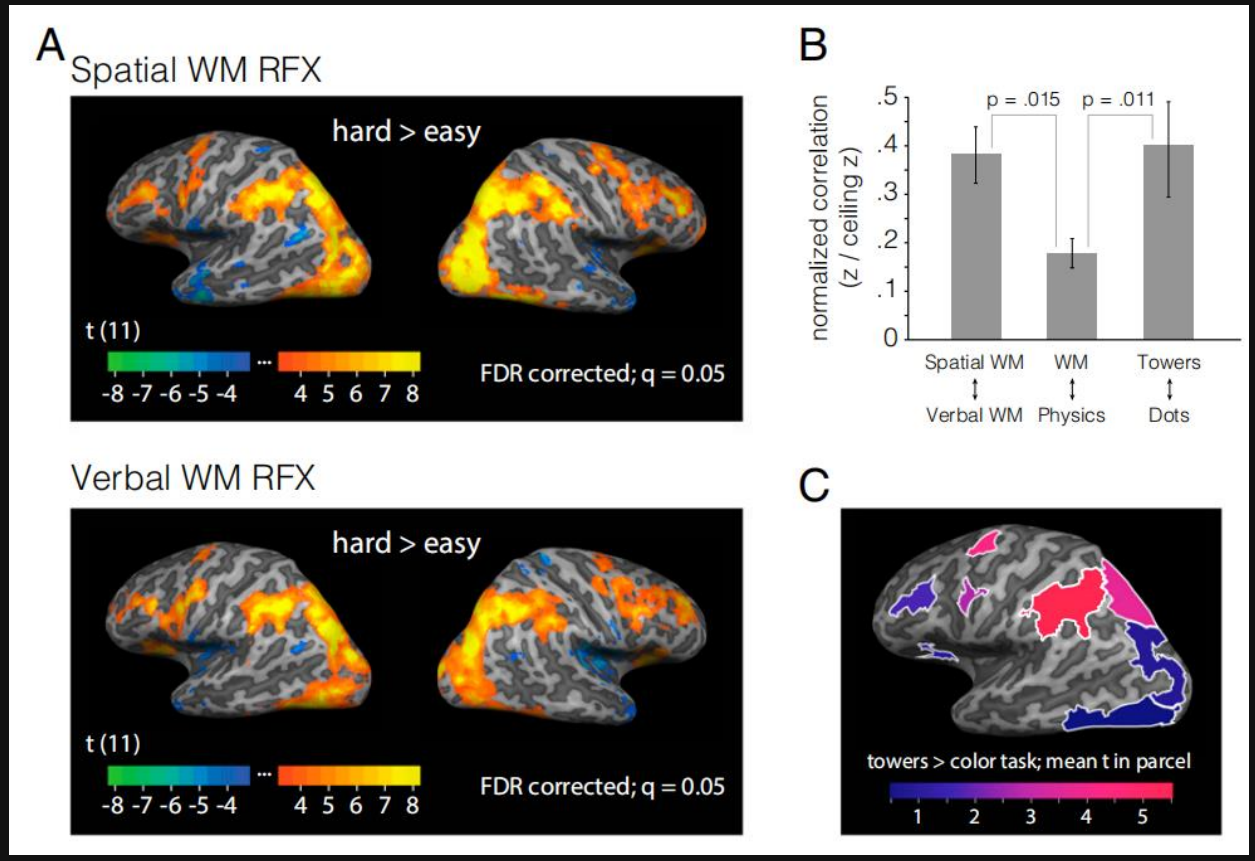
- These regions are selective for physics content: they activate not only during explicit tasks but also when passively watching videos rich in physics content.



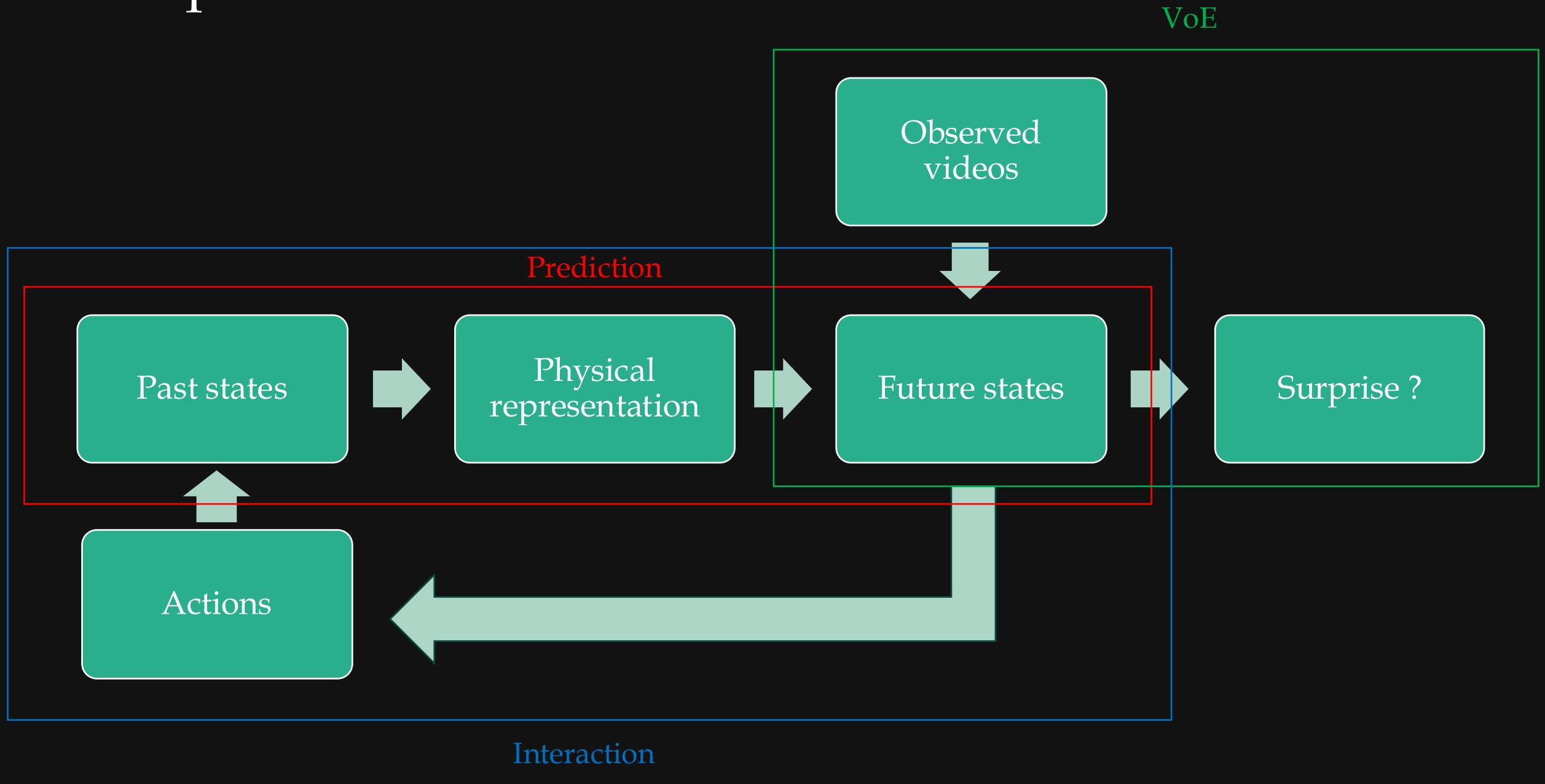


Cognitive basis--Neural basis

- The physical reasoning region overlaps with the MD network responsible for higher-level cognitive tasks such as working memory, but is not completely identical.
- Physical reasoning areas also overlap significantly with brain regions associated with action planning and tool use.



Computational mechanisms



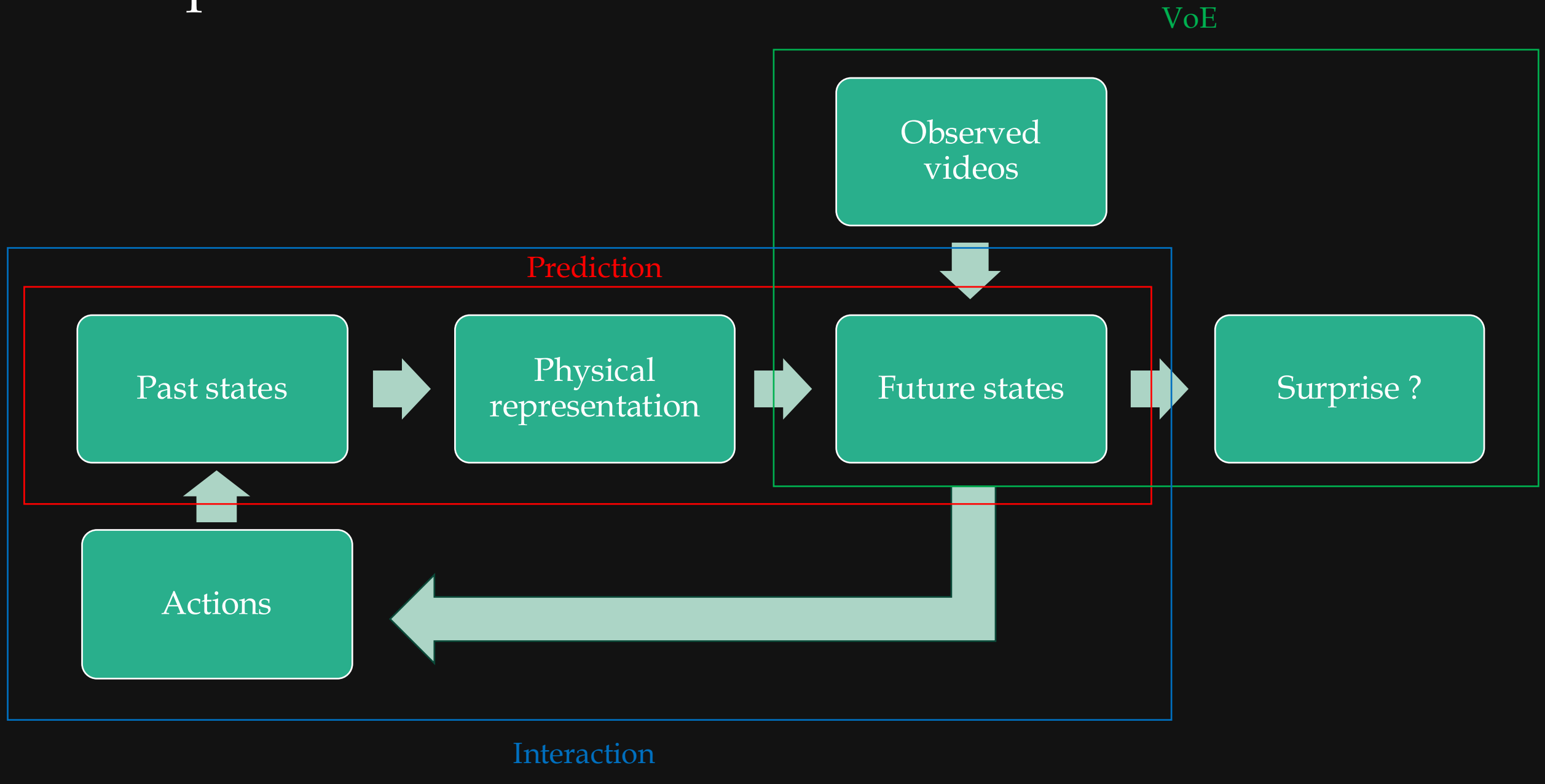
Intuitive physics history summary

Phase	Time	Domain	Key Study / Event	Representative Work	Core Findings / Contributions
Foundational Phase	1980s	Infant Physical Cognition	"Car rolls through an obstacle behind a screen"	Baillargeon (1985, Cognitive Psychology)	Infants look longer at impossible events, showing expectations about object permanence and solidity.
Deeper Exploration	1990s	Infant Physical Cognition	Infants fail to look preferentially at the impossible event stems	Baillargeon (1994)	Infants seek explanations for impossible events.
Animal Cognition	2000s	Animal Cognition	Monkeys show longer attention to impossible stacks	S. A. Jelbert (2014)	Using the Aesop's fable paradigm to investigate causal understanding of water displacement by New Caledonian crows.
Neural Basis	2010s	Cognitive Neuroscience	fMRI/EEG shows activation during physical tasks	Nancy Kanwisher (2016)	A variety of physical inference tasks as well as simply viewing physically rich scenes engage a common brain network in frontal and parietal cortices.
Computational mechanisms	2010-present	Machine Learning	IntPhys and PLATO uses video of physics violations	Riochet et al. (2019) Luis S. Piloto. (2022)	Benchmarks AI models (CNNs, RNNs) on predicting physically consistent video scenes; showing surprise with VoE.

Part 3
Computational models

VoE model, prediction model, interaction models

Computational mechanisms



VoE



PLATO

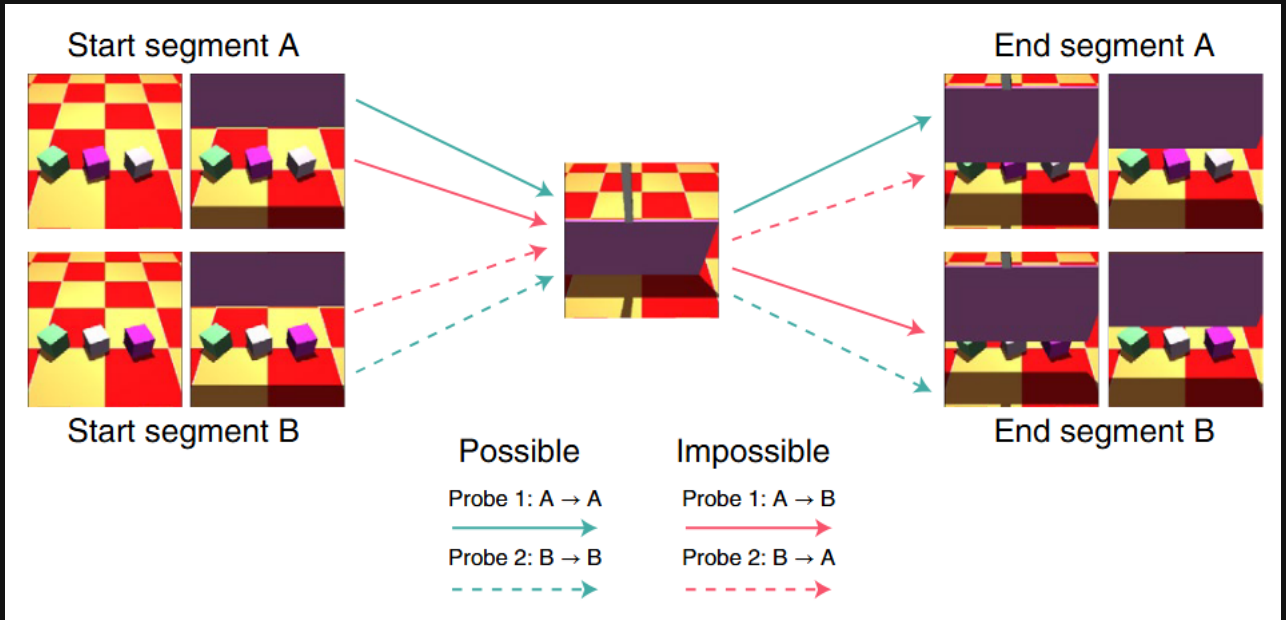
- Physical concepts dataset: contains VoE probe videos targeting five physical concepts:
  - Continuity
  - Object persistence
  - Solidity
  - Unchangeableness
  - Directional inertia

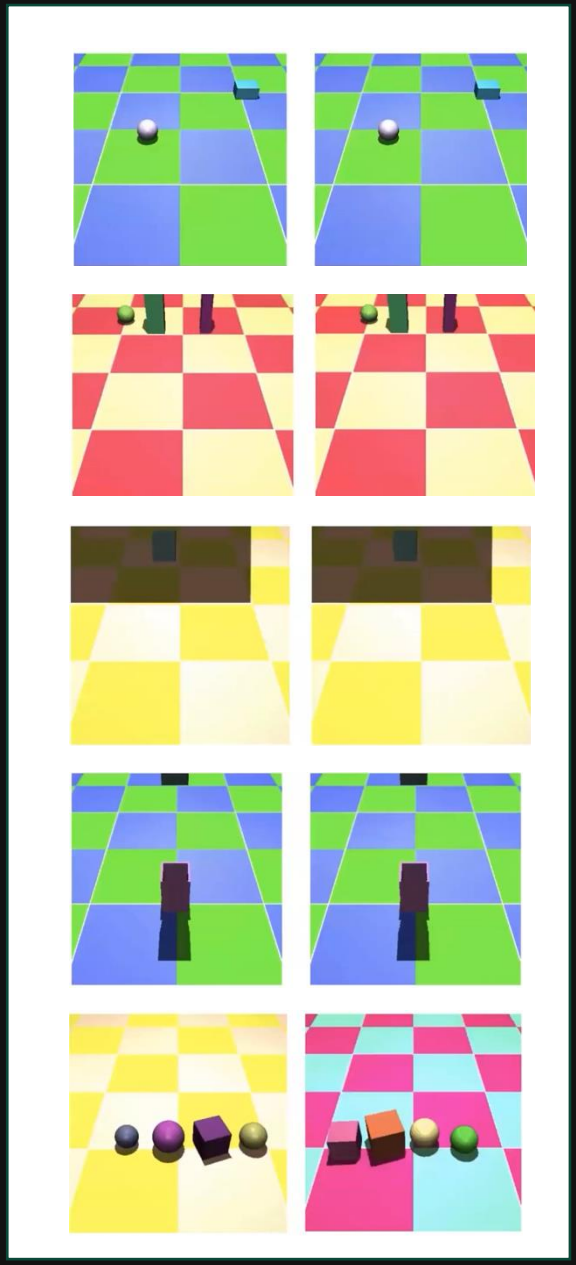


Piloto, L. S., Weinstein, A., Battaglia, P., & Botvinick, M. (2022). Intuitive physics learning in a deep-learning model inspired by developmental psychology. Nature human behaviour, 6(9), 1257-1267.

PLATO

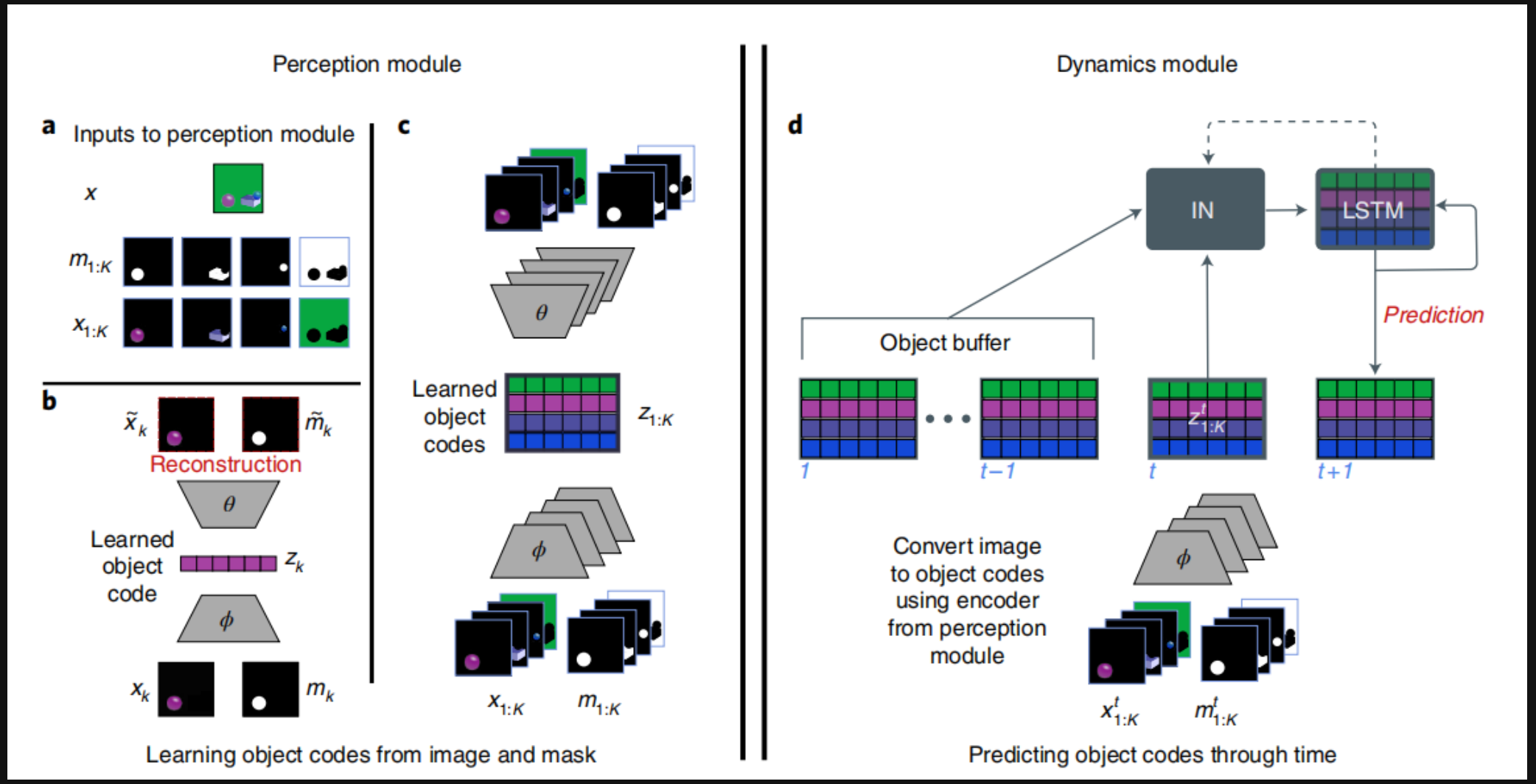
Construct VoE dataset





PLATO

- PLATO: Physics Learning through Auto-encoding and Tracking Objects



PLATO

- Object-centric parsing outperforms pixel-based parsing.
- History memory is important to the occluded tasks.

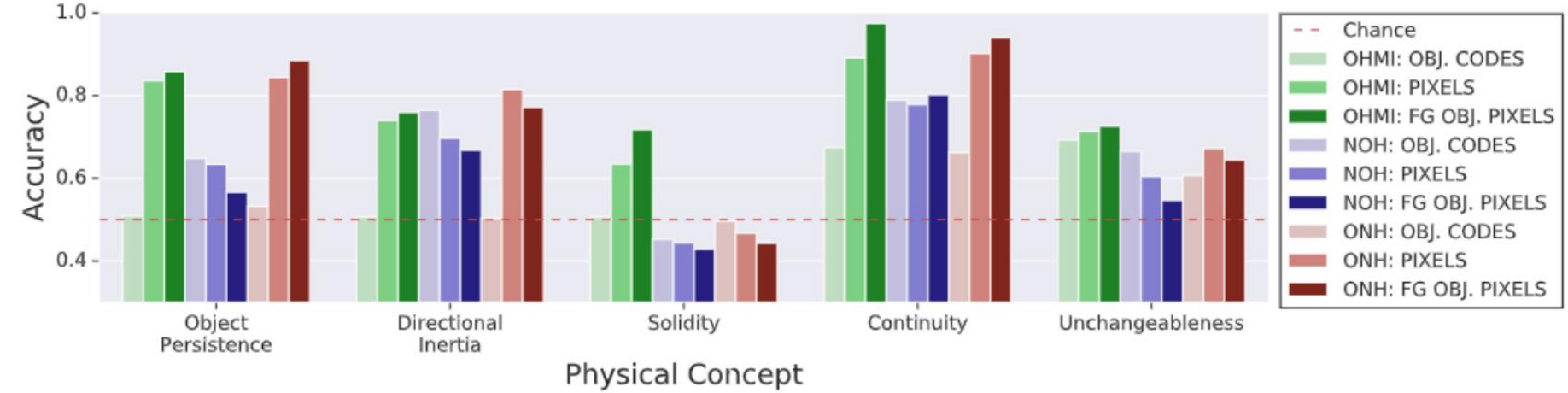
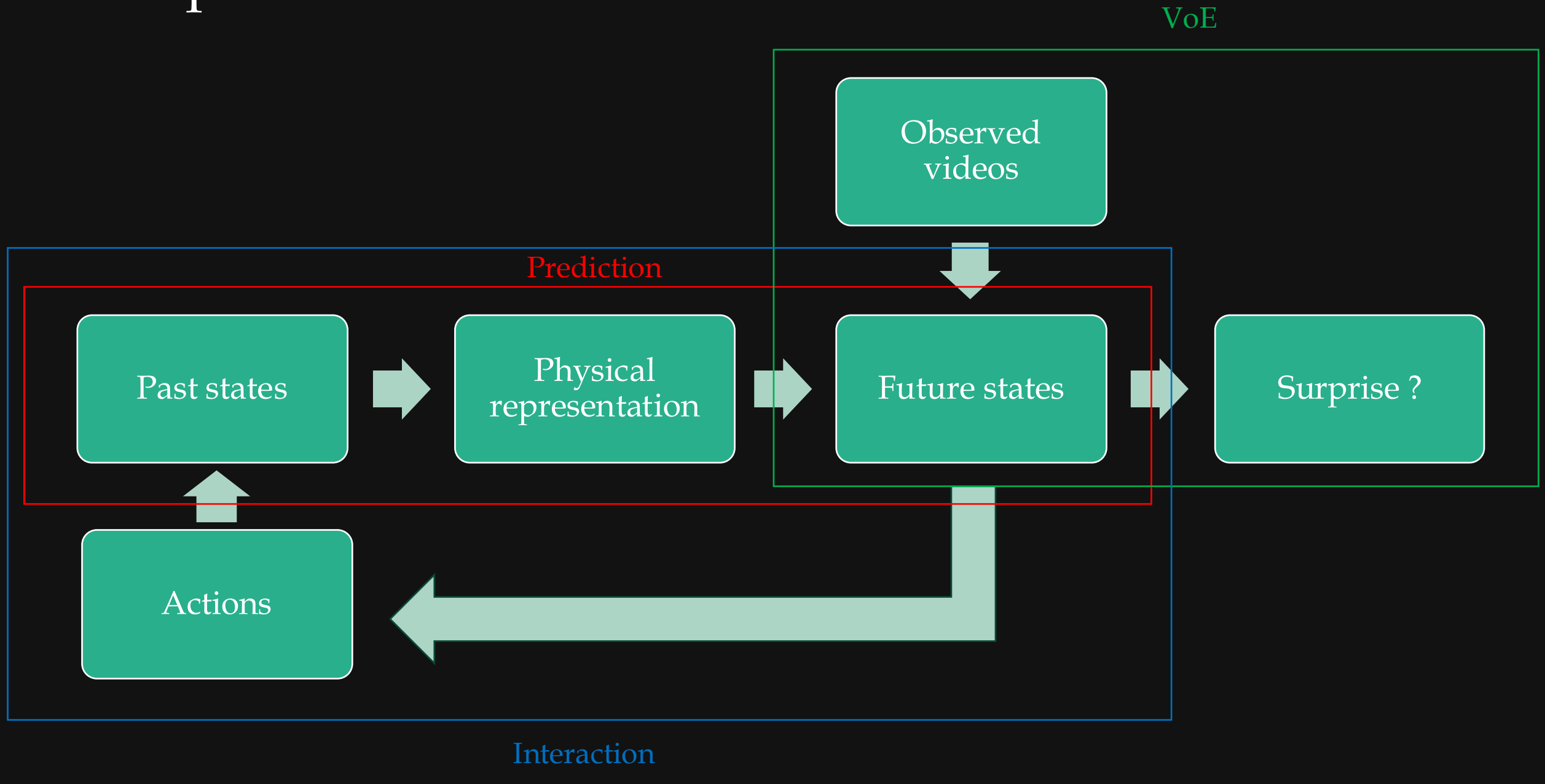


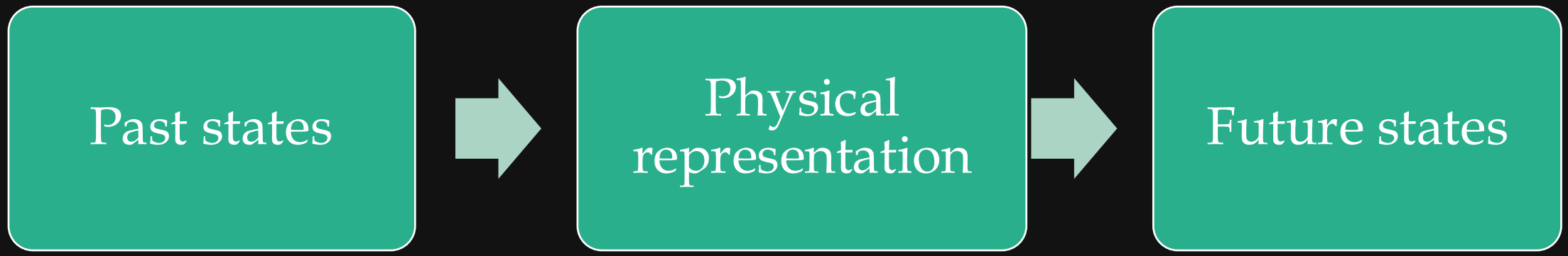
Figure 1: Comparison of model performance on VOE tests for five different physical concepts measured via classification accuracy. Models compared: **OHMI** uses interactions between object histories and memories; **NOH** *No Objects/With History*; **ONH** *Objects/No History*.

Piloto, L. S., Weinstein, A., Battaglia, P., & Botvinick, M. (2022). Intuitive physics learning in a deep-learning model inspired by developmental psychology. Nature human behaviour, 6(9), 1257-1267.

Computational mechanisms



Prediction



Prediction

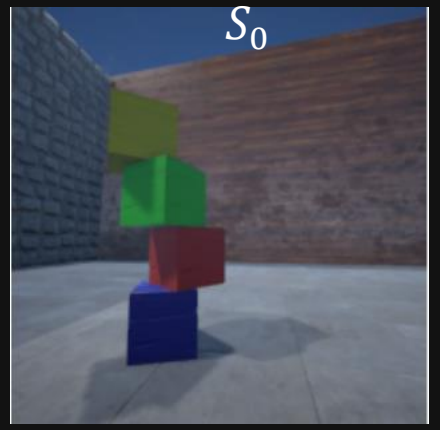
$$P(J|S_0) = \int_{S_{1:T}} P(J|S_0, S_1, \dots, S_T) \prod_{t=0}^{T-1} P(S_{t+1}|S_t)\, dS_{1:T}$$

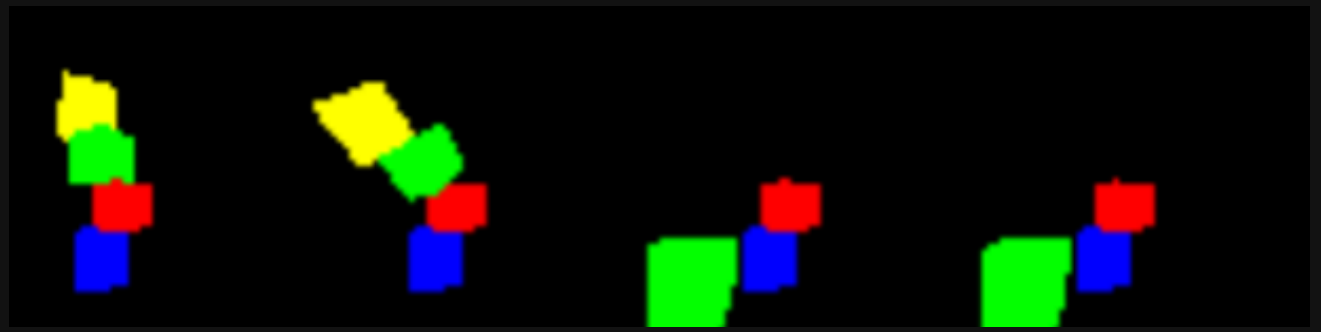
$P(S_0)$
Initial physical scenes:
S_0

$\prod_{t=0}^{T-1} P(S_{t+1}|S_t)$
Physical predictions:
$S_1, S_2, \dots, S_T$

Judgments: J

Dynamic



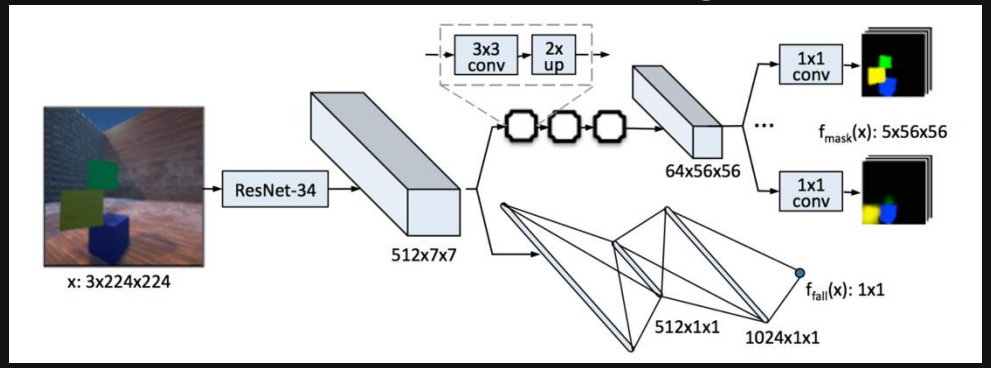


The block tower will fall down

Heuristic



$P(J|S_0) = f(S_0; \theta)$
A direct mapping

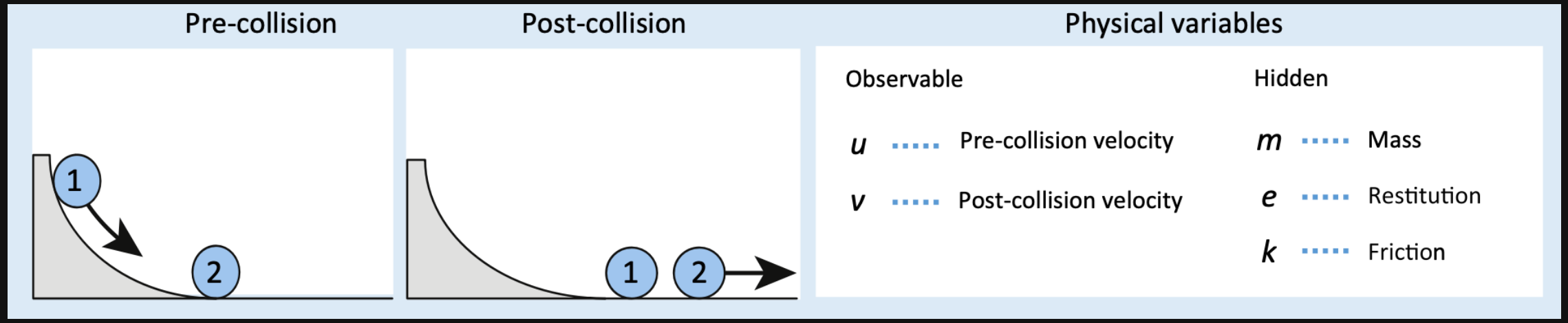


The block tower will fall down

Lerer, A., Gross, S., & Fergus, R. (2016). Learning physical intuition of block towers by example. ICML

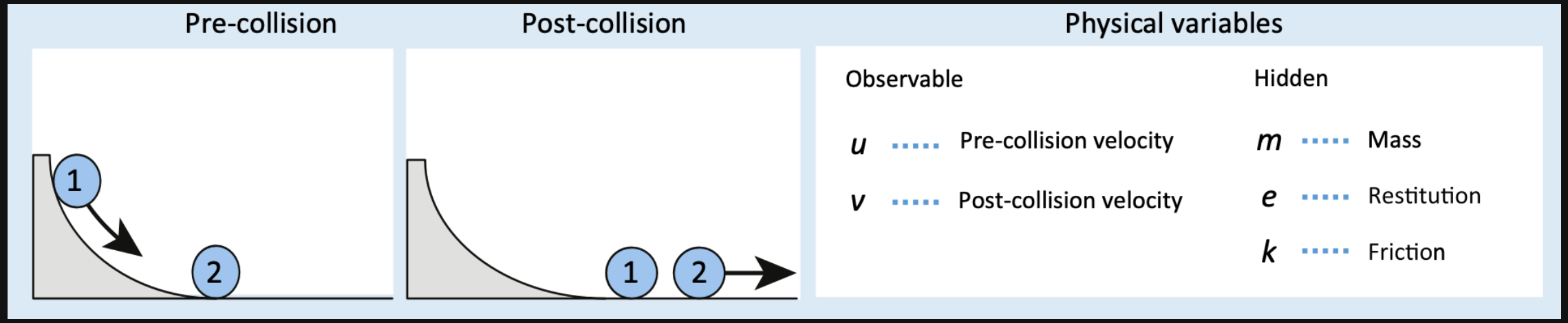
Heuristics

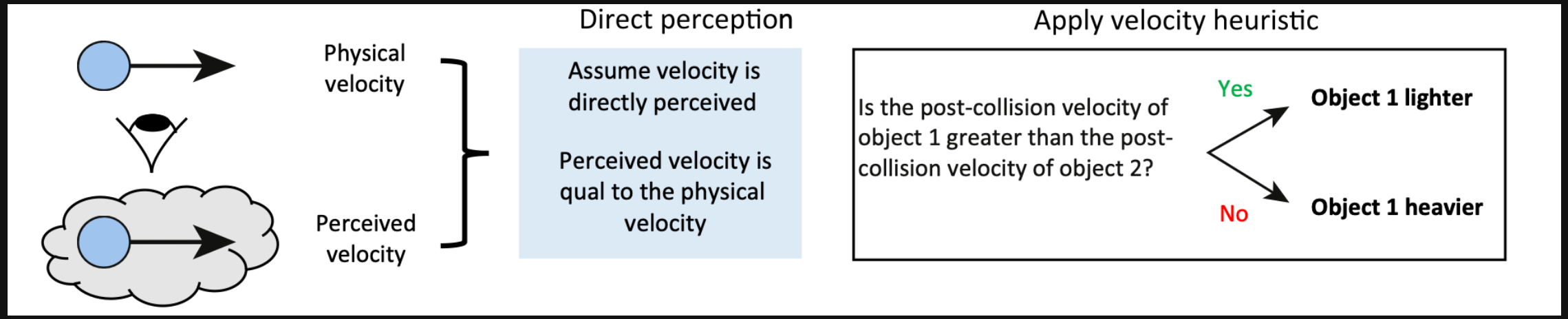
- Problem: Judging the relative masses of two colliding objects



Heuristics

- Problem: Judging the relative masses of two colliding objects





Heuristic

- Direct perception (law of conservation of momentum)

$$m_a v_a + m_b v_b = m_a u_a + m_b u_b \quad \Rightarrow \quad \frac{m_a}{m_b} = \frac{u_b - v_b}{v_a - u_a}.$$

- Heuristic

The three hypotheses about information for relative mass compared in experiment 2 can be summarized in the following way. The final-speed hypothesis can be stated in the form of a biconditional about the judgment of relative mass:

$$m_a > m_b \longleftrightarrow |v_a| < |v_b| ;$$

the direction hypothesis can be similarly stated as

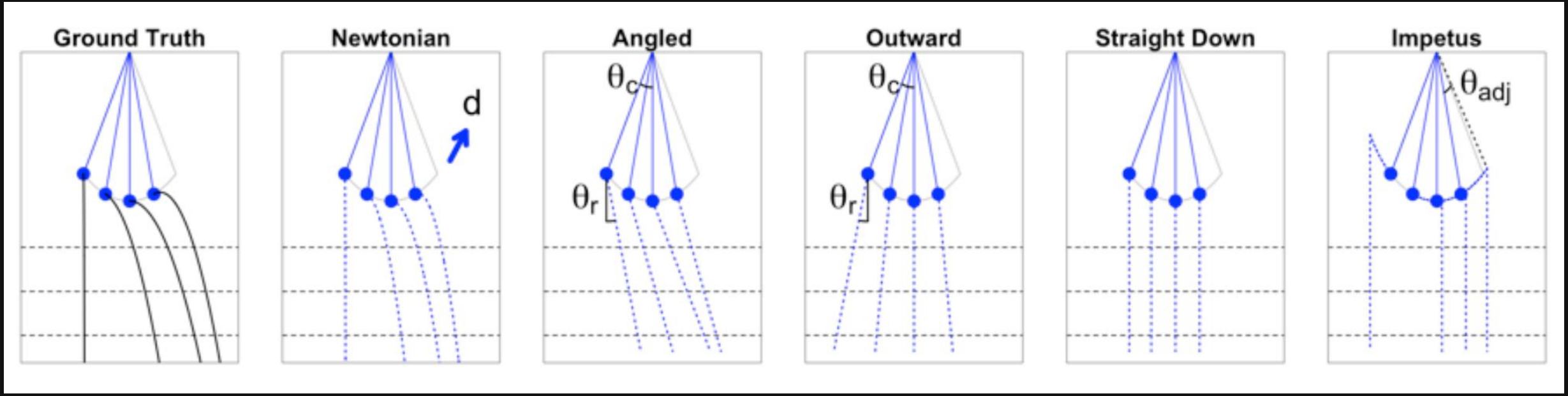
$$m_a > m_b \longleftrightarrow v_a > 0 ;$$

and the initial/final speed hypothesis can be stated as

$$m_a > m_b \longleftrightarrow v_b > u_a .$$

Heuristics

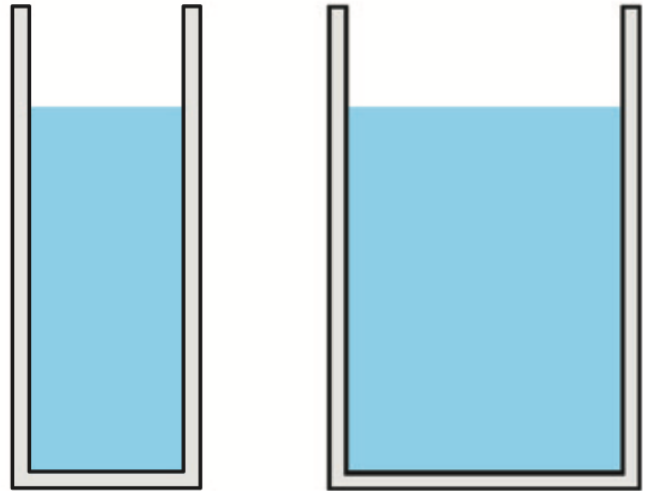
- Problem: Predict the falling position of a released ball



Heuristics

- Problem: Pouring water test

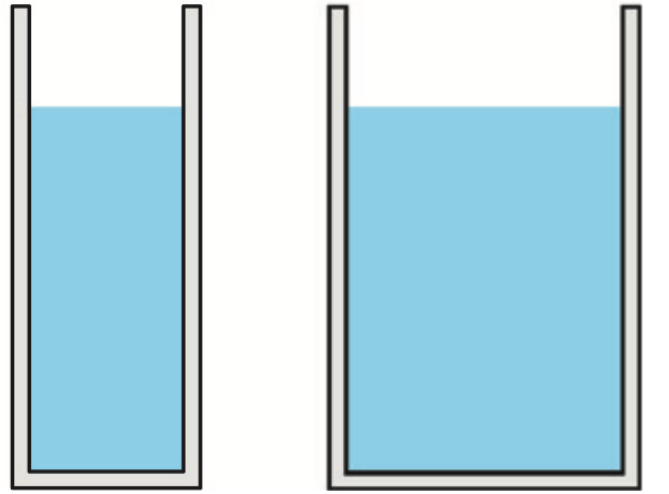
Two containers are depicted. If the containers are tilted, will they pour at the same angle? If not, which will pour first?



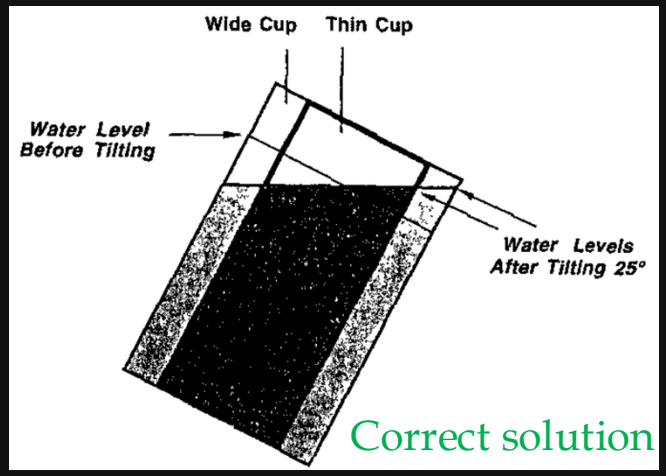
Heuristics

- Problem: Pouring water test

Two containers are depicted. If the containers are tilted, will they pour at the same angle? If not, which will pour first?

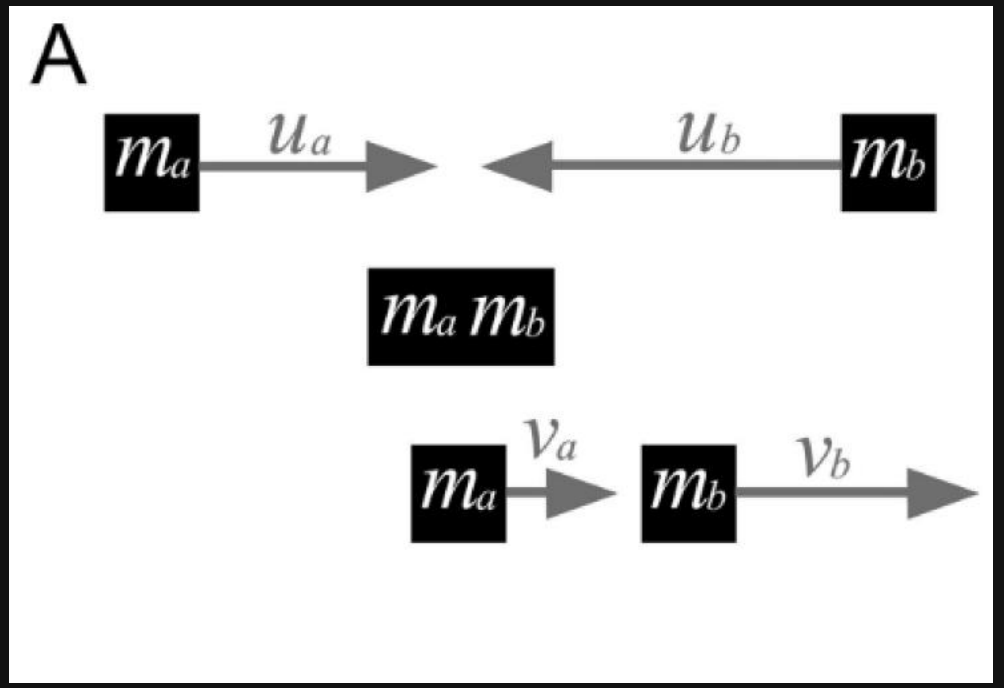


- People may rely on a wrong heuristic that the thin cups are more prone to spill out water due to smaller capacities.



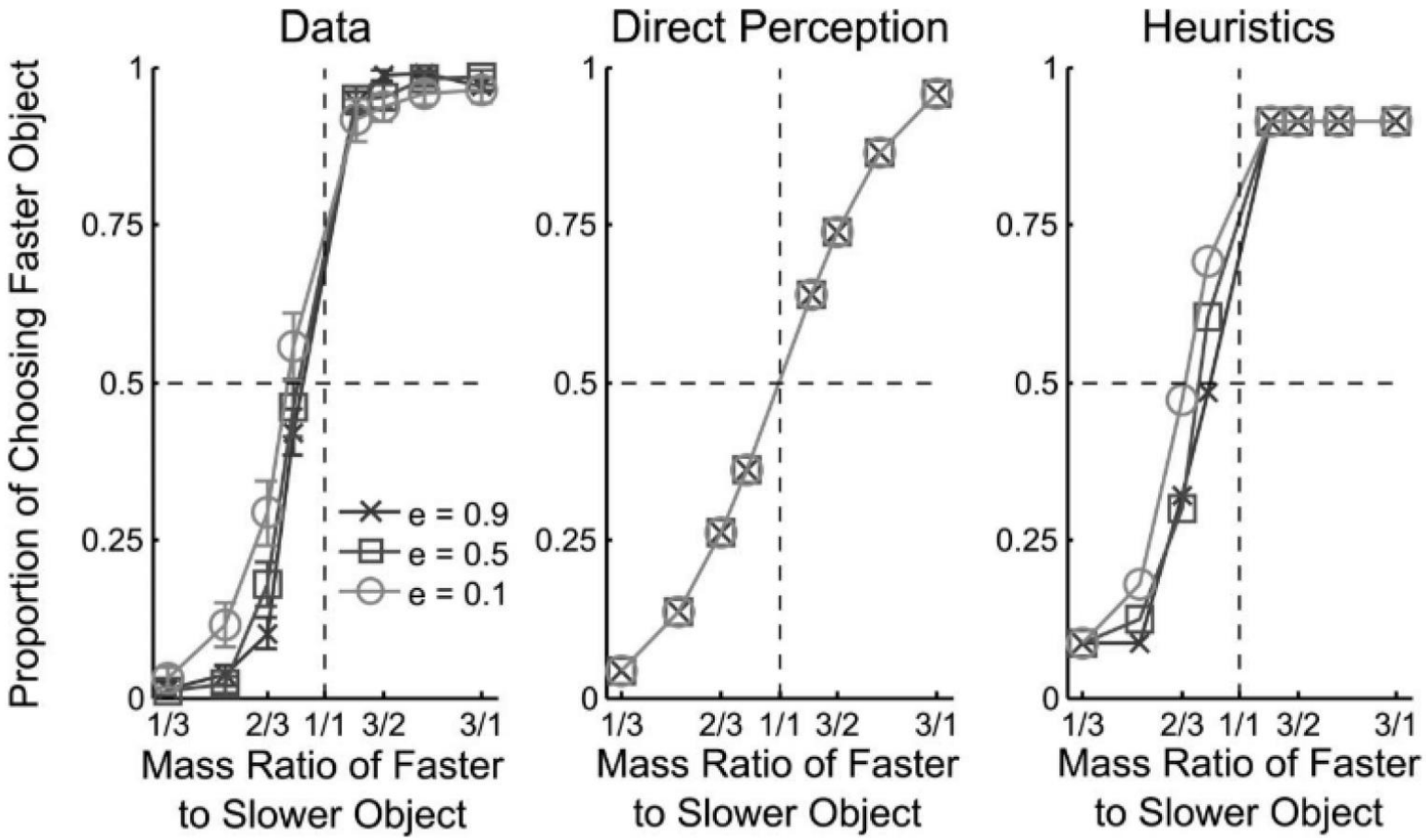
Dynamic

- Problem: Judging the relative masses of two colliding objects



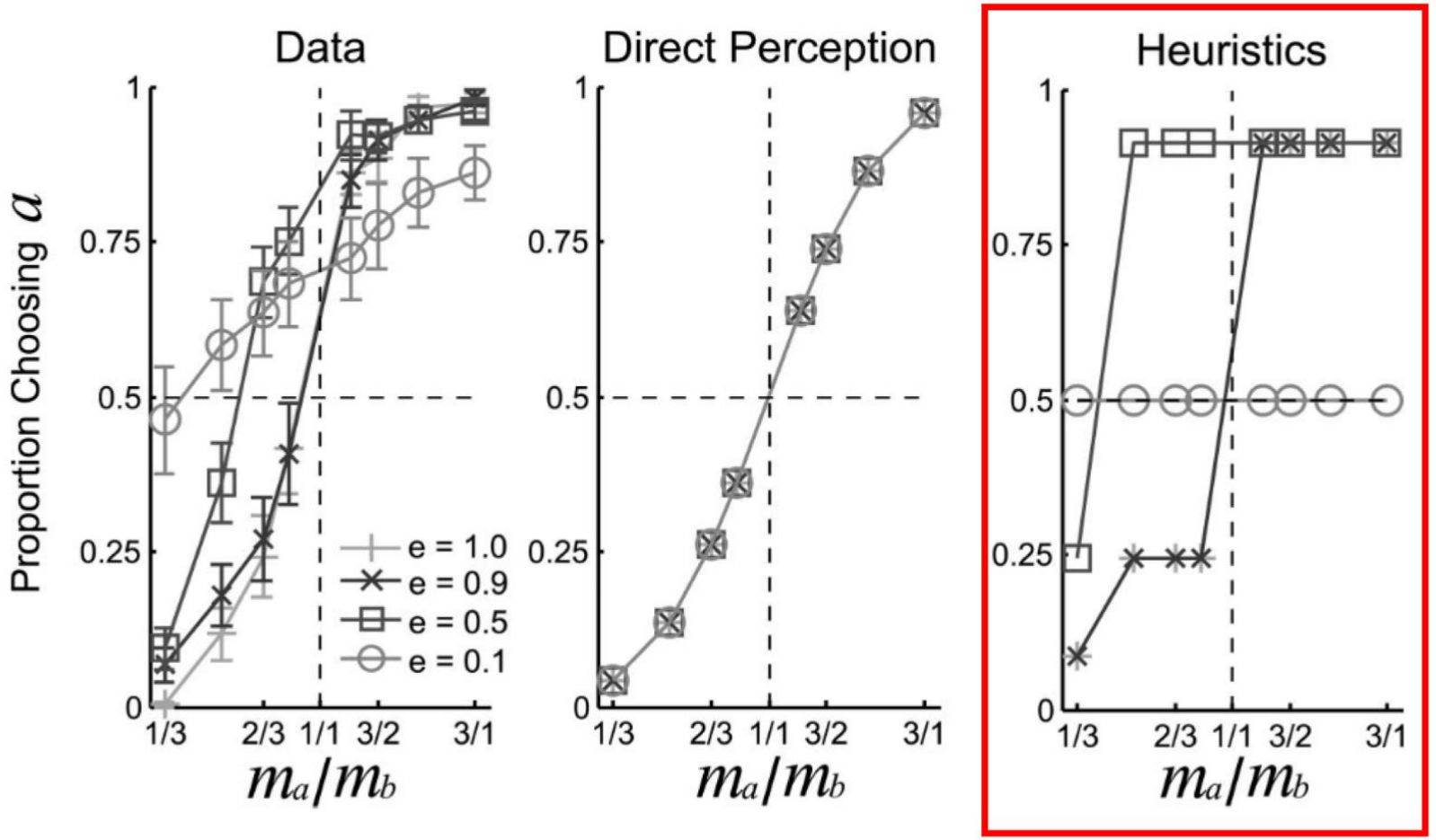
Dynamic

- Experiment 1: Two balls moving towards each other



Dynamic

- Experiment 2: Ball A moves towards Ball B



Dynamic

- Noisy Newton Model

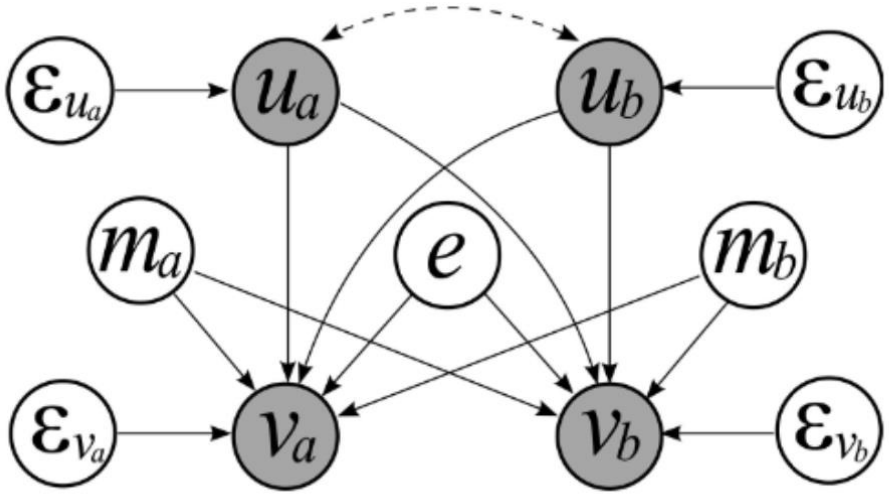


Figure 2. Graphical model showing the dependency between variables for the noisy Newton model of mass judgments. Observable variables are shaded gray. The observed variables, initial velocities u_a and u_b, final velocities v_a and v_b, each depend on an independent source of noise ε. The final velocities v_a and v_b also depend on the object masses m_a and m_b and coefficient of restitution e. The dashed double-arrowed line between u_a and u_b restricts the noiseless values of these variables to values for which the objects will make contact: $u_b < u_a$.

physical equation

$$v_a = \frac{m_a u_a + m_b(u_b + e(u_b - u_a))}{m_a + m_b}$$

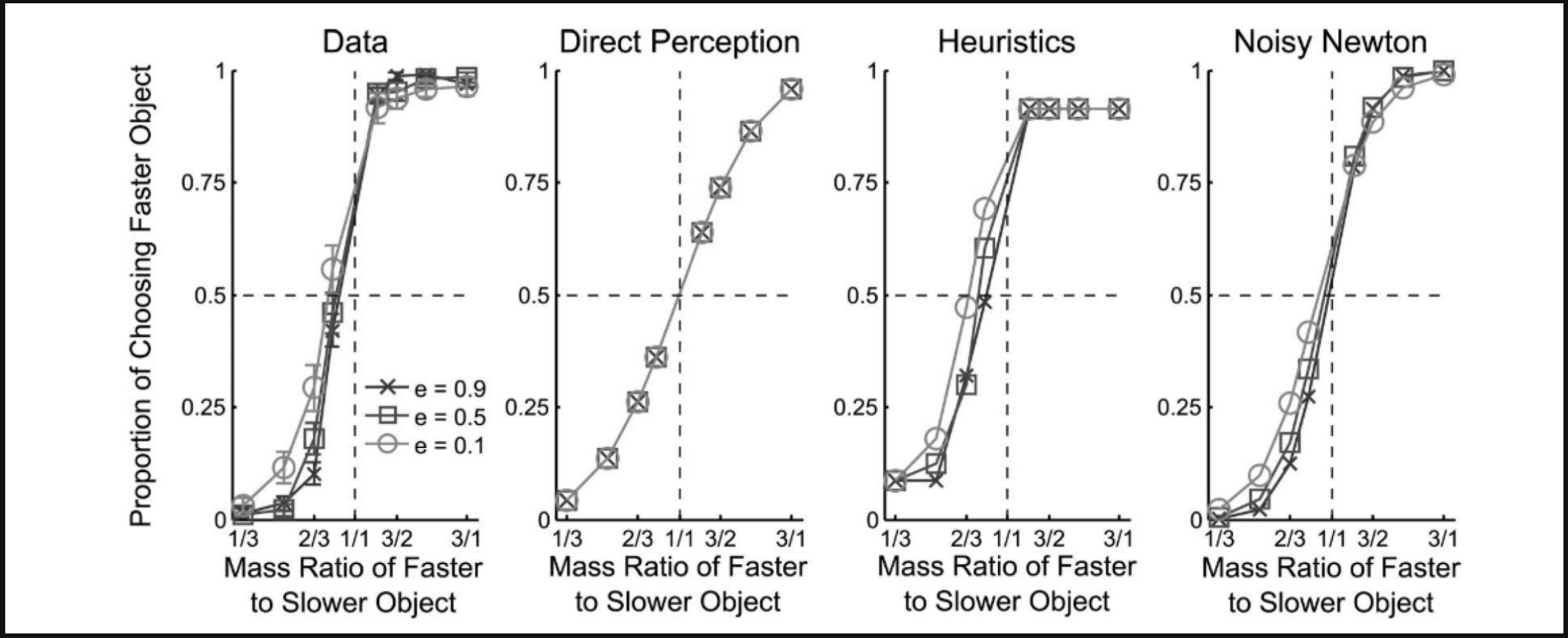
$$v_b = \frac{m_b u_b + m_a(u_a + e(u_a - u_b))}{m_a + m_b},$$

bayes equation

$$p(S \mid O) = \frac{p(O \mid S)p(S)}{p(O)} \approx \frac{p(O \mid S)}{\sum_{S'} p(O \mid S')}$$

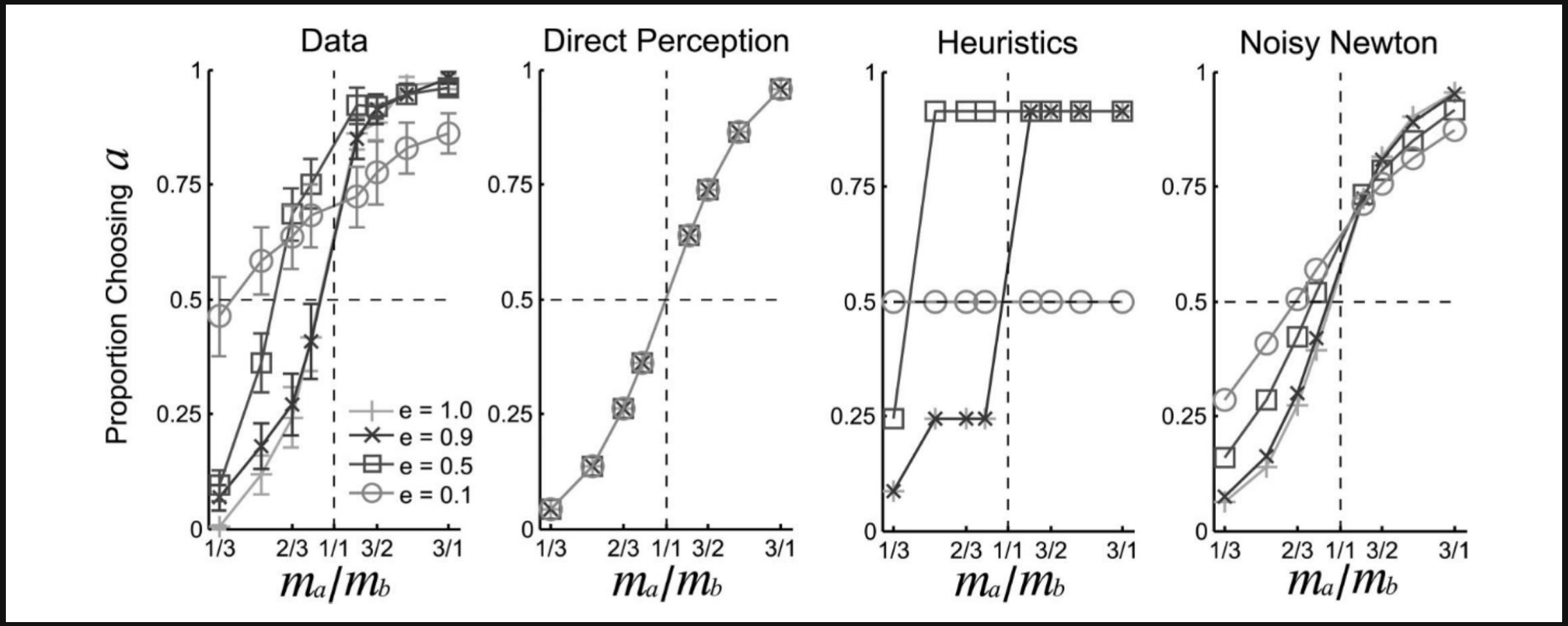
Dynamic

- Experiment 1: Two balls moving towards each other



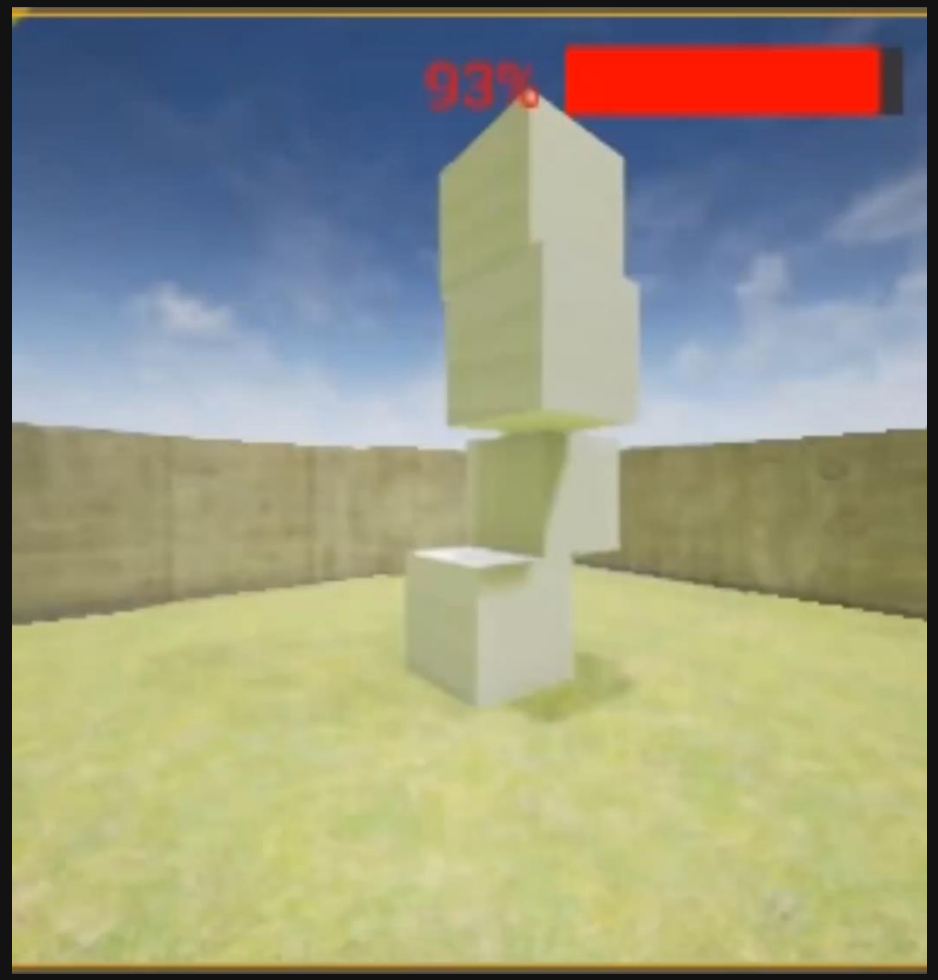
Dynamic

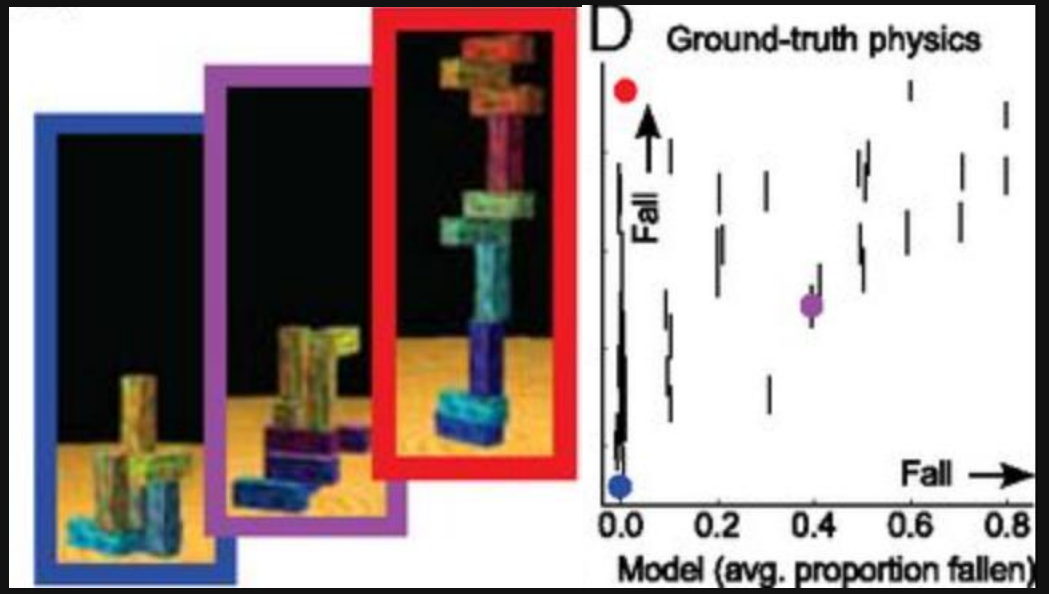
- Experiment 2: Ball A moves towards Ball B



Dynamic

- Task 1: Judging the stability of block towers

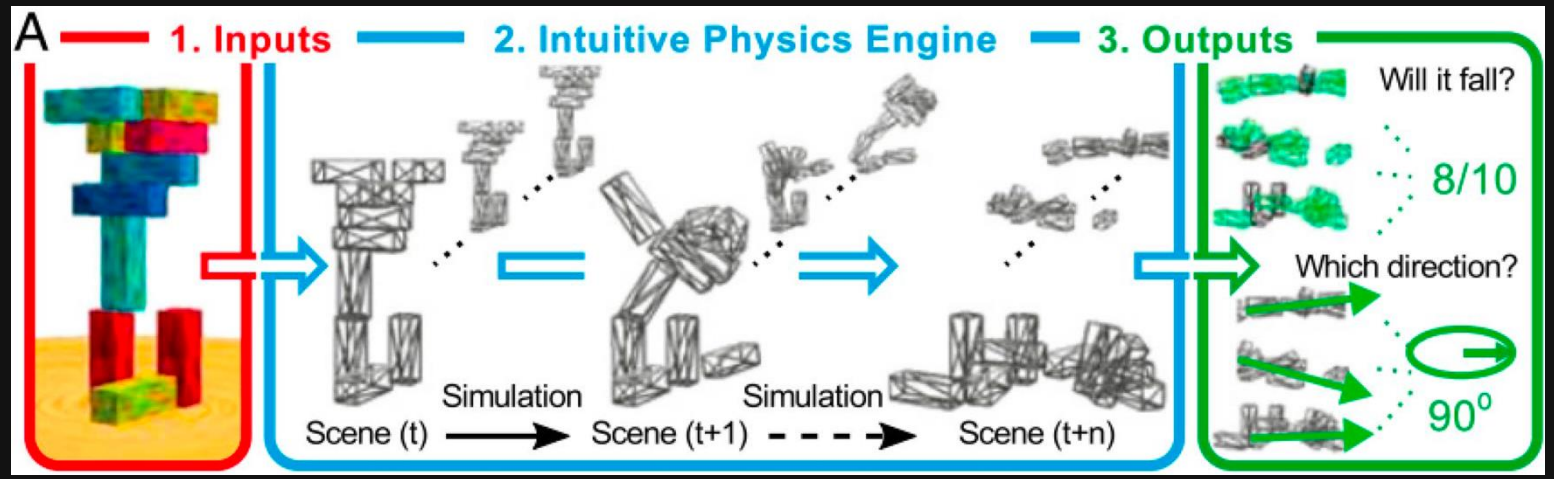




Some of the human predictions are inconsistent with ground-truth physics.

Dynamic

- Intuitive physics engine (IPE)



$$\Pr(S_{0:T} \mid I_{S_0}, I_f) = \int_{f_{0:T-1}} \Pr(S_T \mid S_{T-1}, f_{T-1}) \cdots \Pr(S_1 \mid S_0, f_0) \Pr(S_0 \mid I_{S_0}) \Pr(f_{0:T-1} \mid I_f) \mathrm{d}f_{0:T-1}$$
$$= \int_{f_{0:T-1}} \Pr(\psi(S_{T-1}, f_{T-1}, 1) \mid S_{T-1}, f_{T-1}) \cdots \Pr(\psi(S_0, f_0, 1) \mid S_0, f_0) \Pr(S_0 \mid I_{S_0}) \cdots \Pr(f_{0:T-1} \mid I_f) \mathrm{d}f_{0:T-1}$$

$$\Pr(S_T, S_0 \mid I_{S_0}, I_f) = \int_{f_{0:T-1}} \int \Pr(S_{T:T-1} \mid S_{T-1}, f_{T-1}) \ldots \Pr(S_1 \mid S_0, f_0) \Pr(S_0 \mid I_{S_0}) \Pr(f_{0:T-1} \mid I_f) \mathrm{d}S_{1:T-1} \, \mathrm{d}f_{0:T-1}$$
$$= \int_{f_{0:T-1}} \Pr(S_T \mid S_0, f_{0:T-1}) \Pr(S_0 \mid I_{S_0}) \Pr(f_{0:T-1} \mid I_f) \mathrm{d}f_{0:T-1}$$
$$= \int \Pr(\Psi(S_{t_0}, f_{0:T-1}, 0:T) \mid S_0, f_{0:T-1}) \int_{f_{0:T-1}} \Pr(S_0 \mid I_{S_0}) \Pr(f_{0:T-1} \mid I_f) \mathrm{d}f_{0:T-1}.$$

Definitions.

- S_t : Scene state at time, t.
- $S_{t_0:t_1} = (S_{t_0}, S_{t_0+1}, \ldots, S_{t_1-1}, S_{t_1})$: Sequence of scene states from time t_0 to t_1.
- f_t : Extrinsic force applied beginning at time t.
- $f_{t_0:t_1}$: Sequence of extrinsic forces applied from time t_0 to t_1.
- I_{S_t} : Observed information about S_t.
- I_f : Observed information about $f_{t_0:t_1}$.
- $\psi(\cdot)$: Deterministic physical dynamics from t_0 to t_1, which maps S_{t_0} to a new state at time S_{t_1} : $S_{t_1} = \psi(S_{t_0}, f_{t_0}, t_1 - t_0)$. The force, f_{t_0}, is applied for a duration $t_1 - t_0$. The dynamics can be applied recursively,

$$S_{t_2} = \psi(\psi(S_{t_0}, f_{t_0}, t_1 - t_0), f_{t_1}, t_2 - t_1).$$

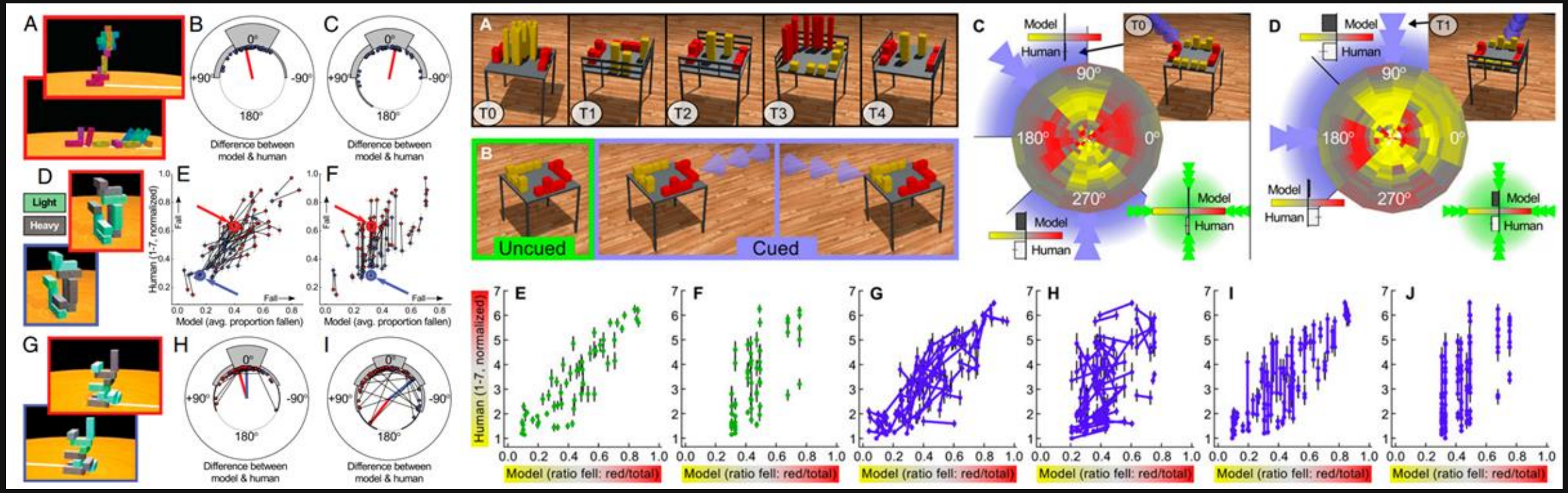
- We denote the repeated application of $\psi(\cdot)$ from $(t_0 : t_n)$ as $\Psi(\cdot)$,

$$S_{t_n} = \psi(\ldots\psi(S_{t_0}, f_{t_0}, t_1 - t_0), \ldots, f_{t_{n-1}}, t_n - t_{n-1})$$
$$= \Psi(S_{t_0}, f_{t_0:t_{n-1}}, t_0 : t_n).$$

- $\mathcal{Q}_q(\cdot)$: Output predicate corresponding to a query, q, which maps an initial (S_0) and a future sequence of scene states $(S_{1:T})$ to a judgment, $J_q = \mathcal{Q}_q(S_{0:T})$. In our experiments, the queries were sensitive only to the initial and final scene states (i.e., for Will the tower fall?, the query reflected how many blocks dropped from $t = 0$ to $t = T$), and so $J_q = \mathcal{Q}_q(S_0, S_T)$.

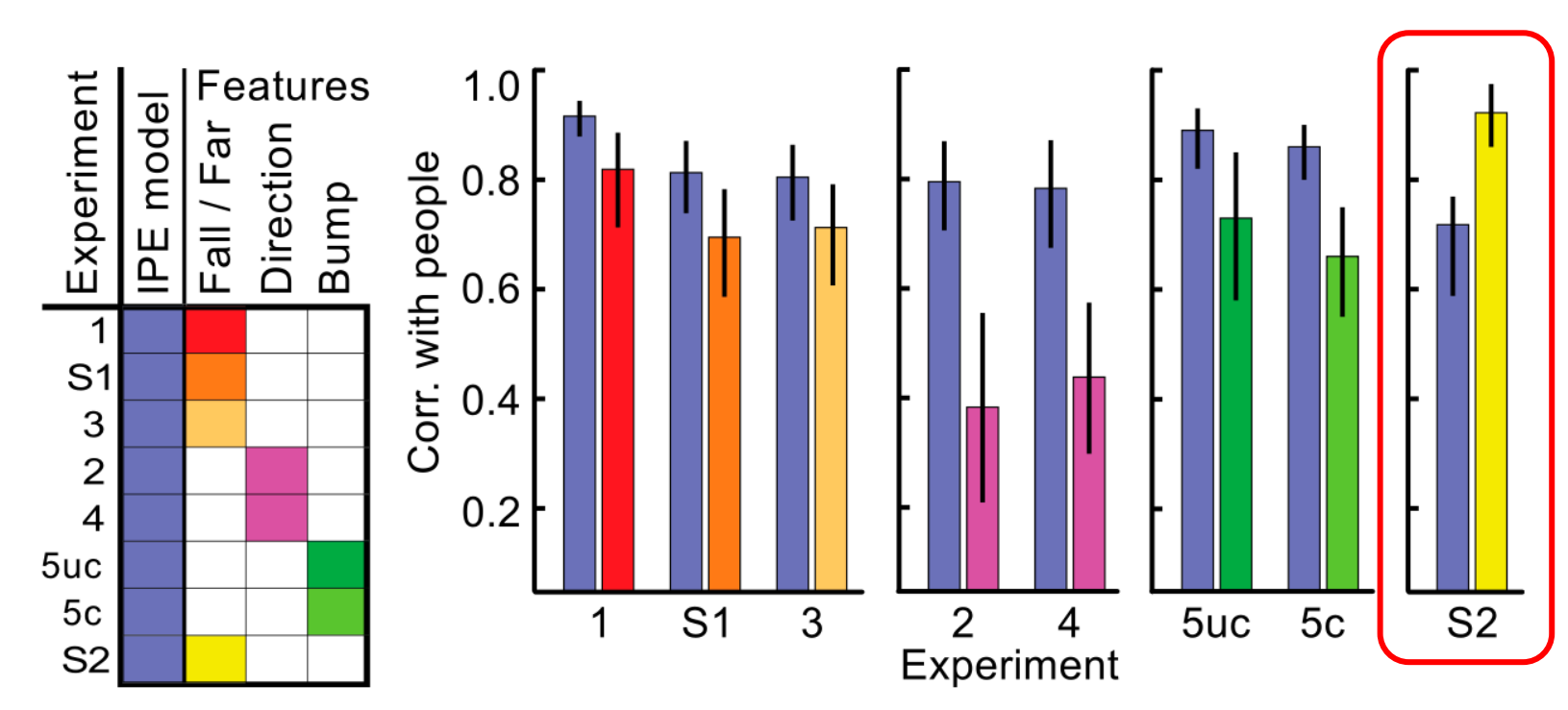
Dynamic

- Task 2: In which direction?
- Task 3 and 4: Varying object masses.
- Task 5: Varying object shapes, physical obstacles, and applied forces.



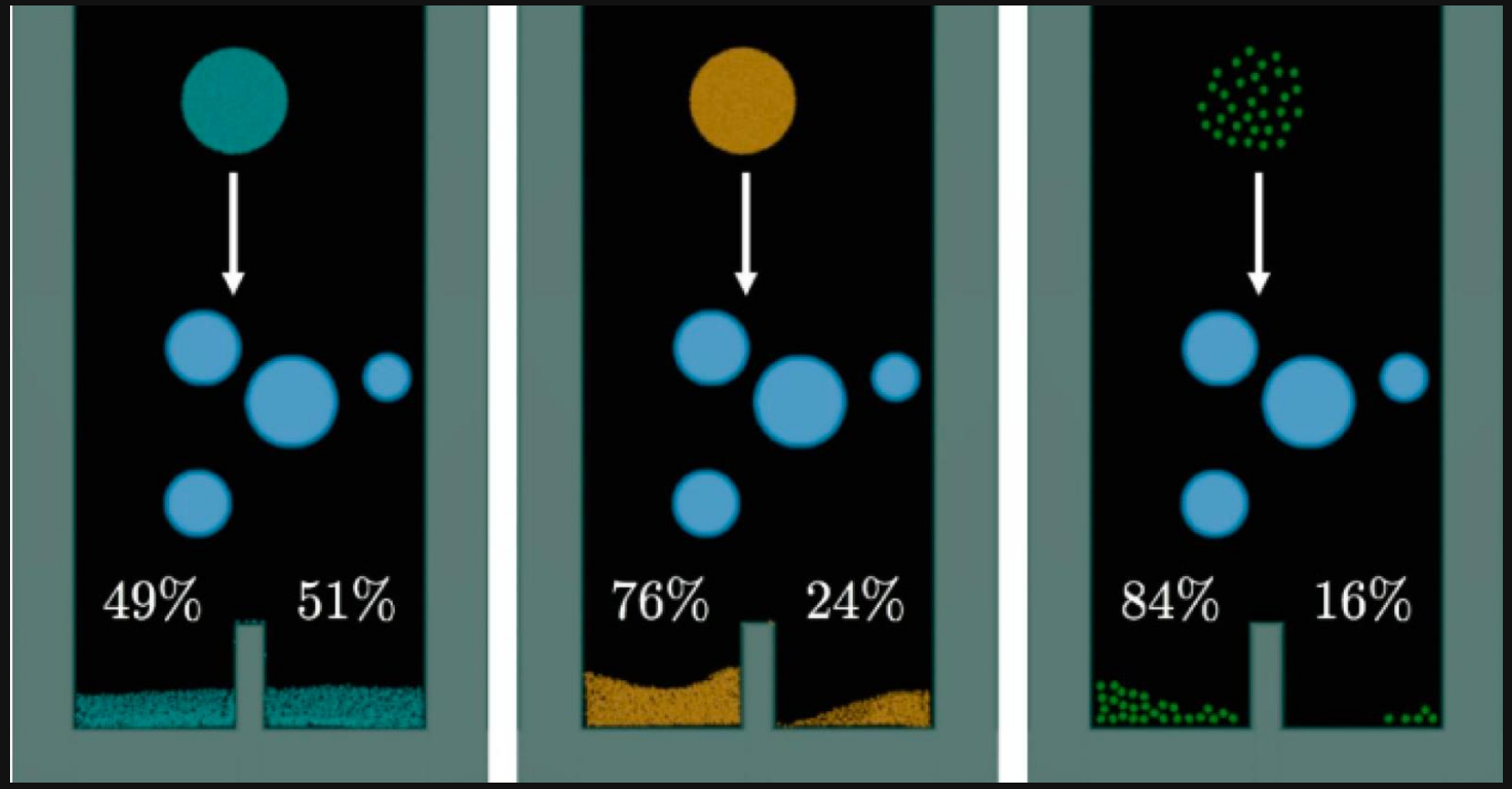
Dynamic

- S2 task: How far will the furthest block come to rest?
- A special task in which a height-based heuristic accounts for human results better that IPE.

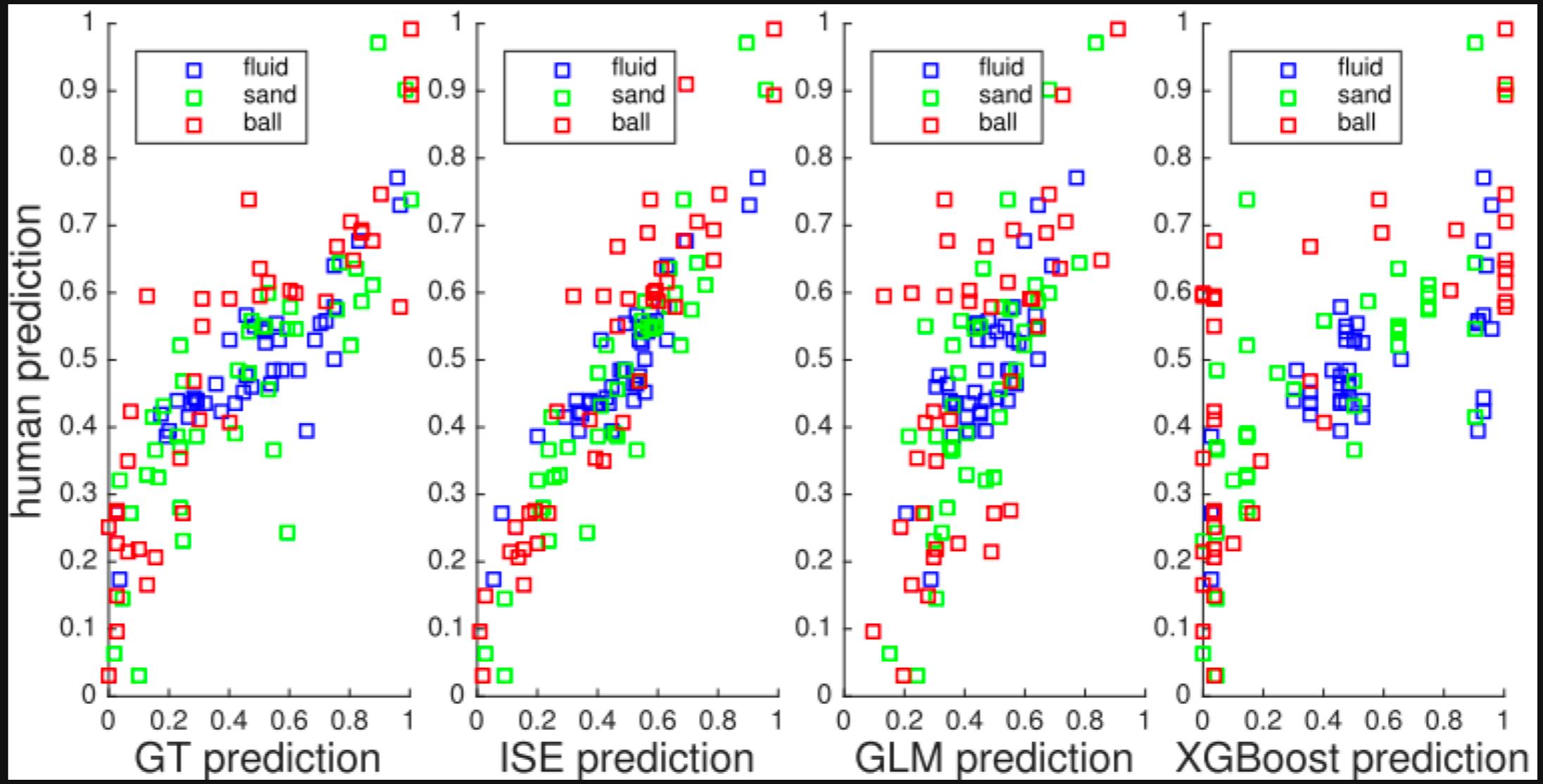


Dynamic

- Task: Predict the percentages of the substance that will fall into the left and right basins.

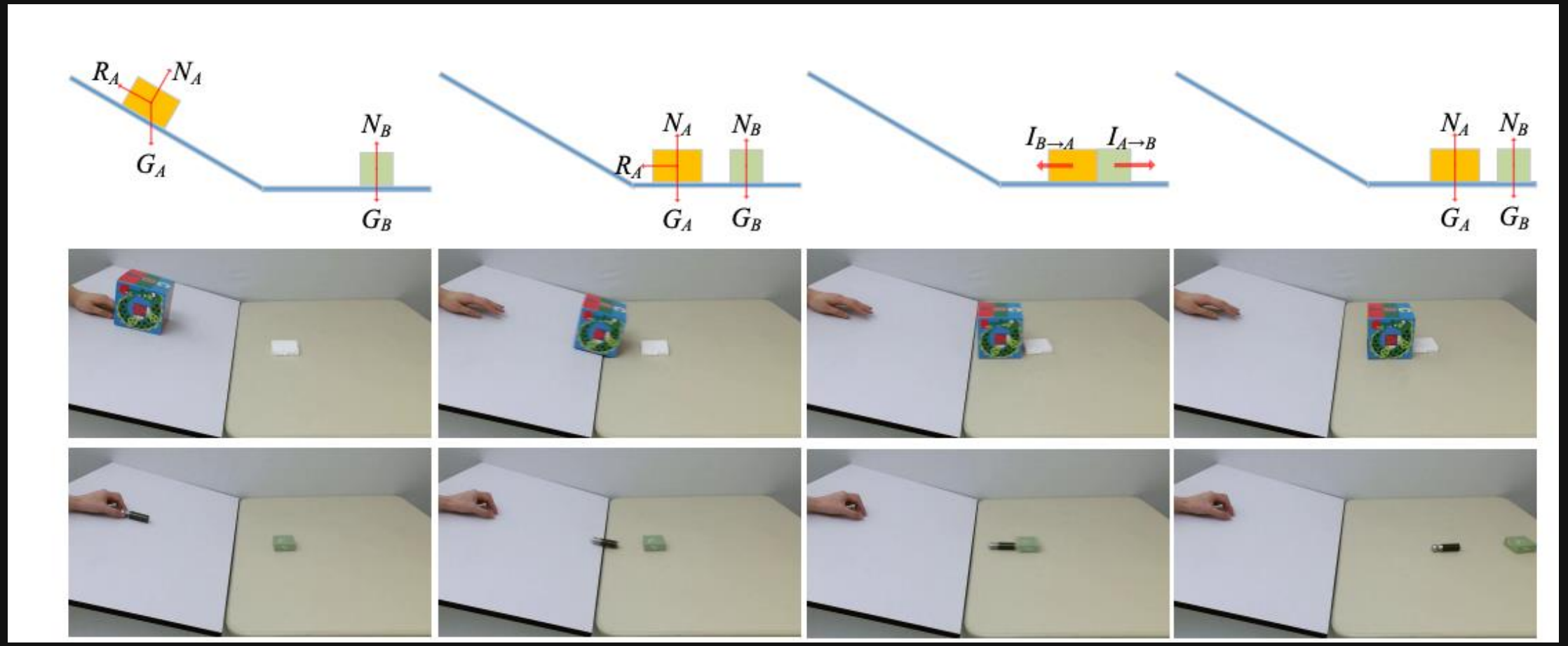


Dynamic


fluid
sand
ball
human prediction
GT prediction
ISE prediction
GLM prediction
XGBoost prediction
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

Dynamic

- Problem: Predict how far the blocks will move



Dynamic

- Analysis by synthesis
- Reducing the sampling space using deep learning

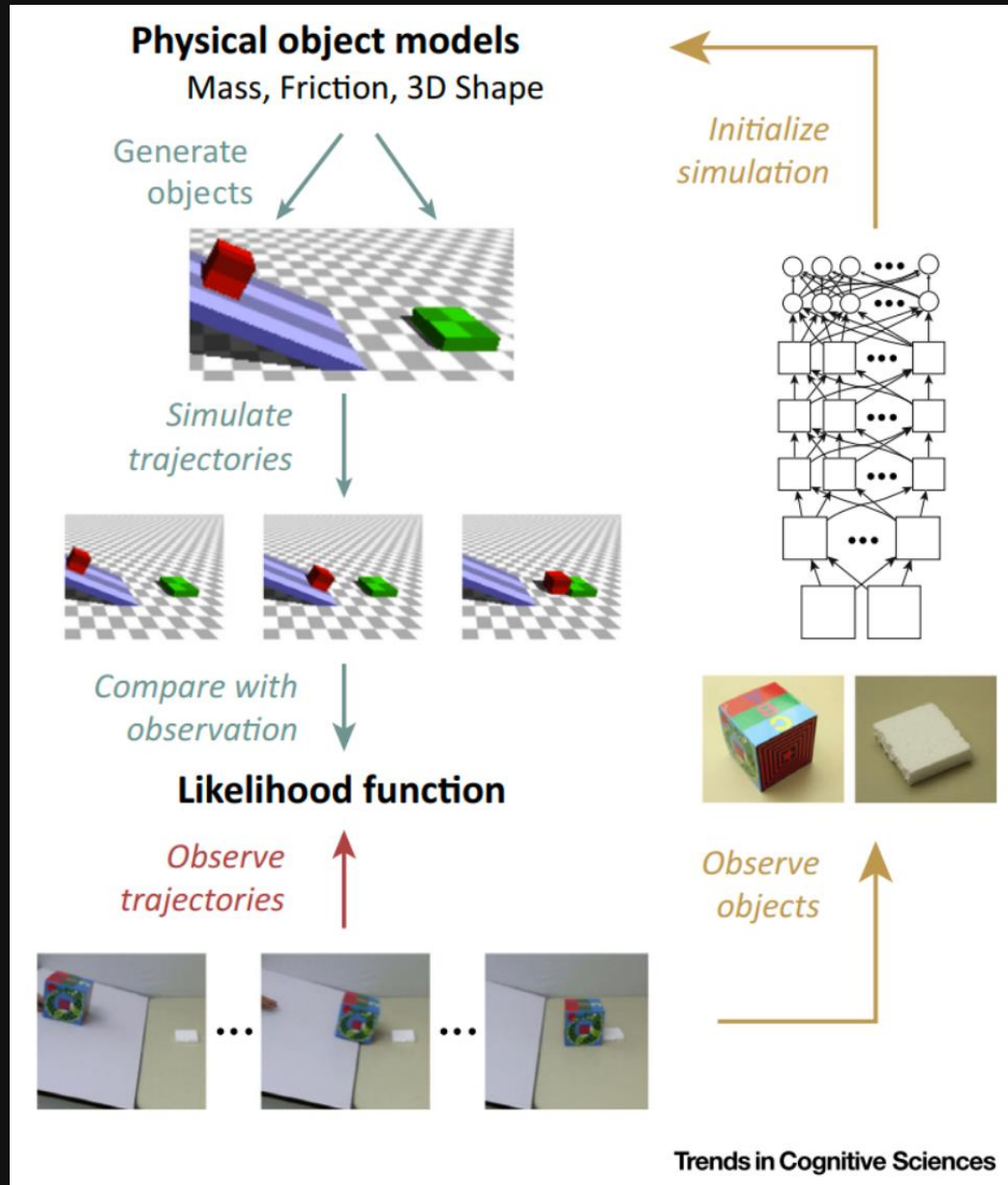
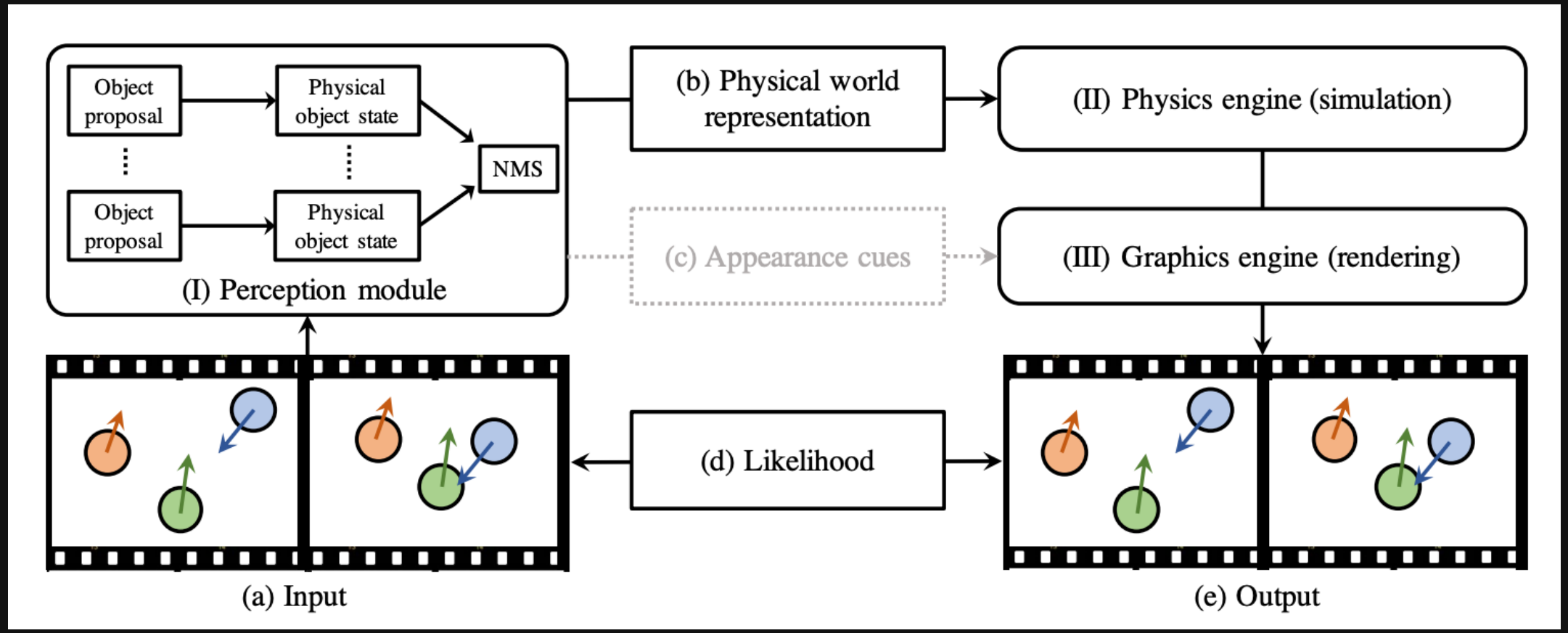


Figure 4. Inferring Physical Parameters. From top left: the Galileo system assumes a probability distribution over physical properties such as mass, and shape. In the forward direction, the system generates objects in space, and simulates their trajectory using a game engine. The simulated trajectory is compared to a real trajectory of objects in motion (bottom left), resulting in a likelihood for the simulated trajectory. The physical parameters are adjusted to maximize this likelihood and better match observation. In parallel, a neural network is trained to predict the physical properties of objects, given their visual appearance (right). The prediction of the network is used as the initial 'guess' for the physical parameters of the game engine, speeding up inference ([66] for full details).

Dynamic

- Learning to see physics via visual de-animation

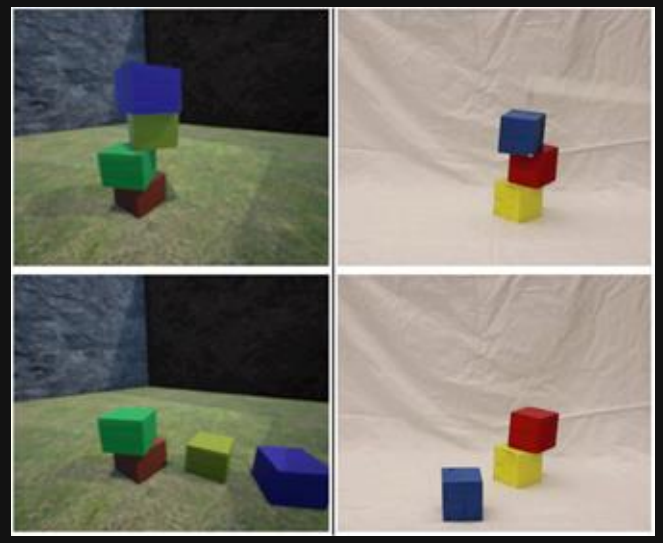


A debate: do dynamic prediction really help physical reasoning?

Heuristics vs Dynamics

- A debate: whether dynamic prediction is helpful in solving physical reasoning tasks?

Two reasoning processes when solving physical puzzles



Physical intuition at a glance without much thinking

Judging whether a stacked block tower will collapse

Adam Lerer, et al. 2016



Unfolding of states under the assumed physical dynamics

Solving complex physical tasks such as PHYRE

Anton Bakhtin, et al. 2019

Heuristics vs Dynamics

Motivation: From an interesting finding in PHYRE

A simple binary-classification Vision Transformer reaches or outperforms SOTA dynamic-based RPIN.

1. Within: Evaluate on similar but different tasks.
2. Cross: Evaluate on unseen tasks (more difficult).

Model	Mechanism	Input	Supervision	Within	Cross
RPIN	LfD	Initial scenes, bboxs	Bboxs, masks, outcomes	85.49	50.86
ViT	LfI	Initial scenes	Outcomes	84.16±0.30	56.31±1.95

Is dynamics prediction indispensable for physical reasoning?

If so, what kind of role do the dynamics prediction modules play?

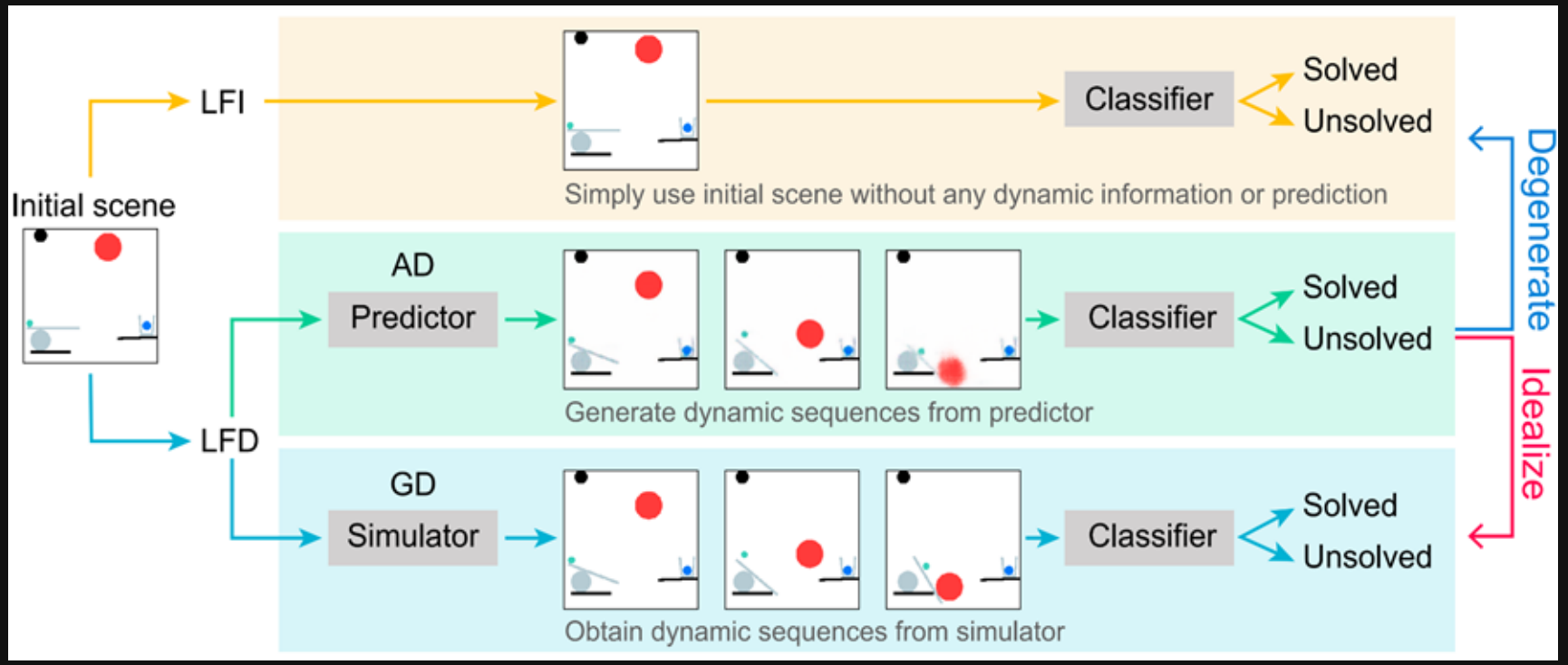
Heuristics vs Dynamics

Definition of two learning mechanisms

1. Learning from intuition (LfI)
2. Learning from dynamics (LfD)

$$P(y|X_0) = f(X_0;\theta)$$

$$P(y|X_0) = f(D;\theta),\ \text{where}\ D = g(X_0;\phi),$$



Heuristics vs Dynamics

Experimental Design

1. LfD vs.LfI: Challenge the previous thought

   ViT and RPIN

Heuristics vs Dynamics

Experiment 1: LfD v.s. LfI
A simple binary-classification Vision Transformer reaches or outperforms SOTA dynamic-based RPIN.

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Heuristics vs Dynamics

Experimental Design

1. LfD vs.LfI: Challenge the previous thought

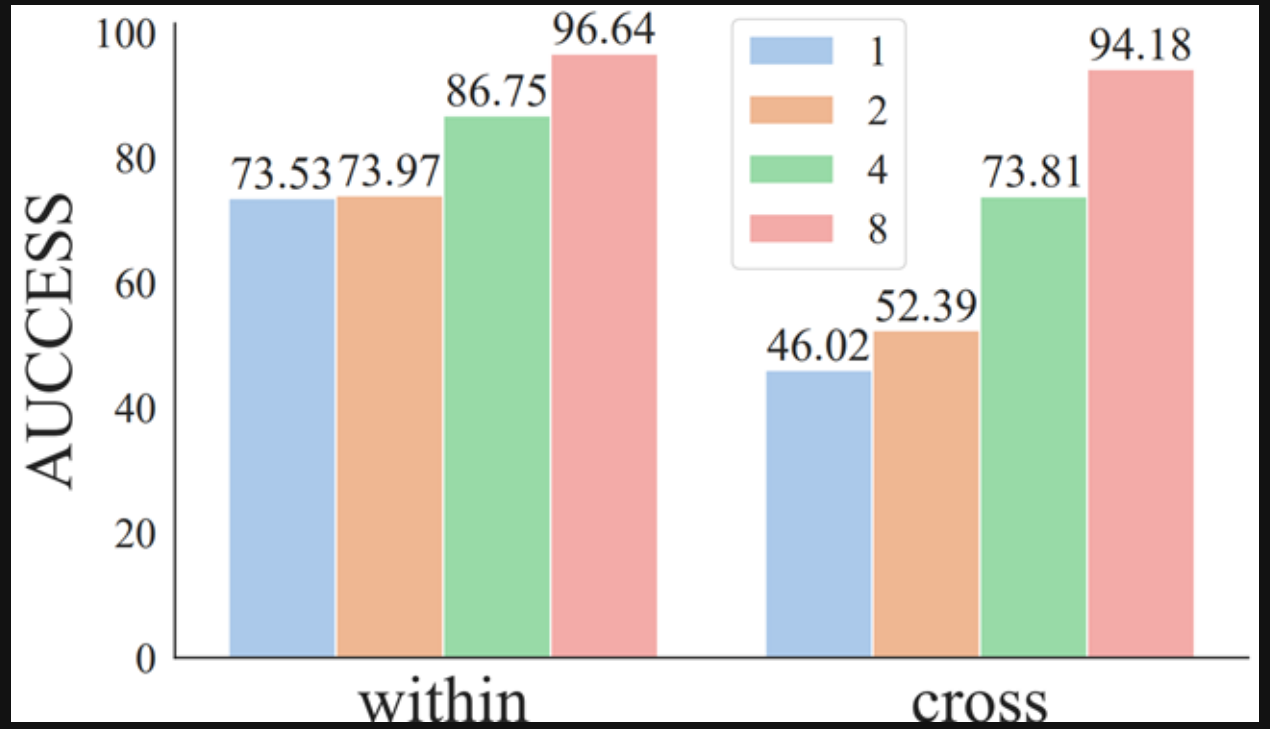
   ViT and RPIN

2. LfD under GD: Do ground-truth dynamics help make better decisions than intuition?

   Ground-truth dynamics from simulator + Video classifier

Heuristics vs Dynamics

Experiment 2: LfD under GD
Dynamics do help problem-solve in physical reasoning, especially in unseen scenarios.



Heuristics vs Dynamics

Experimental Design

1. LfD vs.LfI: Challenge the previous thought

    ViT and RPIN

2. LfD under GD: Do ground-truth dynamics help make better decisions than intuition?

    Ground-truth dynamics from simulator + Video classifier

3. LfD under AD: How does approximate dynamics perform?

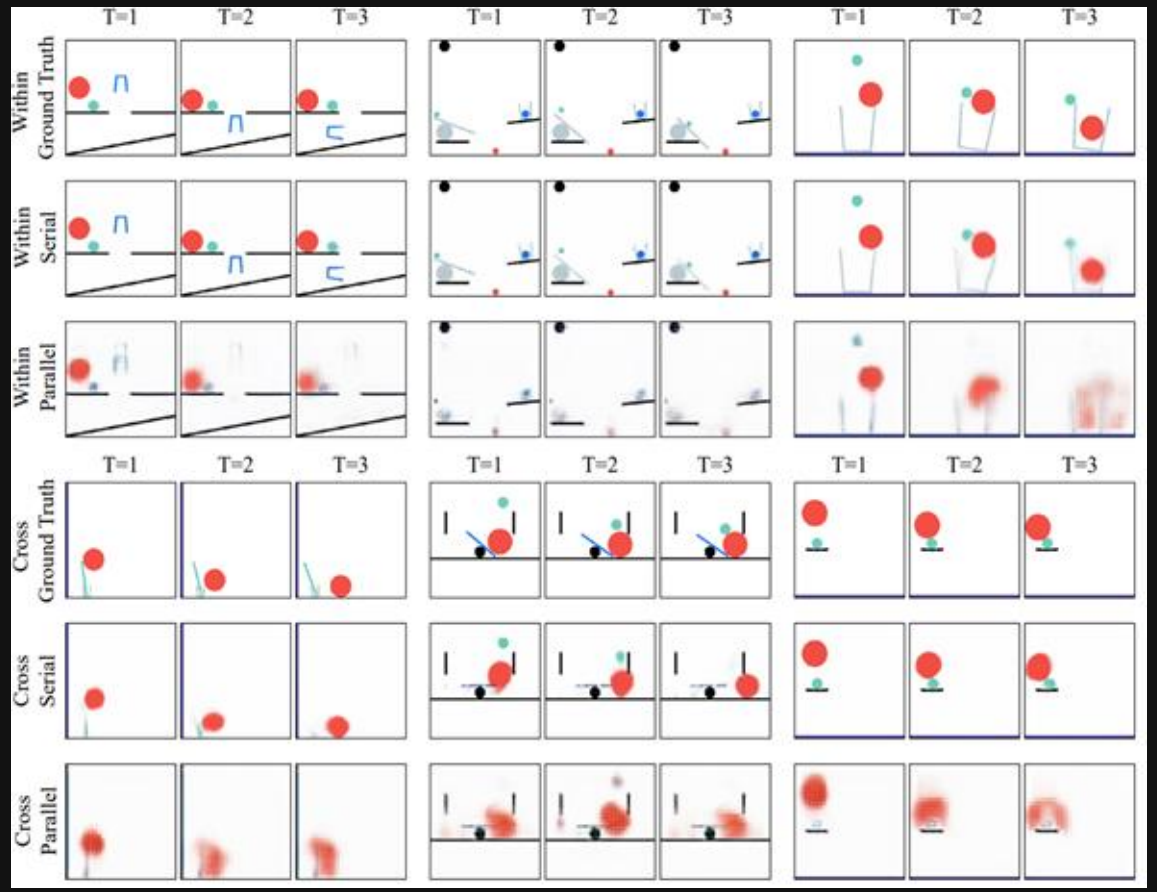
    Predicted dynamics + Video classifier

Heuristics vs Dynamics

Experiment 3: LfD under AD
LfD under AD degenerates to LfI or even worse.

Prediction	NF	Within	Cross
PredRNN (parallel)	4	75.22	46.42
PredRNN (serial)	4	64.90	44.33
/	1	73.53	46.02
Simulator	4	86.75	73.81

Opt	Loss	Within	Cross
Parallel	entropy	0.0638	0.5726
	dynamics	0.0039	0.0049
Serial	entropy	0.1285	0.6554
	dynamics	0.0003	0.0021

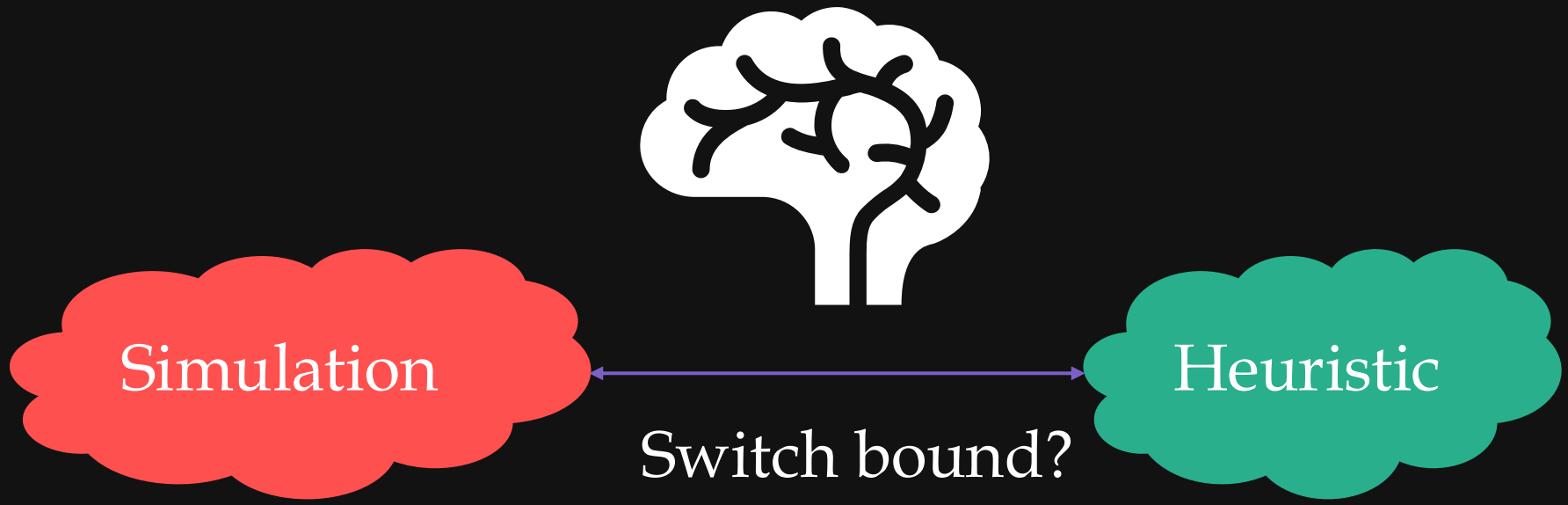


Model	Generalizable	Efficient	Learnable
Heuristic account	✗	✓	✓
Probabilistic simulation	✓	✗	✗

How about a dual-process model?

Dual-process model

- Do humans consistently rely on mental simulation?
- Any alternative heuristic strategies?
- What is the switch point?

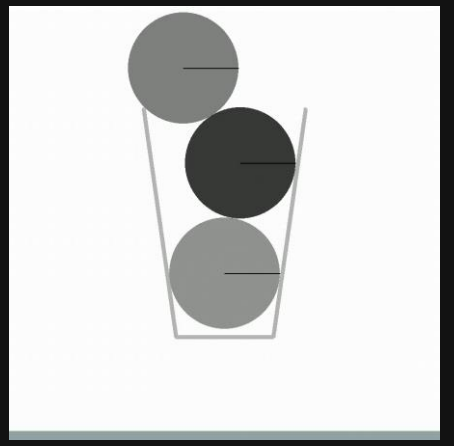


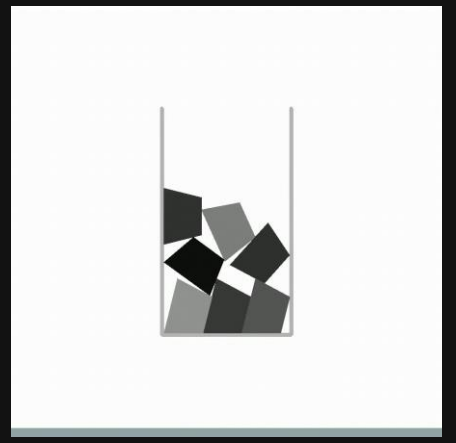
Dual-process model

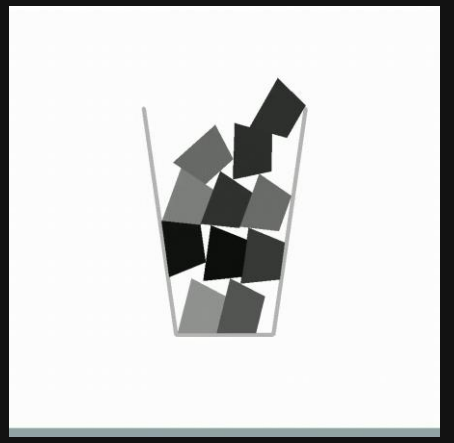
Pouring marble task: Judge the tilting angle at which the cup begins to pour out marbles

Cup shape			
Object shape			
Object size			
Filling height			
Example1	Example2	Example3	



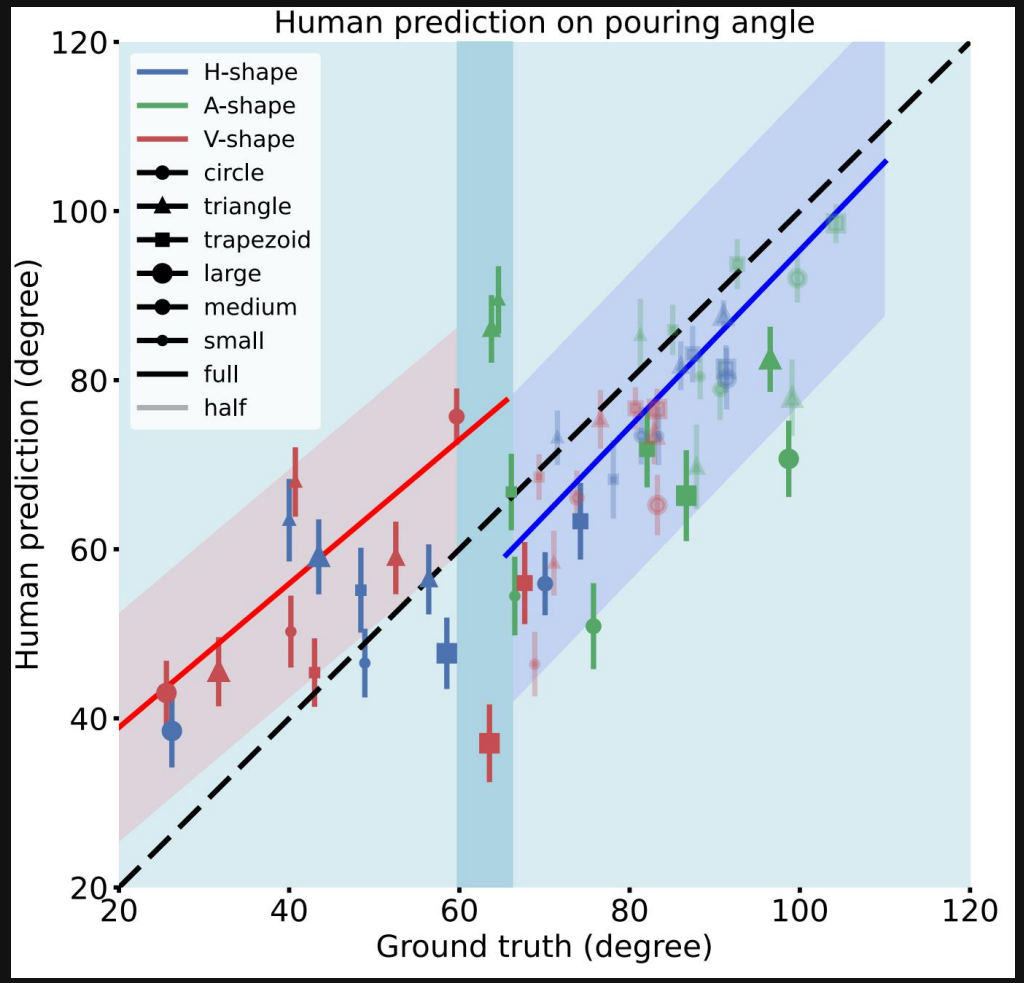






Dual-process model

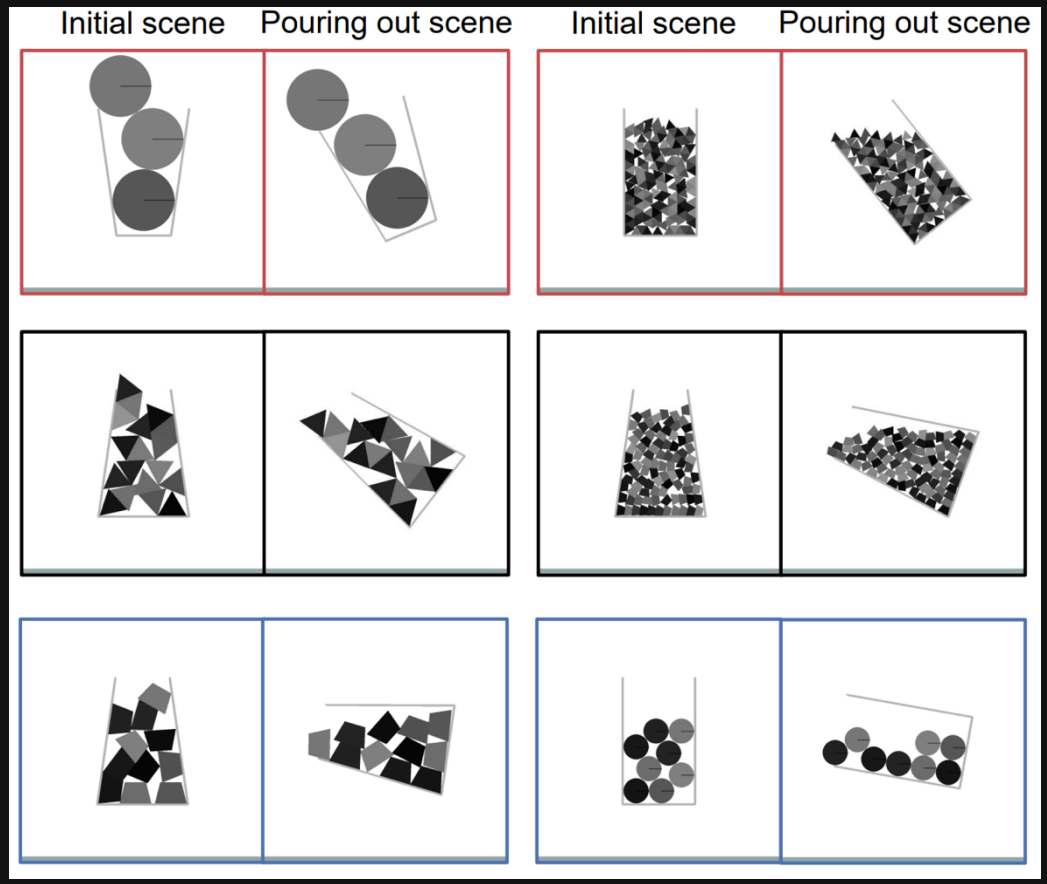
People show two different patterns (overestimation and underestimation) in their judgments.



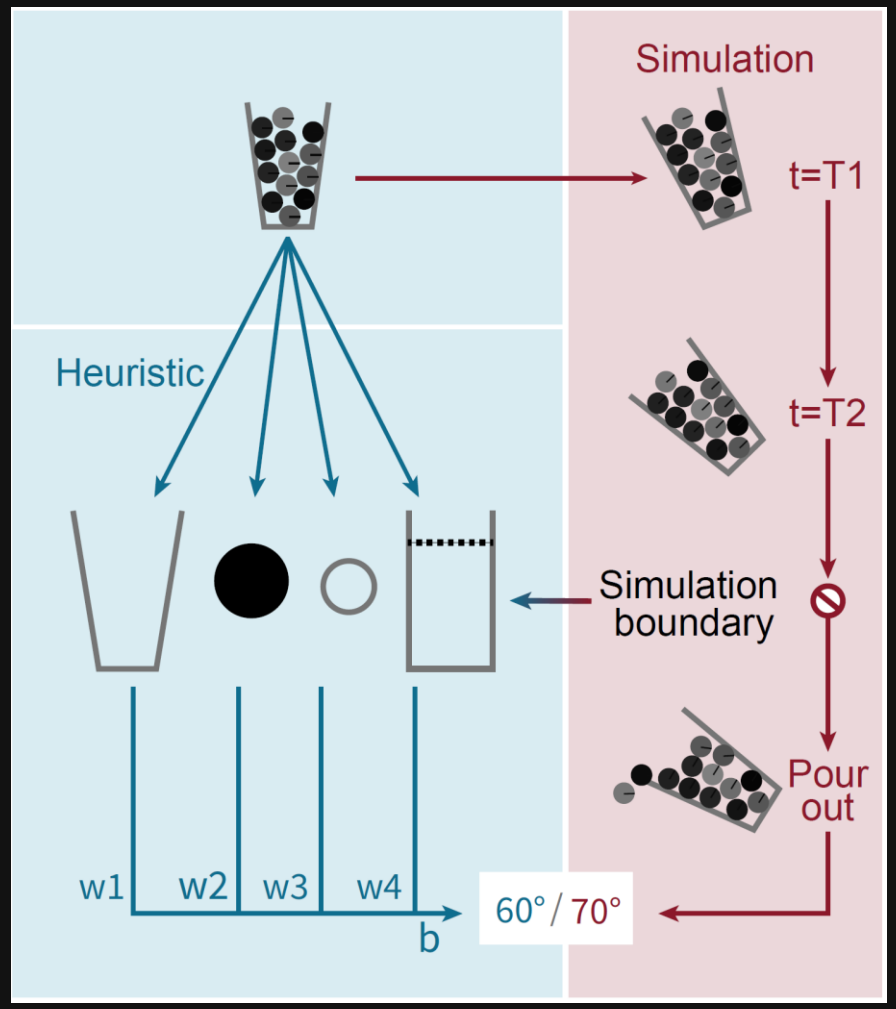
Overestimation scenes

Boundary

Underestimation scenes



Dual-process model



People have two mechanisms in physical reasoning, switching at a certain boundary:

Mechanism 1: probabilistic simulation

$$P(J|S_0, f_{0:T-1}) = \int_{S_{1:T}} P(J|S_{1:T}) P(S_{1:T}|S_0, f_{0:T-1}) dS_{1:T}, \quad (1)$$

Mechanism 2: heuristic method

$$J = g(S_0^1, ..., S_0^n) = \sum_{i=1}^{n} \omega_i S_0^i + b, \quad (2)$$

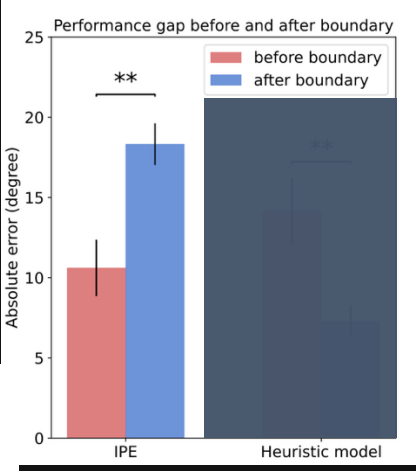
$$\begin{cases} J = E_\epsilon[M(S_0; \epsilon)], & \text{if } T \leqslant \theta, \\ J = \sum_{i=1}^{n} \omega_i S_0^i + b, & \text{if } T > \theta \end{cases}. \quad (3)$$

Switch: when simulation reaches resource boundary.

where $S_{t+1} = \phi(S_t + \epsilon, f_t)$ with noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$, and $\phi(\cdot)$ deterministic physical dynamics. We simplify the mapping from the initial state to the final judgment as $M(S_0; f, \epsilon)$.

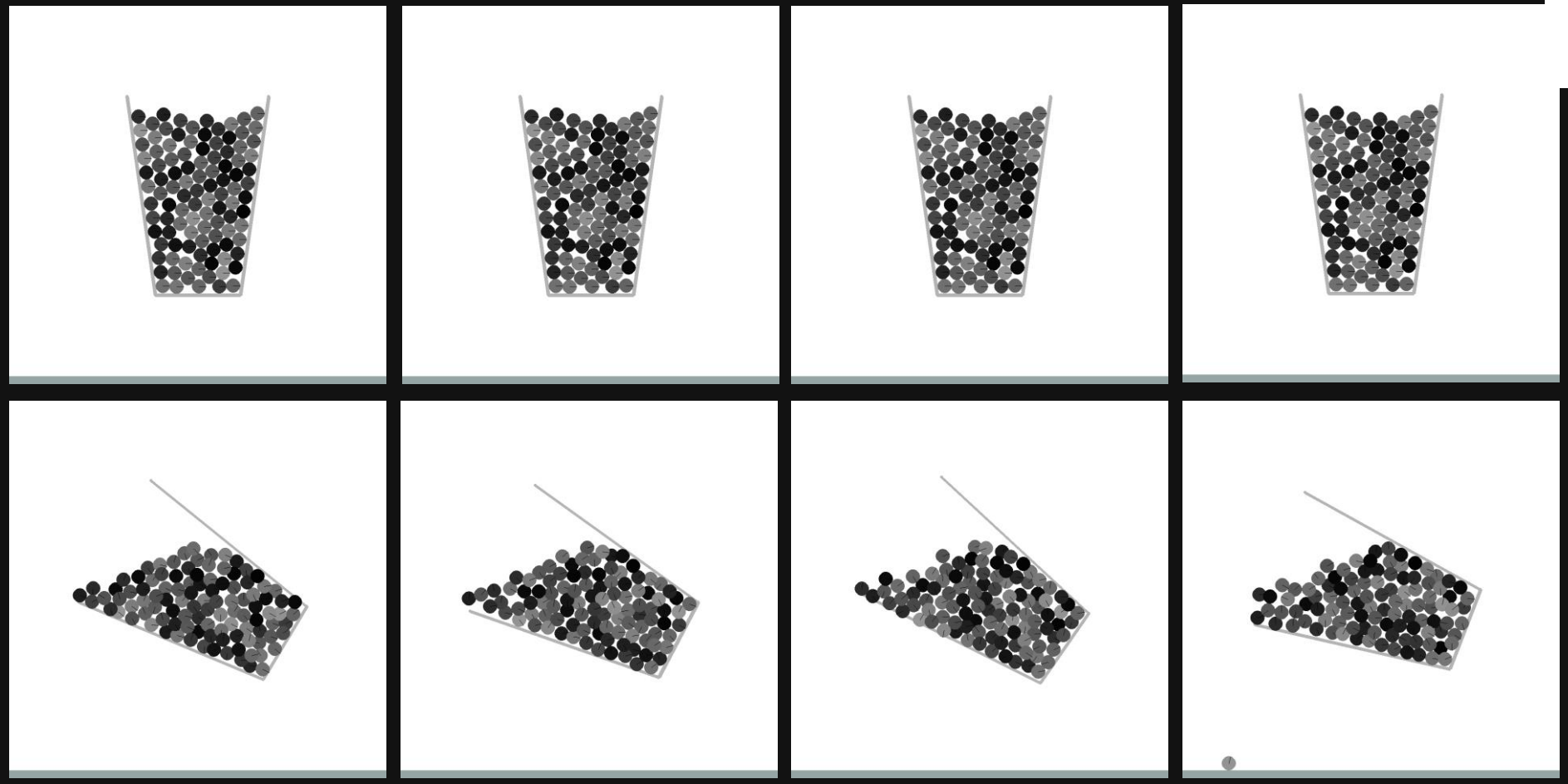
Dual-process model

Position Gaussian noise (mean=0, var=2), angle Gaussian noise (mean=0, var=0.1), and cup angle Gaussian noise (mean=0, var=0.02).



Noisy simulation

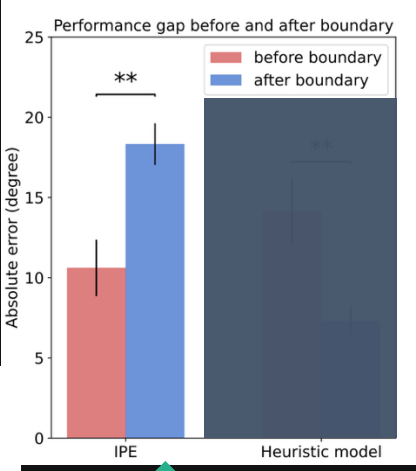
Pouring moment



61.63° 62.53° 56.53° 67.99°

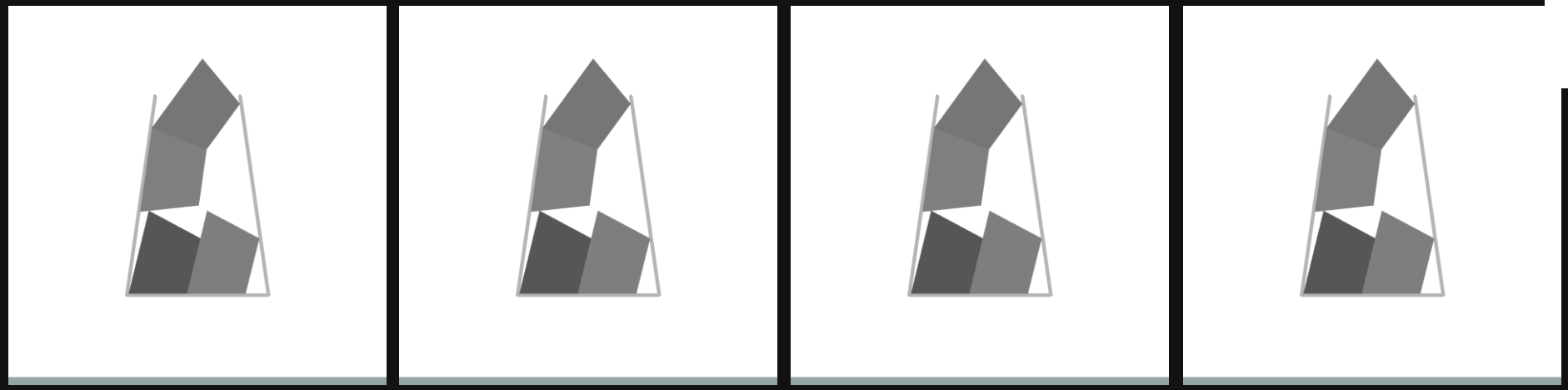
Dual-process model

Position Gaussian noise (mean=0, var=2), angle Gaussian noise (mean=0, var=0.1), and cup angle Gaussian noise (mean=0, var=0.02).

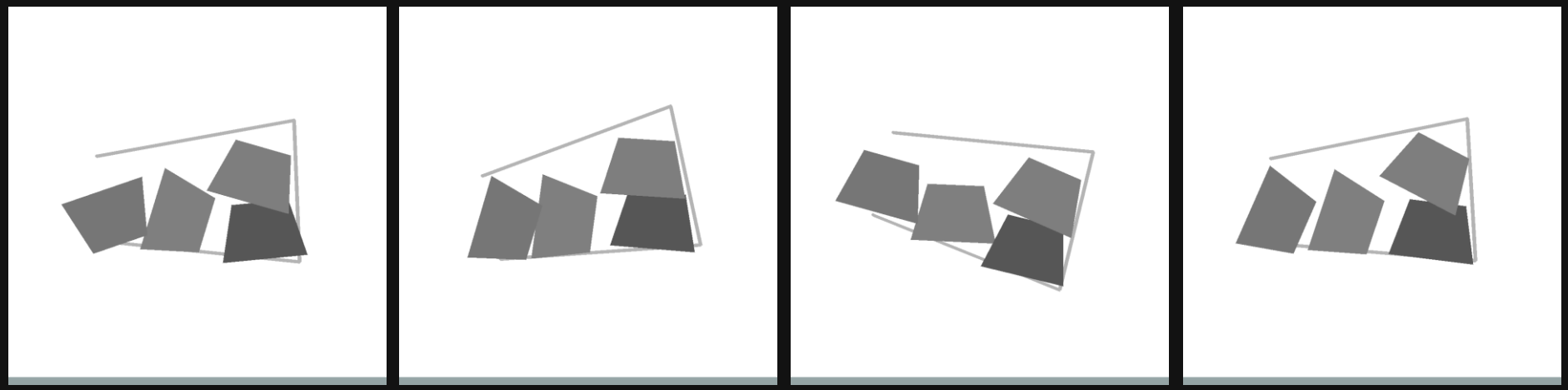


Error increases

Noisy simulation



Pouring moment



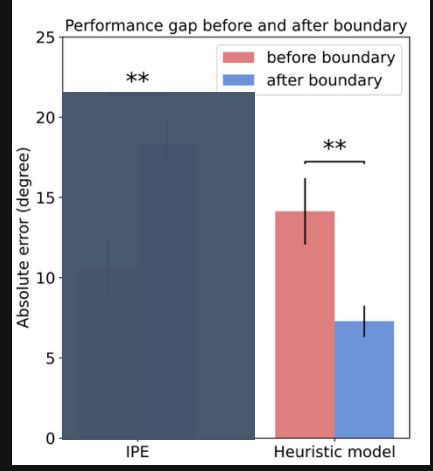
94.41°

107.18°

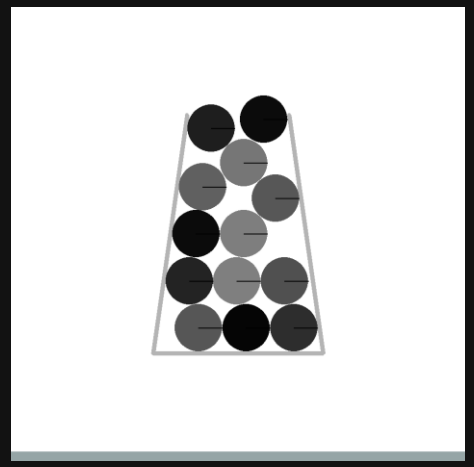
78.91°

93.75°

Dual-process model



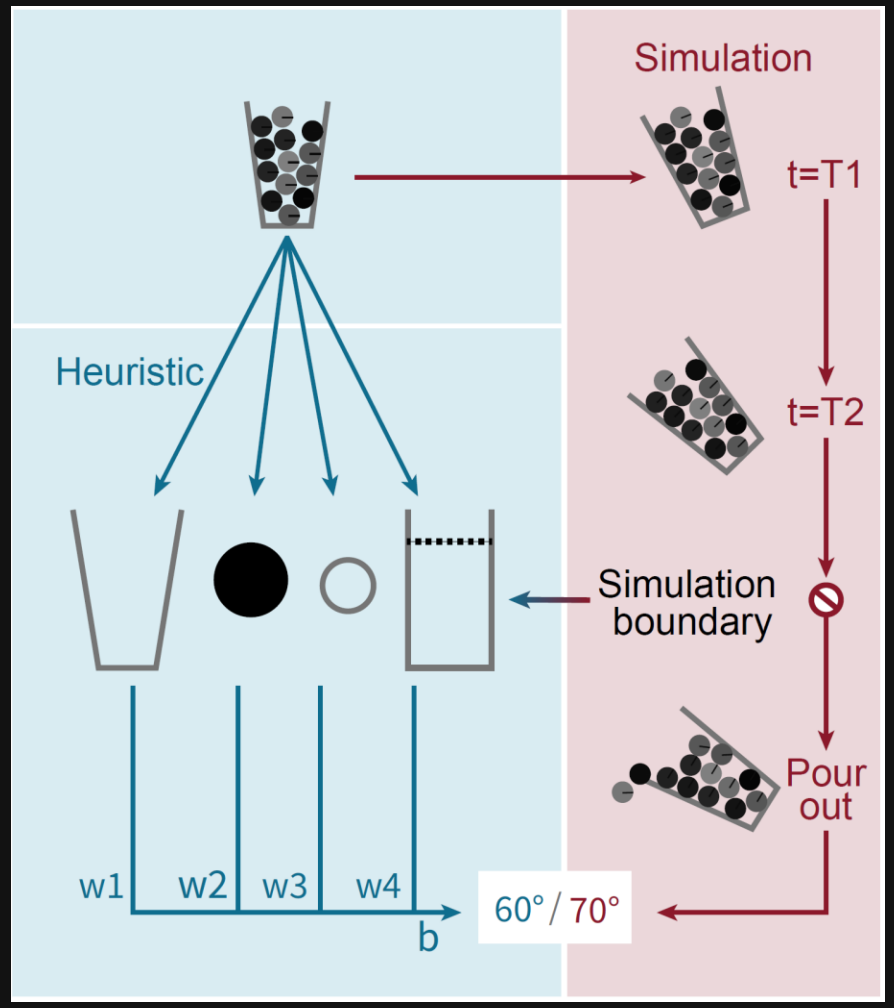

```
sizes_encoding = {'small': 0, 'medium': 1, 'large': 2}
filling_heights_encoding = {'half': 1, 'full': 2}
object_shapes_encoding = {'circle': 0, 'triangle': 3, 'trapezoid': 4}
cup_shapes_encoding = {'erlenmeyer': -1, 'ierlenmeyer': 1, 'regular': 0}
```

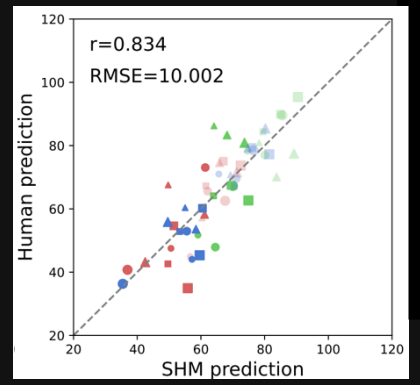


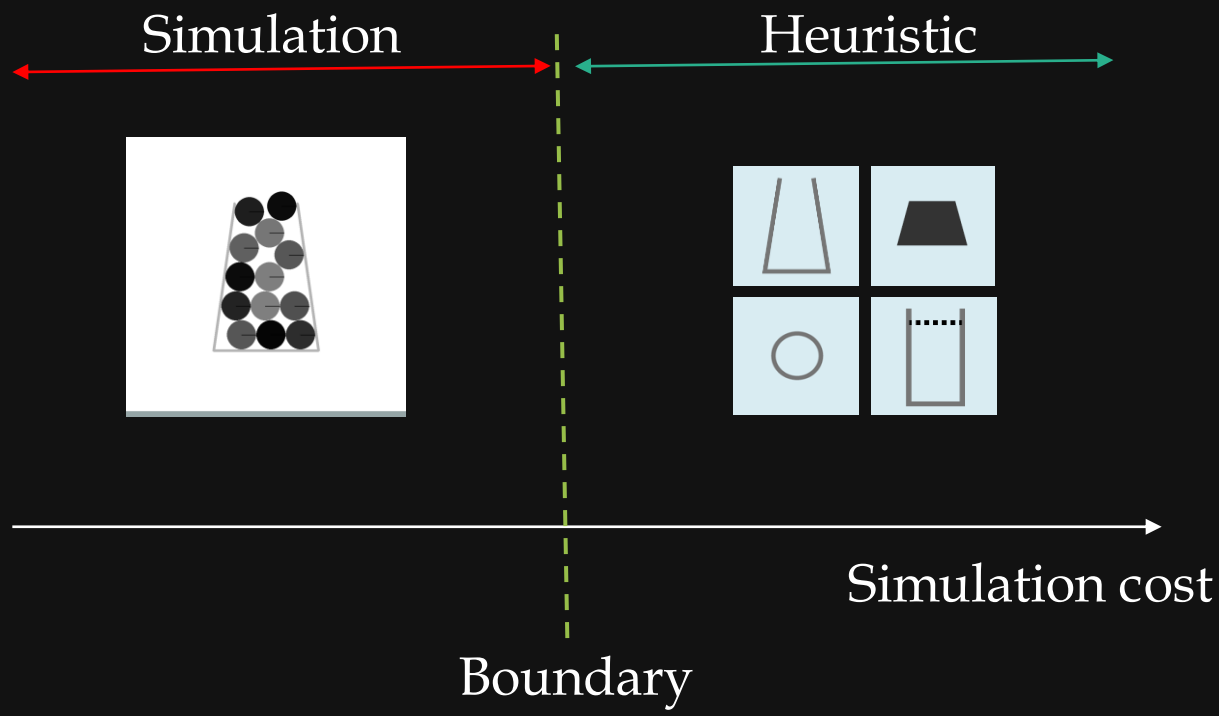
Physical features	Encoding		Learned weights
Cup_shape=erlenmeyer	-1	*	-11.5281999
Object_shape=circle	0	*	1.57705401
Object_size=medium	1	*	7.02898528
Filling_height=full	2	*	-19.95479845

Bias: + 90 = Result: 68.6476

Dual-process model

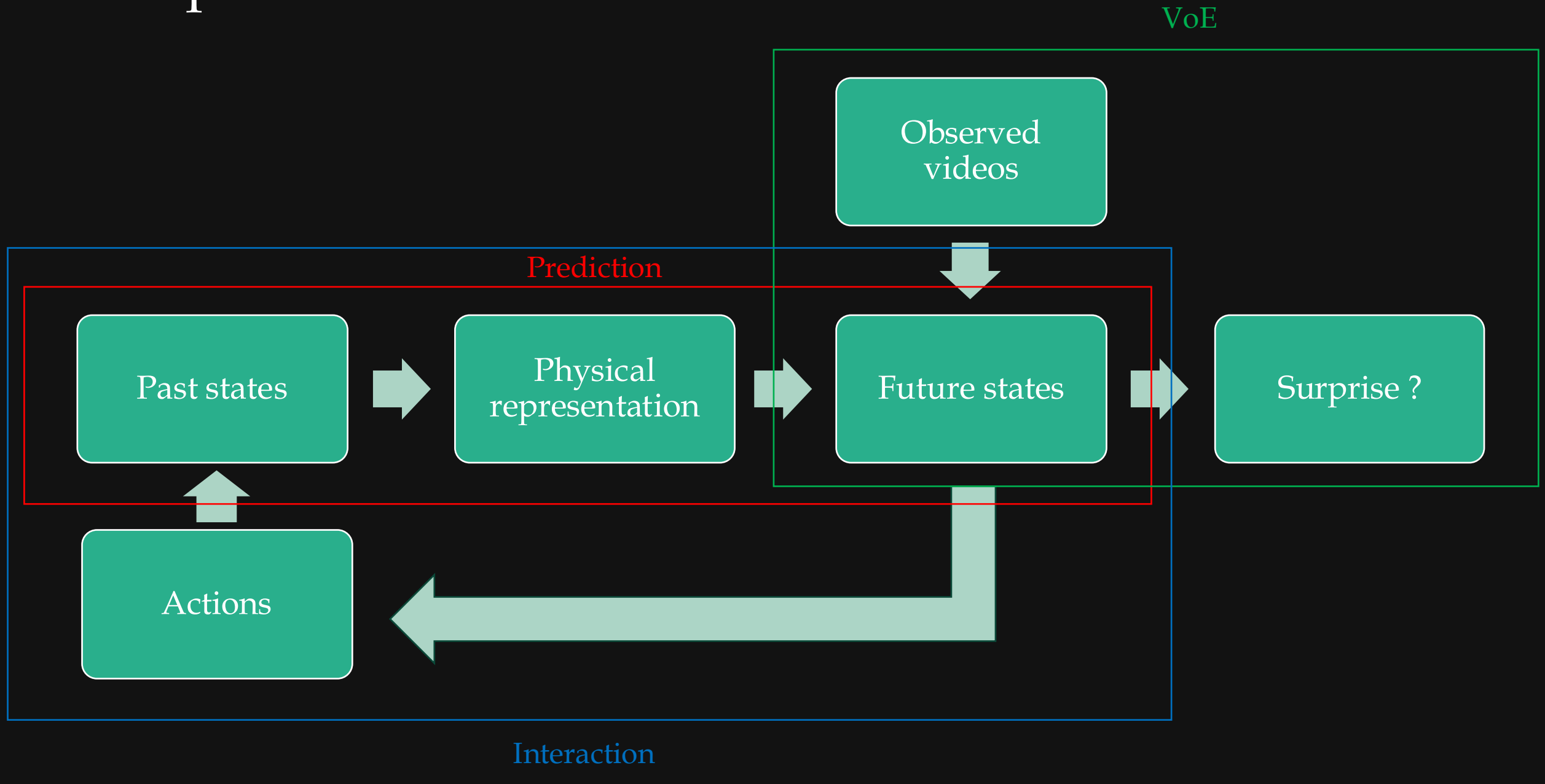




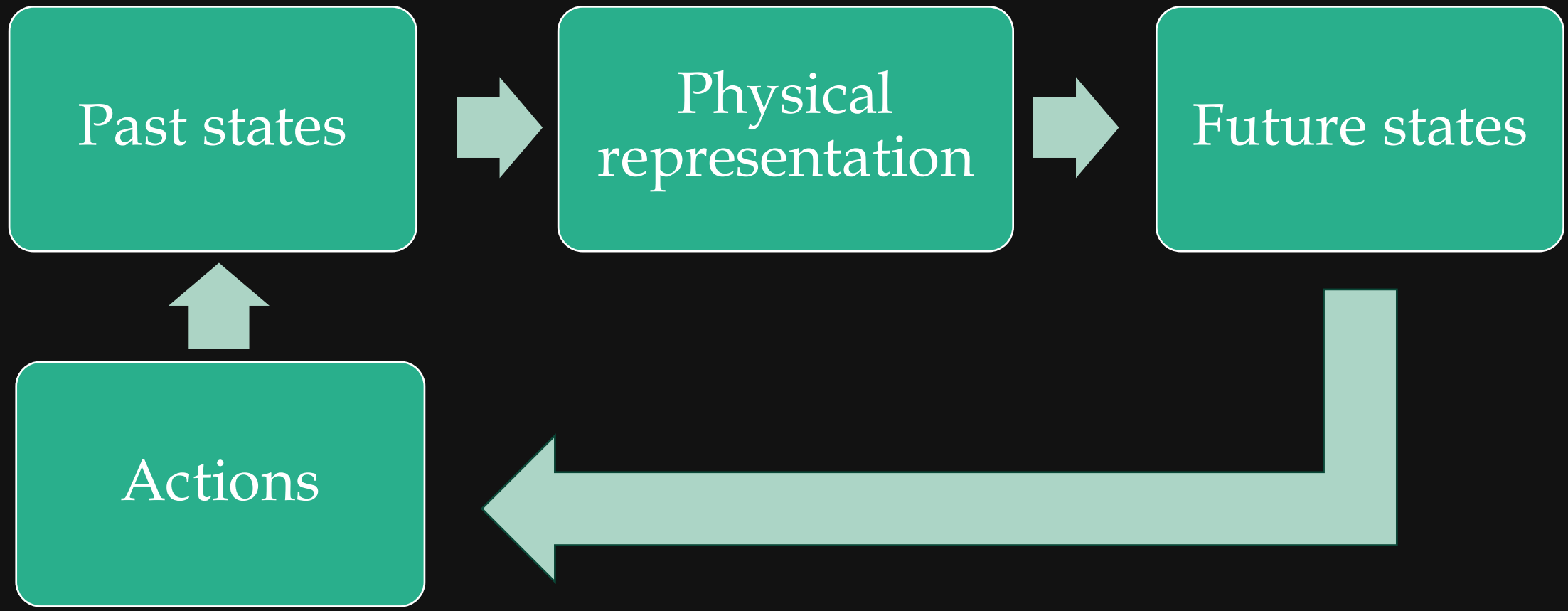



A
B
C

Computational mechanisms



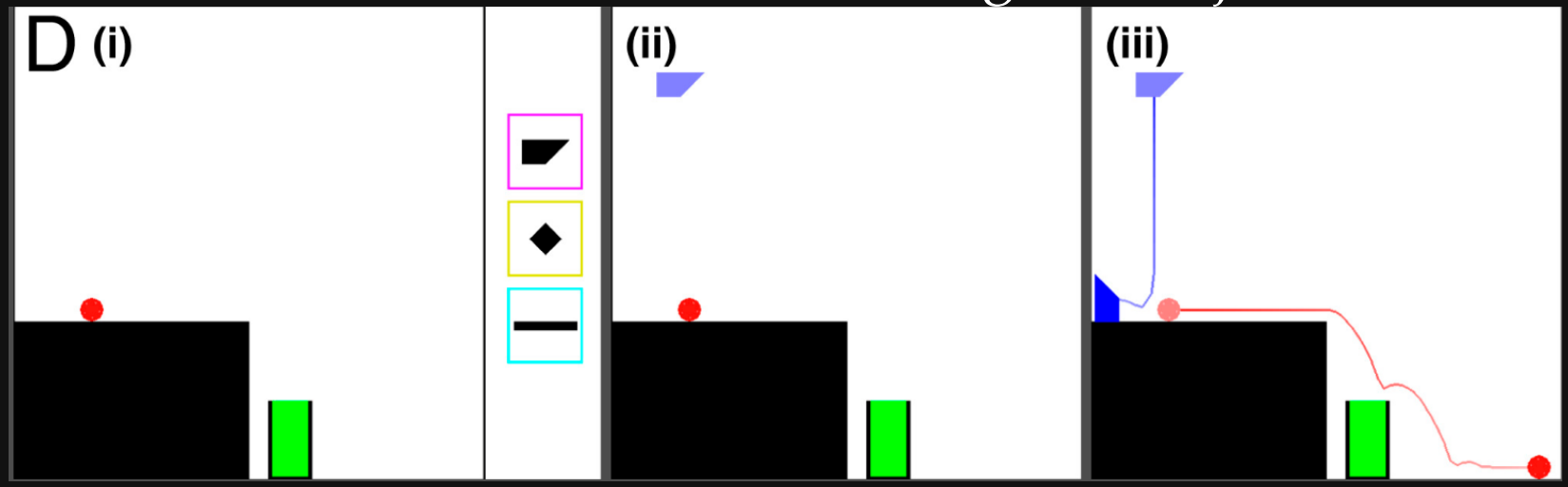
Interaction modeling



Interaction modeling

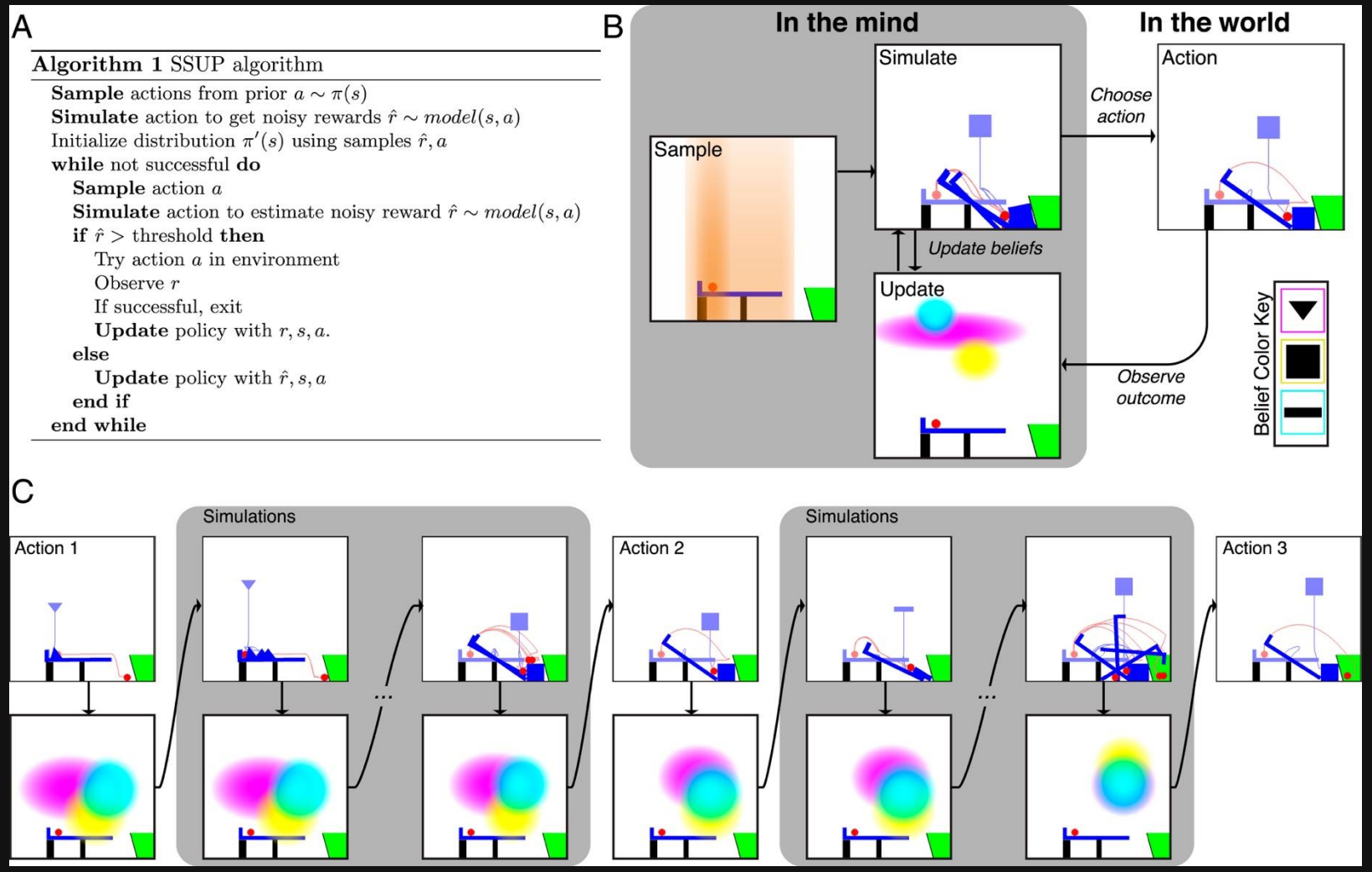
- Problem: Virtual tool game

Make the red ball touch the green object



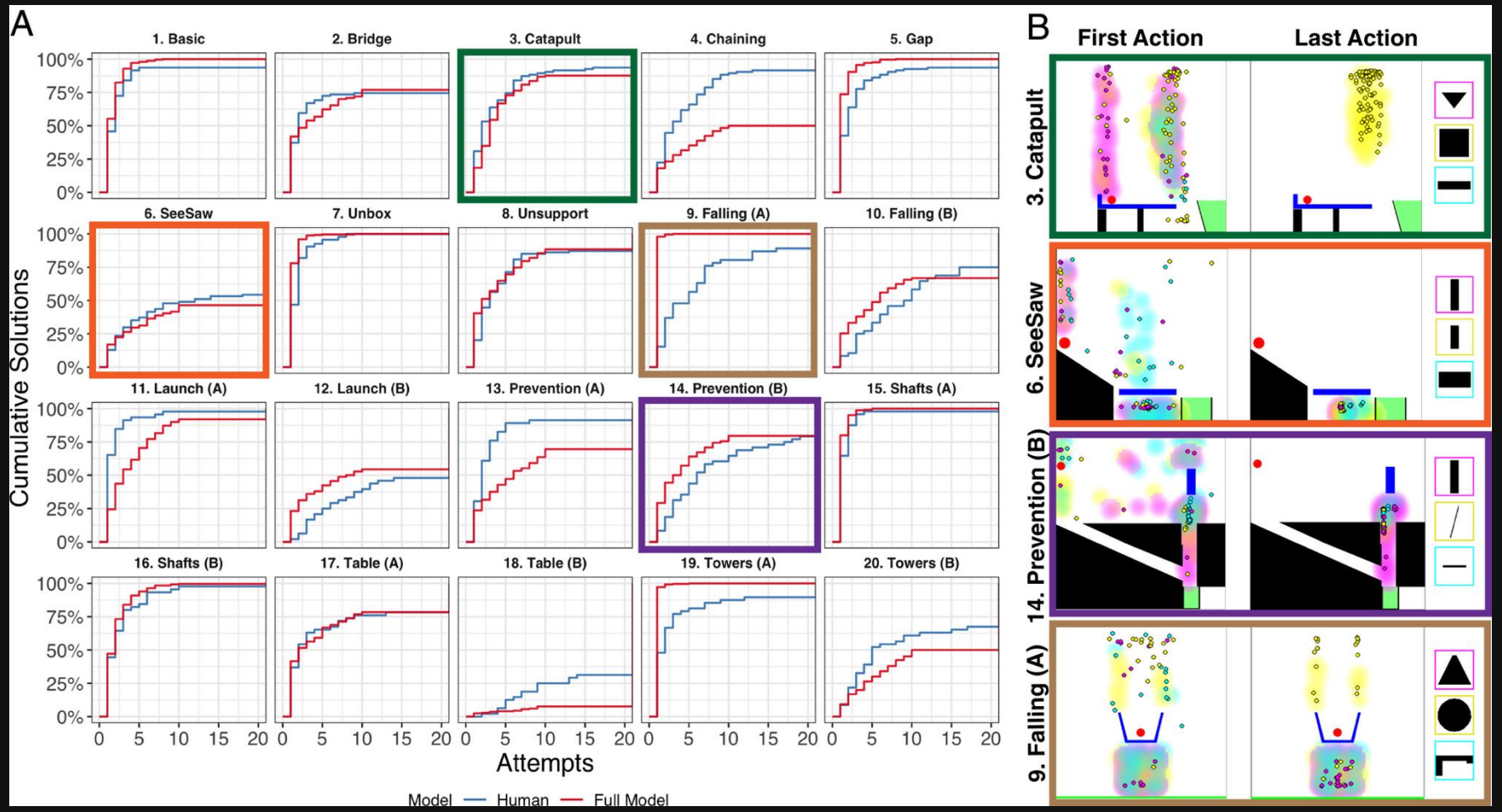
Interaction modeling

- Model: Use RL to solve inverse problem and IPE to do forward simulation



Interaction modeling

Humans solve the problems after trial-and-error learning.
The SSUP model captures the evolution of participants' solutions well.



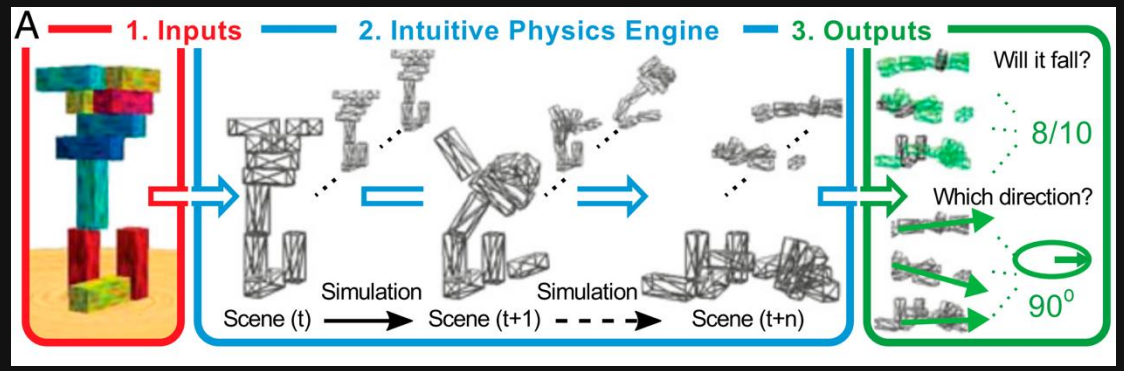
Interaction



"Interactivity"
3D Pinball

Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing

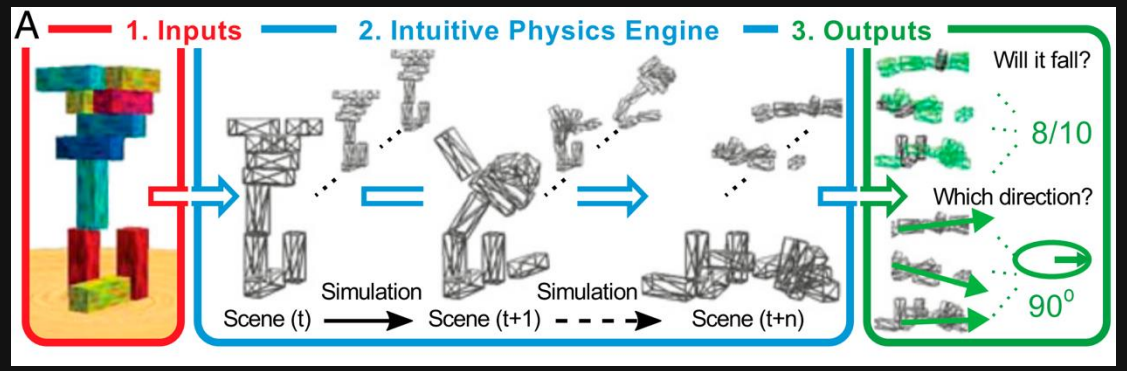
Interaction



Observation
Block Towers

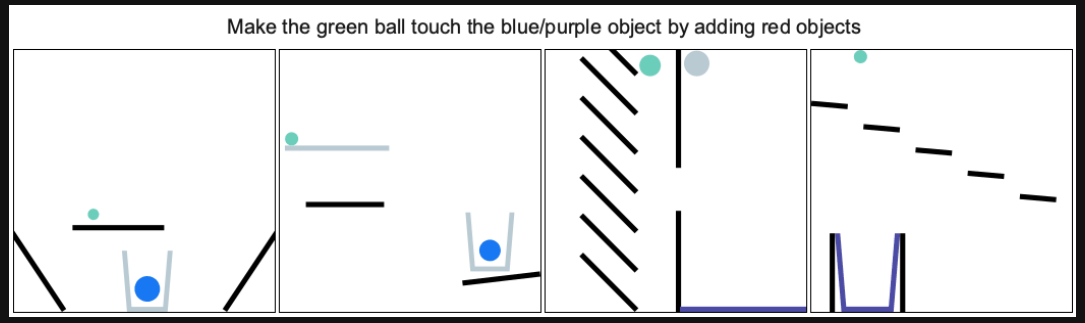
Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing

Interaction



Observation
Block Towers

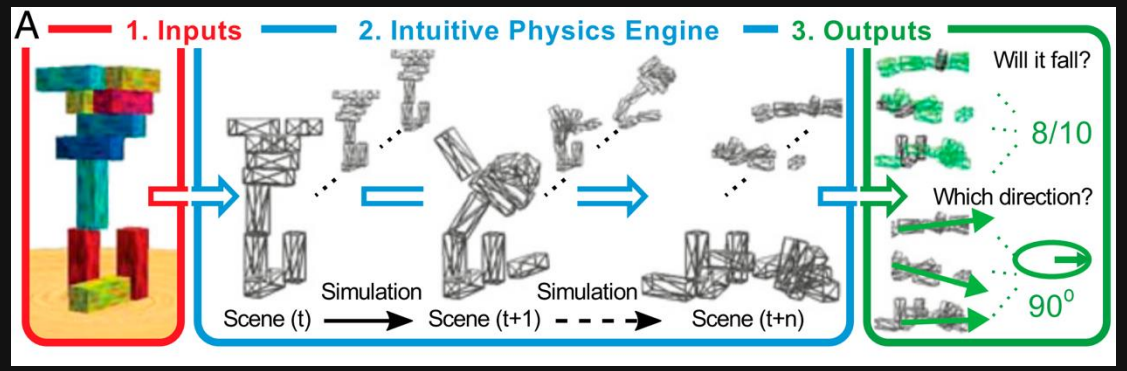
Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing



Single Action
PHYRE

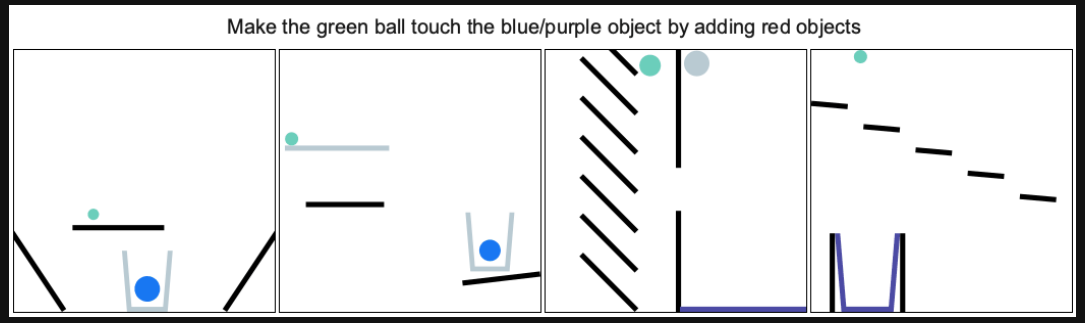
Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing

Interaction



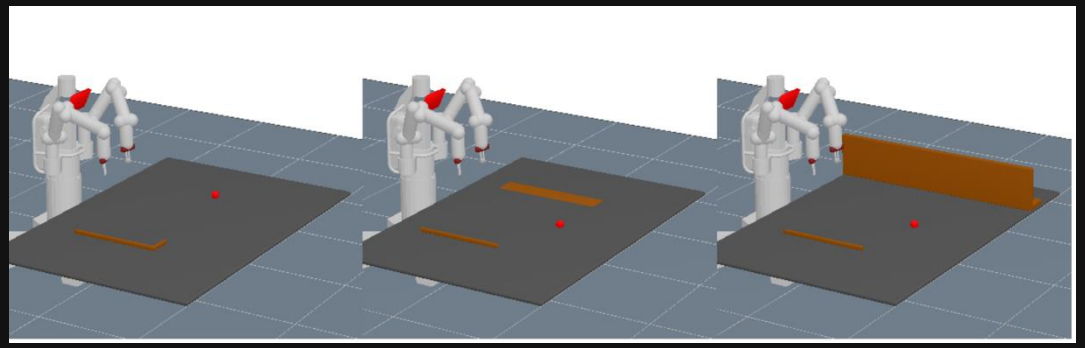
Observation Block Towers

Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing



Single Action PHYRE

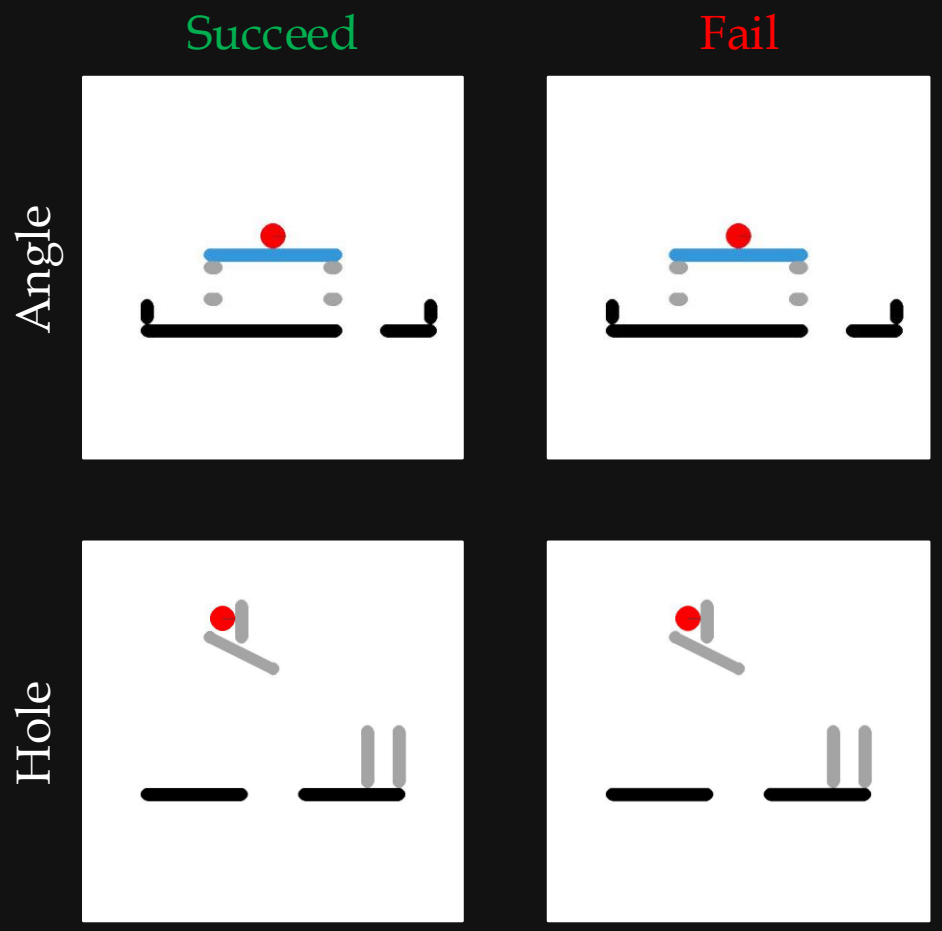
Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing



Simulation Sequential Manipulation Planning

Intuitive	Multi-step
Rich Dynamics	Action Order
Planning	Action Timing

I-PHYRE



Aim: Build human-level interactive agents that can precisely control the physical dynamics by intuitive physical reasoning, multi-step planning, and in-situ intervention.

Task: Drop the red balls into the abyss by eliminating gray blocks.

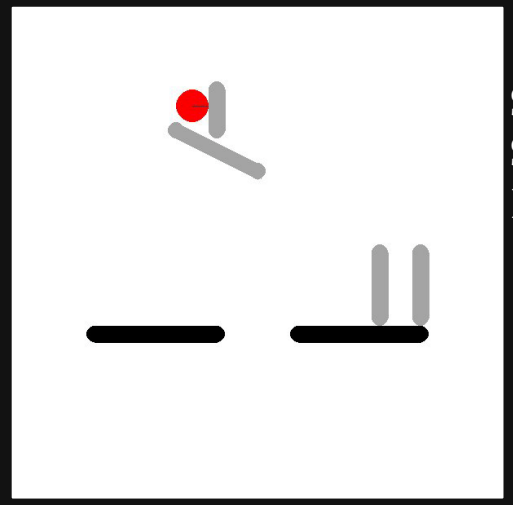
Split: Basic, noisy, compositional, and multi-ball games.

Reward Setting: Reach the goal with fewer elimination steps and less time to get higher scores.

Event	Reward	Account
Forwarding a timestep	-0.1	Encourage time-efficiency
Eliminating a block	-10	Encourage action-efficiency
Reaching the goal	+1000	Encourage problem-solving

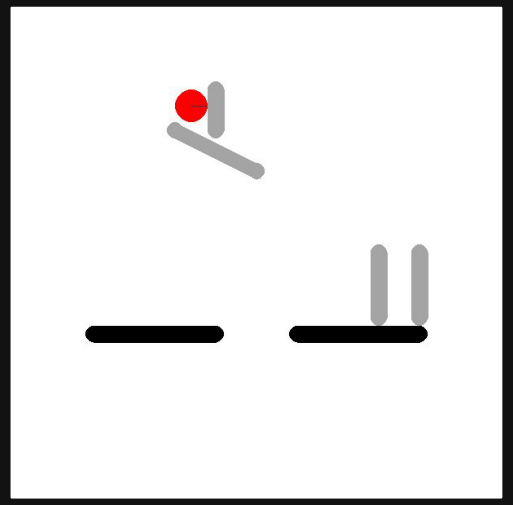
I-PHYRE

Perfect Solution: Use 2 steps



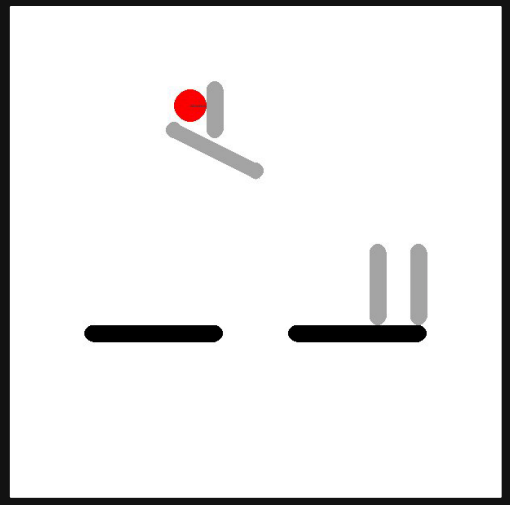
Step 1: The top block at 1.0s
Step 2: The inclined block at 2.0s
Result: Success

Failure: Wrong order



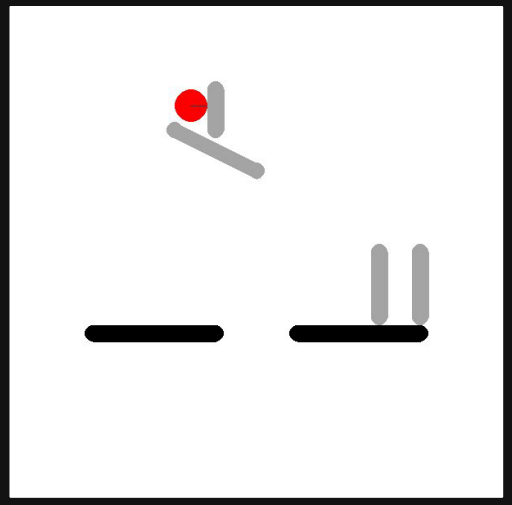
Step 1: The inclined block at 1.0s
Step 2: The top block at 2.0s
Result: Failure

Failure: Miss the perfect timing



Step 1: The top block at 1.0s
Step 2: The inclined block at 3.0s
Result: Failure

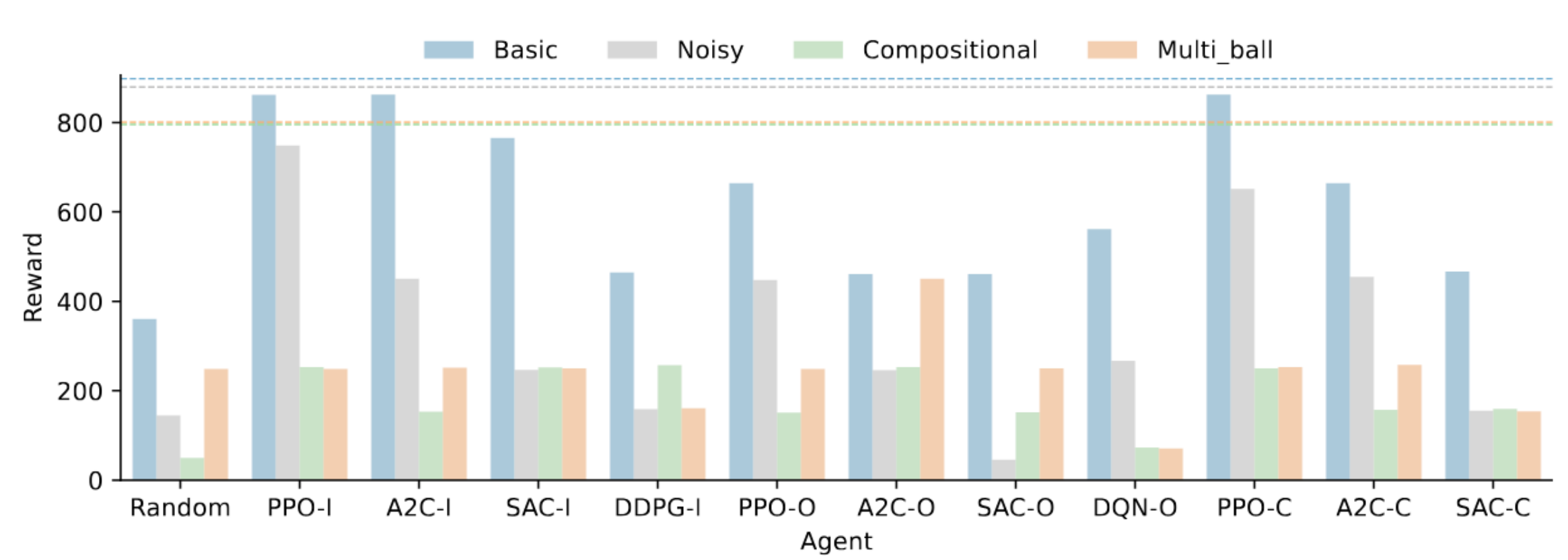
Suboptimal Solution: Use 3 steps



Step 1: The top block at 1.0s
Step 2: The first right block at 2.0s
Step 3: The first right block at 3.0s
Result: Success

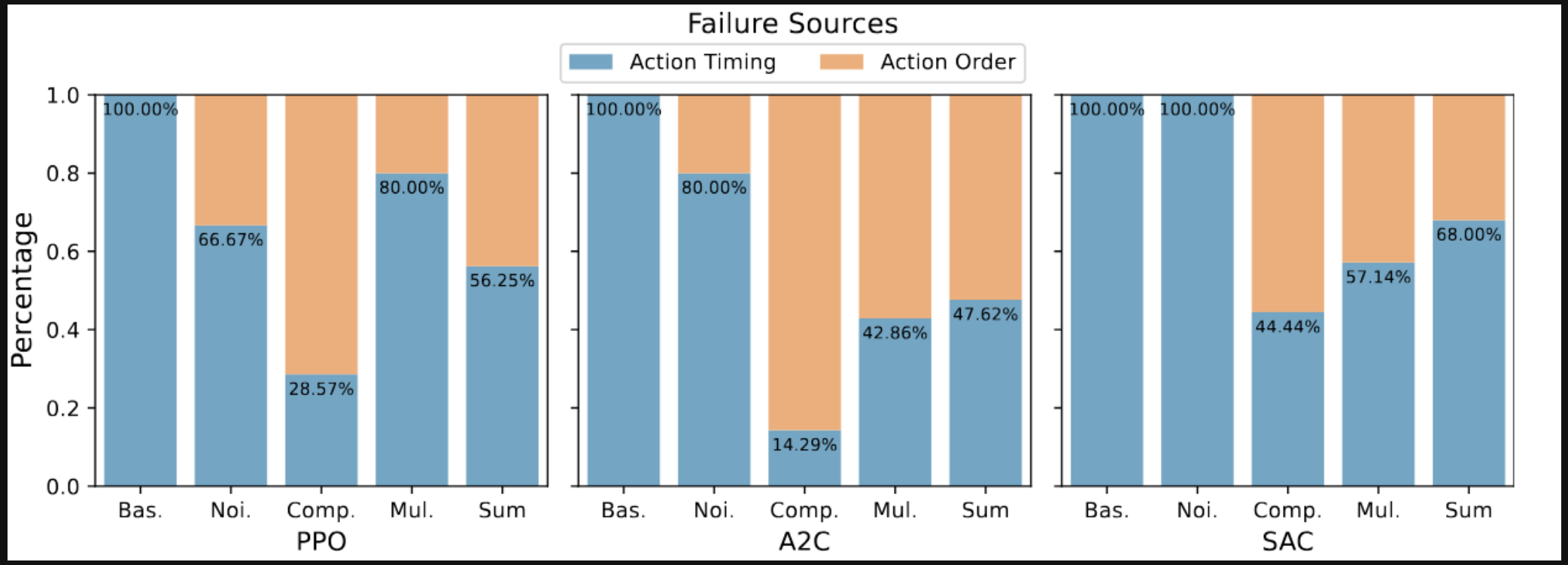
I-PHYRE

Results: Current RL agents don't have human-level generalization ability on I-PHYRE.

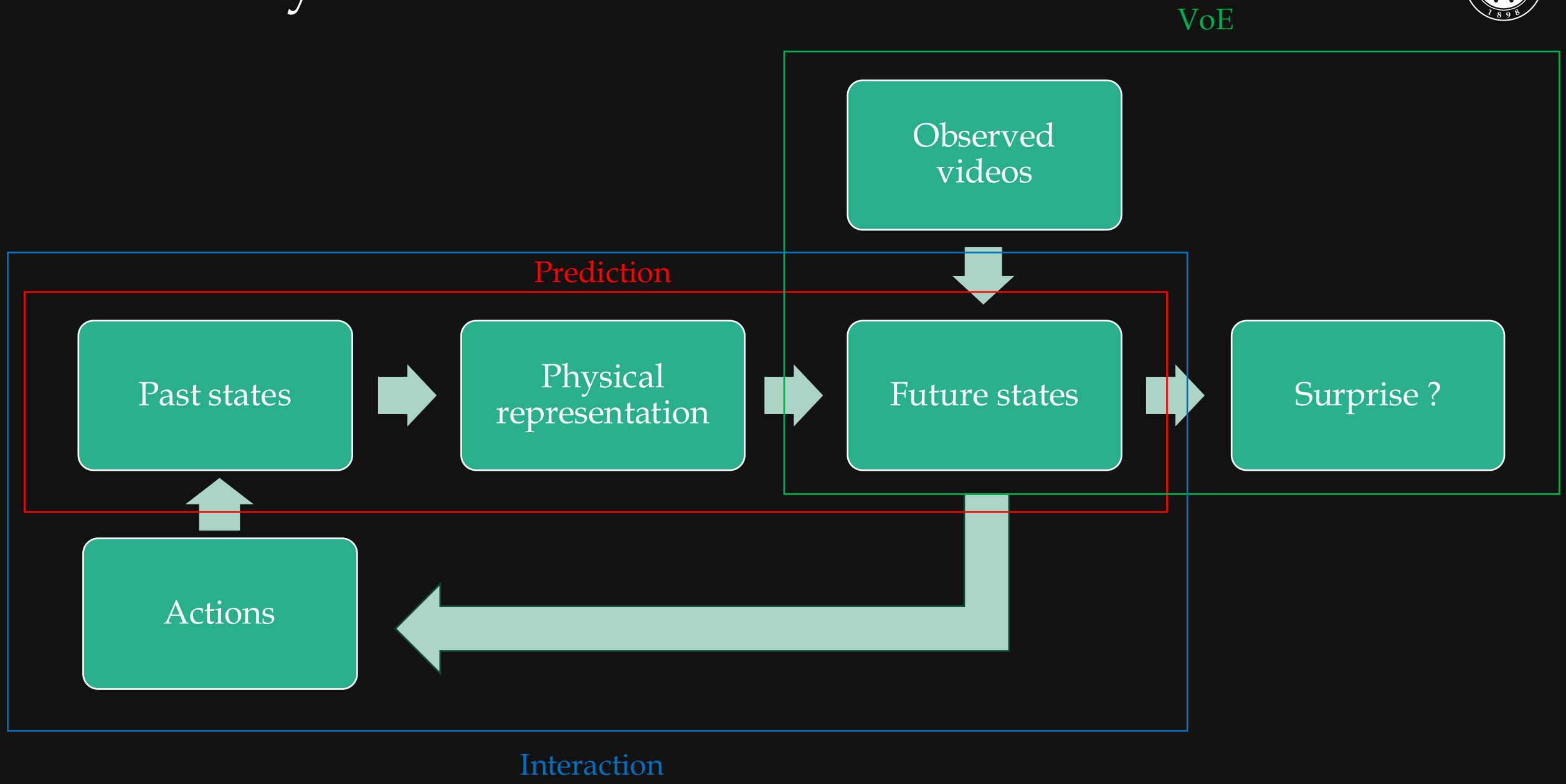


I-PHYRE

Results: Action order and action timing are two crucial failure sources for RL agents. Action timing shows more challenges.



Summary


VoE
Observed videos
Prediction
Past states
Physical representation
Future states
Surprise ?
Actions
Interaction

Part 4
Modern AI systems on physics

Dynamic prediction model, RL, Neural ODEs

Development of modern AI systems on physics

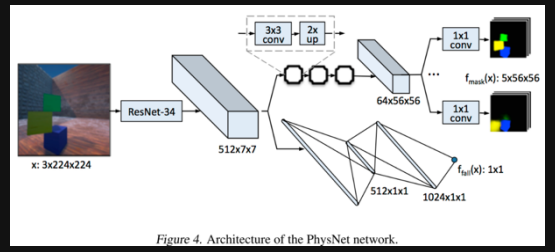
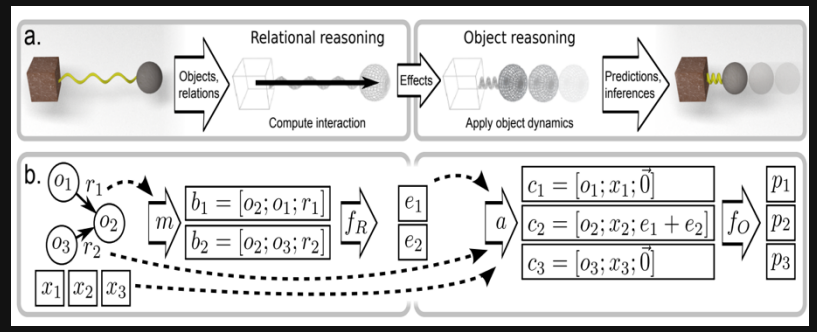
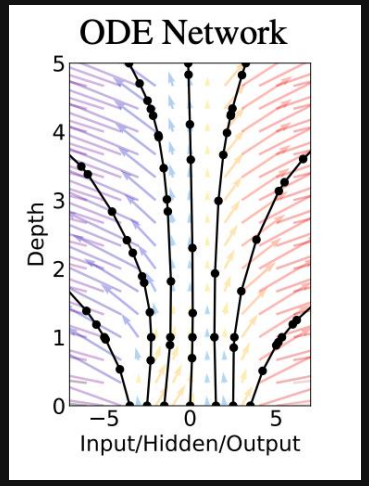


Figure 4. Architecture of the PhysNet network.

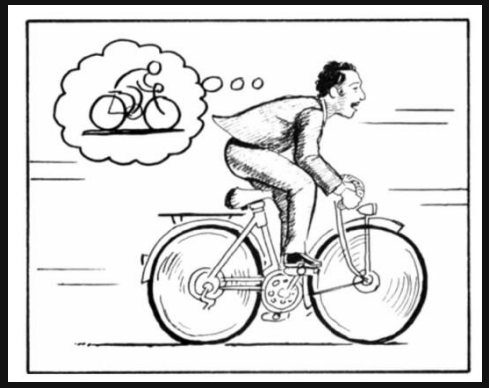
Image-level prediction
2016-now



Object-centric dynamic modeling
2016-now



Continuous dynamic modeling
2018-now

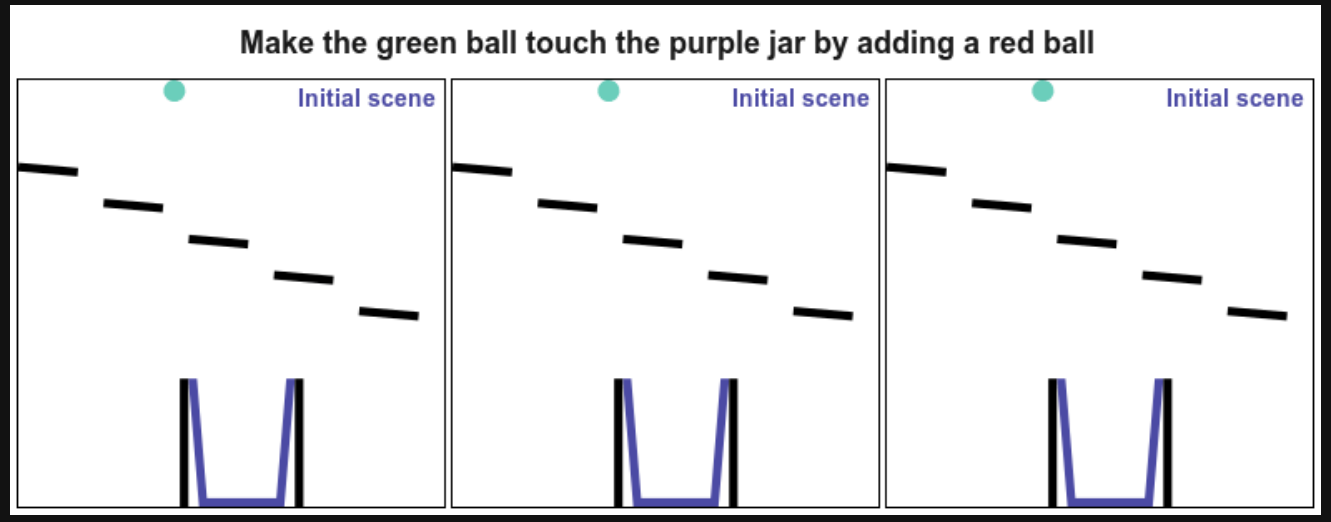


Video generation
2020-now

Physical reasoning dataset

The goal for all tasks within PHYRE is the same: at the end of the simulation the green object must touch the blue or purple object. Agents take actions by placing red objects within the scene to achieve the goal.

Diverse motion: Falling, collision, rotation, containment

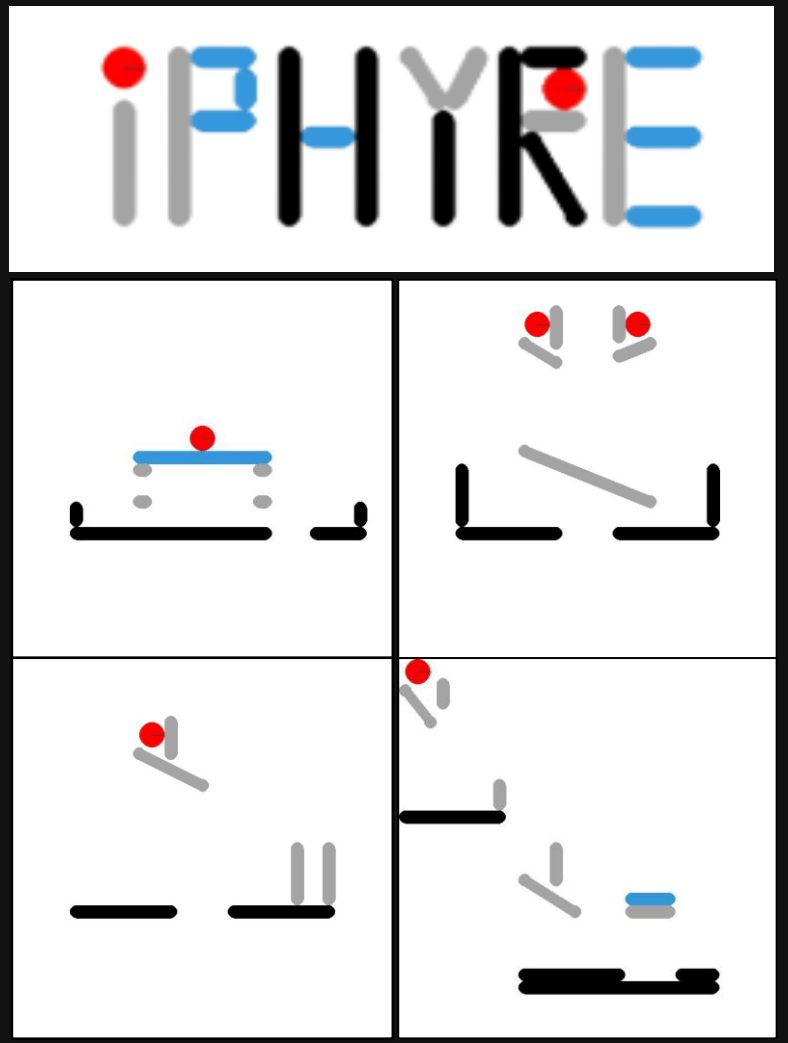




Physical reasoning dataset

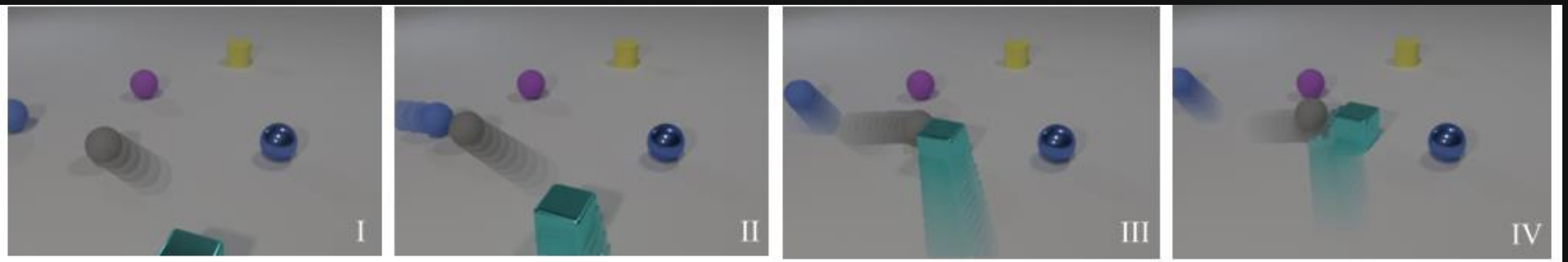
Drop the red balls into the abyss by eliminating gray blocks.

- Intuitive physical reasoning: Our focus is on fostering a rapid yet approximate capability for predicting physical outcomes, rather than an exact simulation or modeling of the physical events.
- Multi-step interventions: The resolution of a problem demands the execution of actions in multiple steps. In contrast to single-action setups, multi-step actions require agents to engage in both long-range prediction and interventional reasoning within the specified scene.
- In-situ interventions: The essence of interaction in physics dynamics lies in timing. Agents that deviate even slightly in timing at any step may potentially fail to complete the task.



Physical reasoning dataset

CLEVRER: CoLlision Events for Video REpresentation and Reasoning



Descriptive:

Q: *How many spheres are moving?*
A: *2*

Q: *What shape is the second object to collide with the gray object?*
A: *Cube*

Q: *Are there any collisions after the cube enters the scene?*
A: *Yes*

Explanatory:

Q: *Which of the following is responsible for the collision between the gray object and the cube?*

a) The presence of the purple object
b) The collision between the blue sphere and the gray sphere
c) The presence of the purple object
d) The presence of the blue object

A: *b), d)*

Predictive:

Q: *What will happen next?*

a) The cube and the gray object collide
b) The gray sphere collides with the purple sphere
c) The metal sphere and the cube collide
d) The gray sphere collides with the blue sphere

A: *b)*

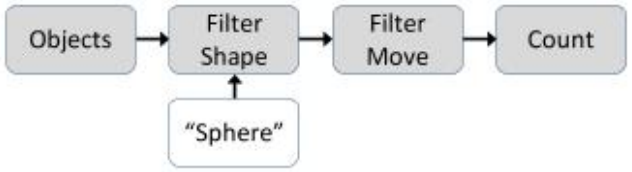
Counterfactual:

Q: *What will happen if the gray sphere is removed?*

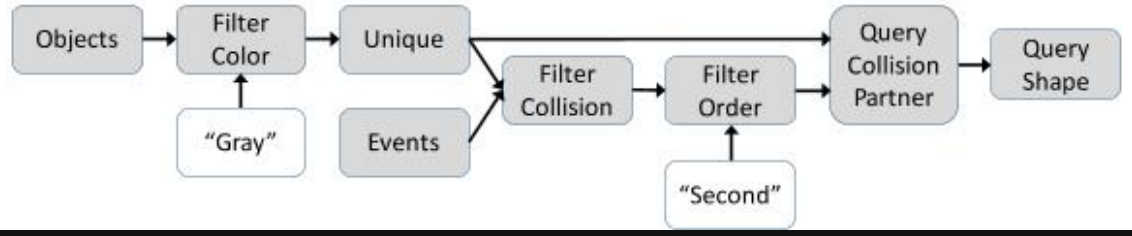
a) The blue sphere collides with the cube
b) The blue sphere and the metal sphere collide
c) The purple object collides with the cylinder
d) The cube and the metal sphere collide

A: *a), d)*

How many spheres are moving?

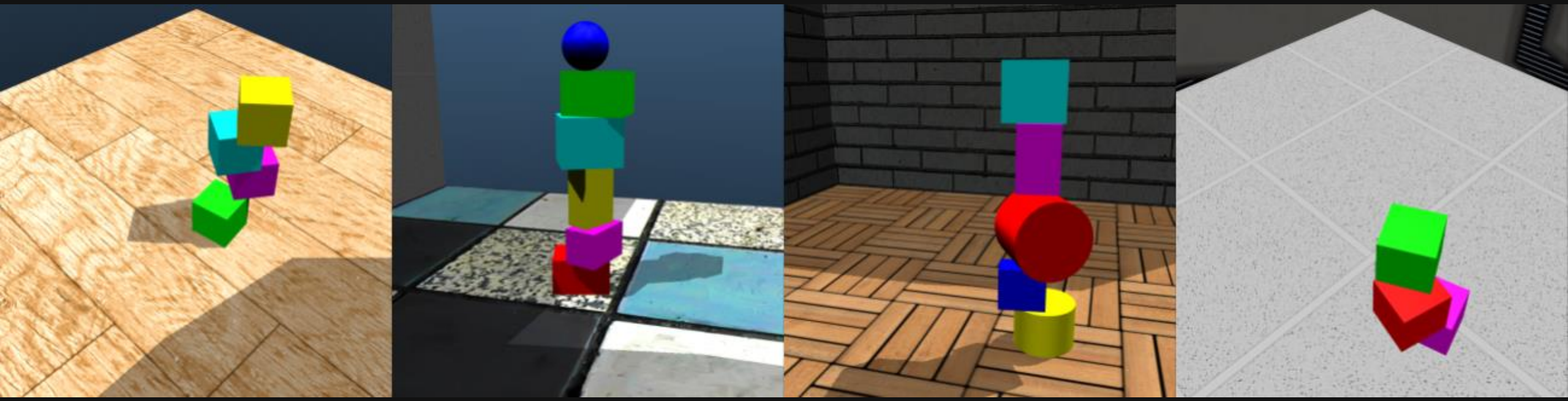


What shape is the second object to collide with the gray object?



Physical reasoning dataset

ShapeStacks

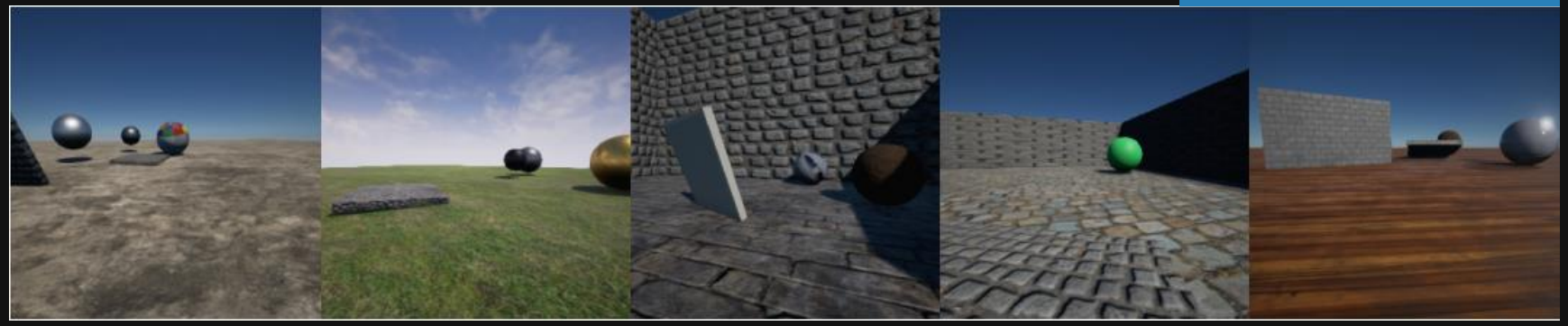


Physical reasoning dataset

Examine agents' understanding of physical concepts such as:

- Object permanence
- Share constancy
- Spatiotemporal continuity
- Energy conservation

IntPhys



Physical reasoning dataset

Physion

Physion is a dataset and benchmark for evaluating AI models against human intuitions about how objects move and interact with one another.



End2end prediction

- Problem: Judge the stability of block towers and predict future states

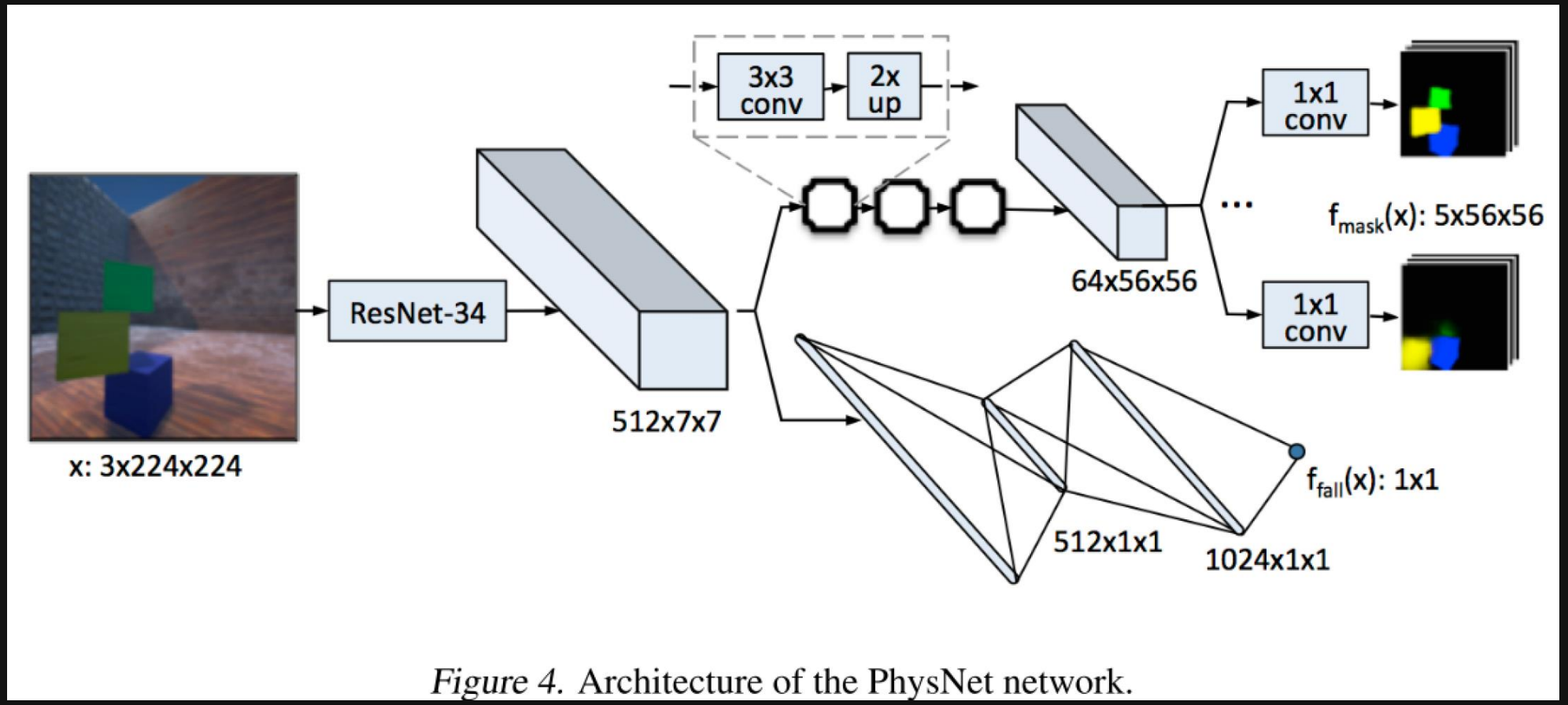
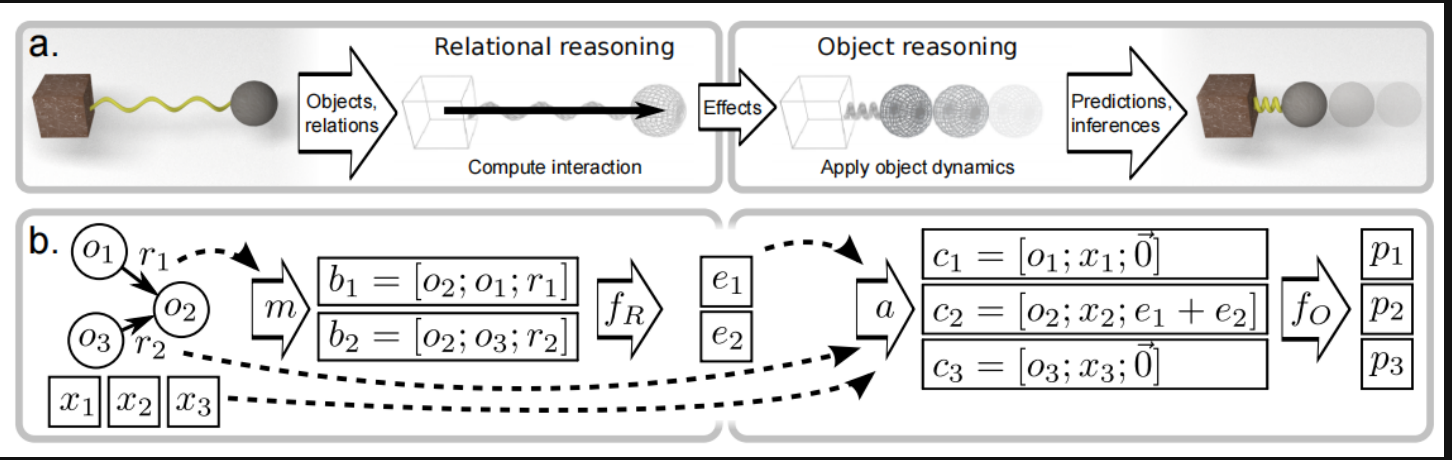


Figure 4. Architecture of the PhysNet network.

Lerer, A., Gross, S., & Fergus, R. (2016, June). Learning physical intuition of block towers by example. In International conference on machine learning (pp. 430-438). PMLR.

Prediction with intermediate dynamics



- Interaction Network: Learning about Objects, Relations and Physics
  - Relation-centric function, MLP: $e_{t+1} = f_R(o_{1,t}, o_{2,t}, r)$
  - Object-centric function, MLP: $o_{2,t+1} = f_O(o_{2,t}, e_{t+1})$
- Task formulation: Given graph $G = < O, R >$, where $O = \{o_j\}, R = \{< i, j, r_k >_k\}$, external effects $X = \{x_j\}$

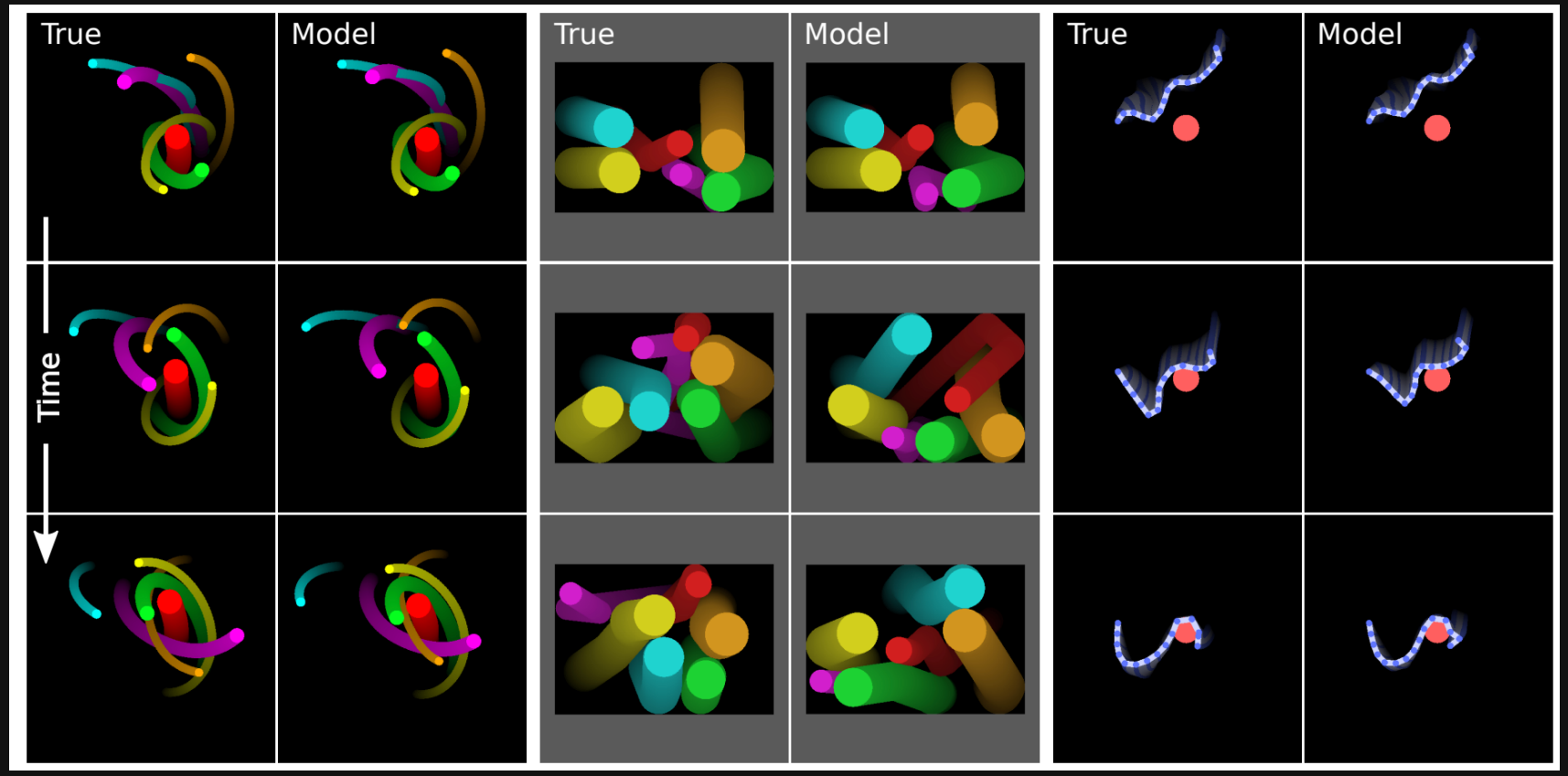
The basic IN is defined as,

$$\text{IN}(G) = \phi_O(a(G,\ X,\ \phi_R(\ m(G)\ )\ ))$$

$$\begin{aligned} m(G) &= B = \{b_k\}_{k=1\ldots N_R} & a(G,X,E) &= C = \{c_j\}_{j=1\ldots N_O} \\ f_R(b_k) &= e_k & f_O(c_j) &= p_j \\ \phi_R(B) &= E = \{e_k\}_{k=1\ldots N_R} & \phi_O(C) &= P = \{p_j\}_{j=1\ldots N_O} \end{aligned}$$

Battaglia, P., Pascanu, R., Lai, M., & Jimenez Rezende, D. (2016). Interaction networks for learning about objects, relations and physics. Advances in neural information processing systems, 29.

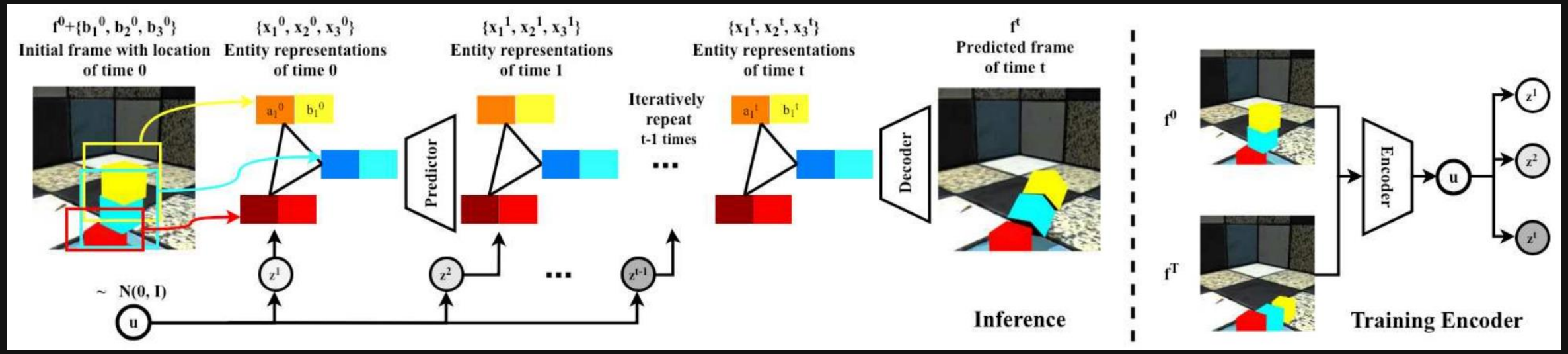
Prediction with intermediate dynamics



Battaglia, P., Pascanu, R., Lai, M., & Jimenez Rezende, D. (2016). Interaction networks for learning about objects, relations and physics. Advances in neural information processing systems, 29.

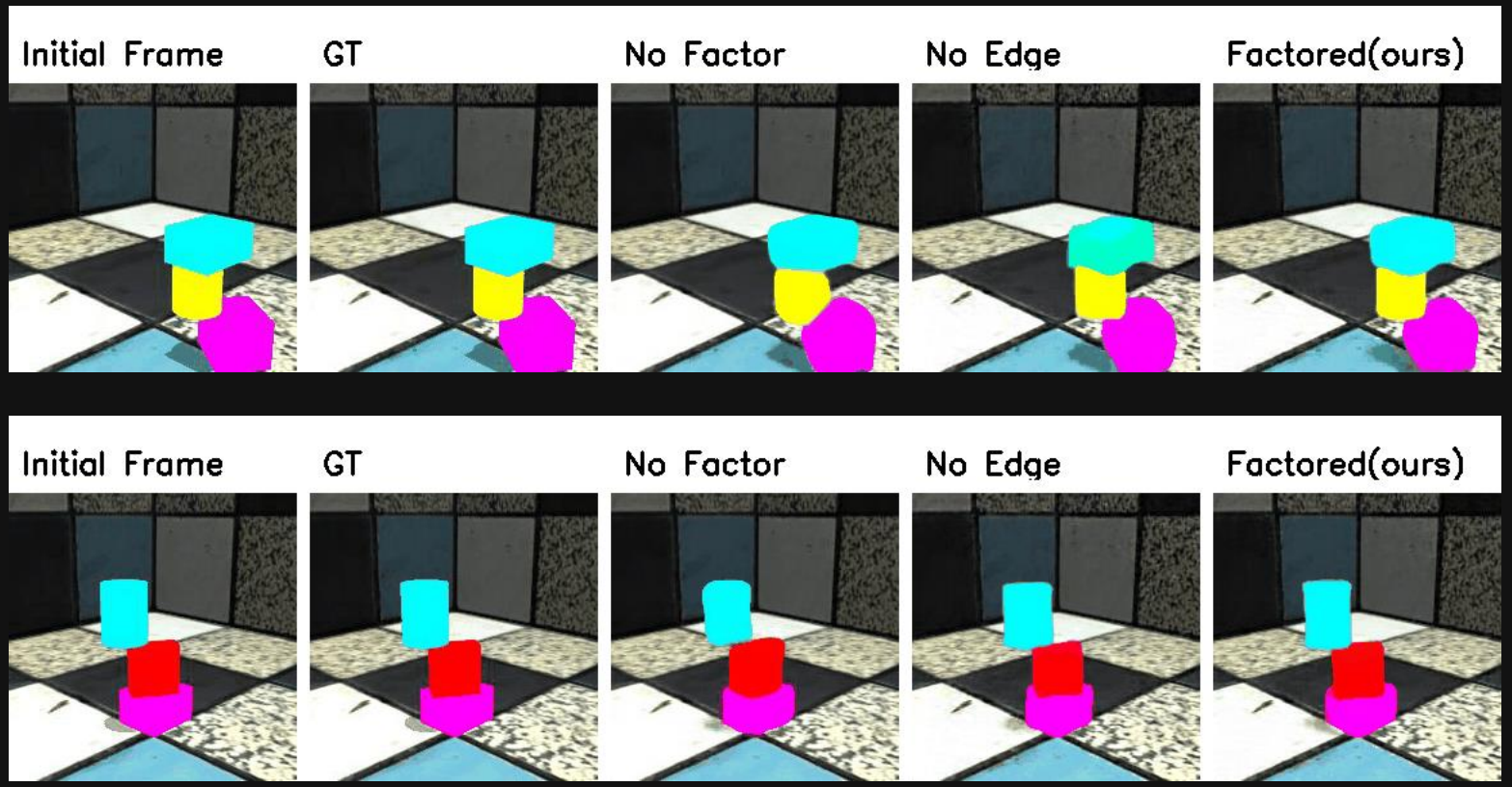
Vision-based prediction

- Compositional Video Prediction. ICCV, 2019
  - Detect objects in images
  - Learn object encoder from images
  - Predict next state in latent space
  - Decode object into pixels



Vision-based prediction

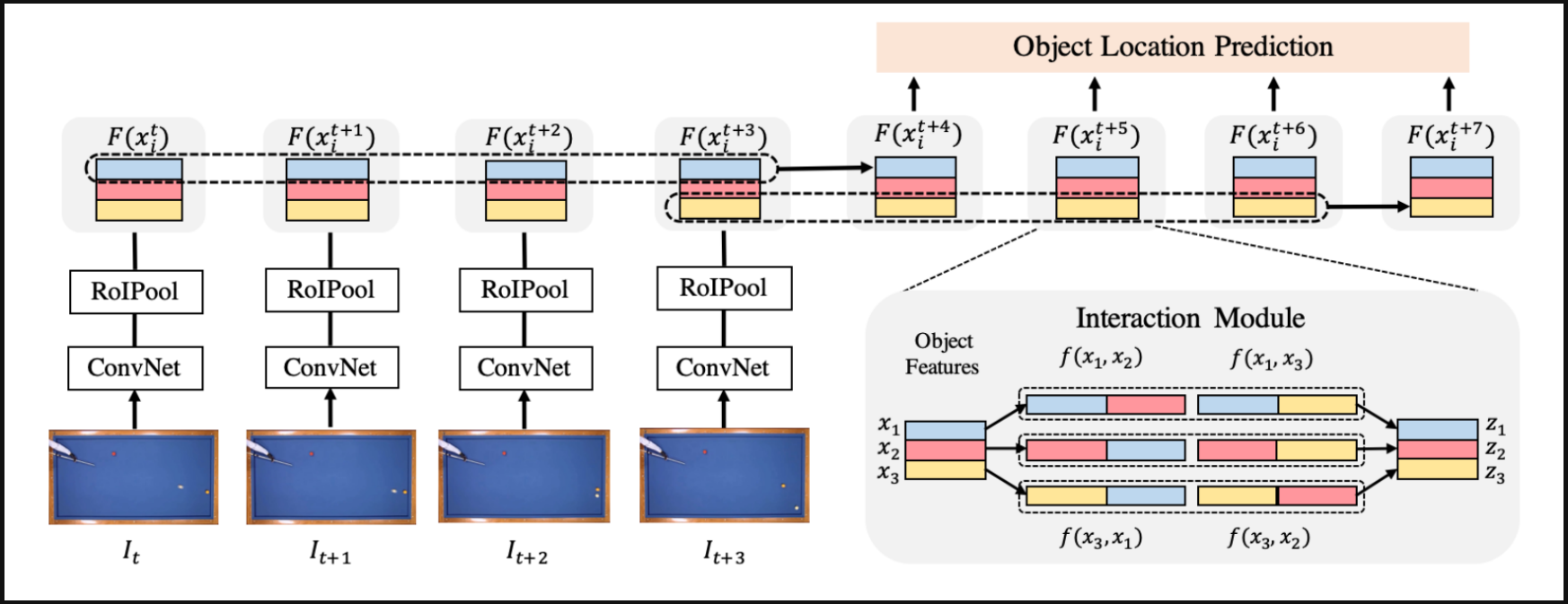
- Results



Vision-based prediction

Region-proposal Interaction network (2021):

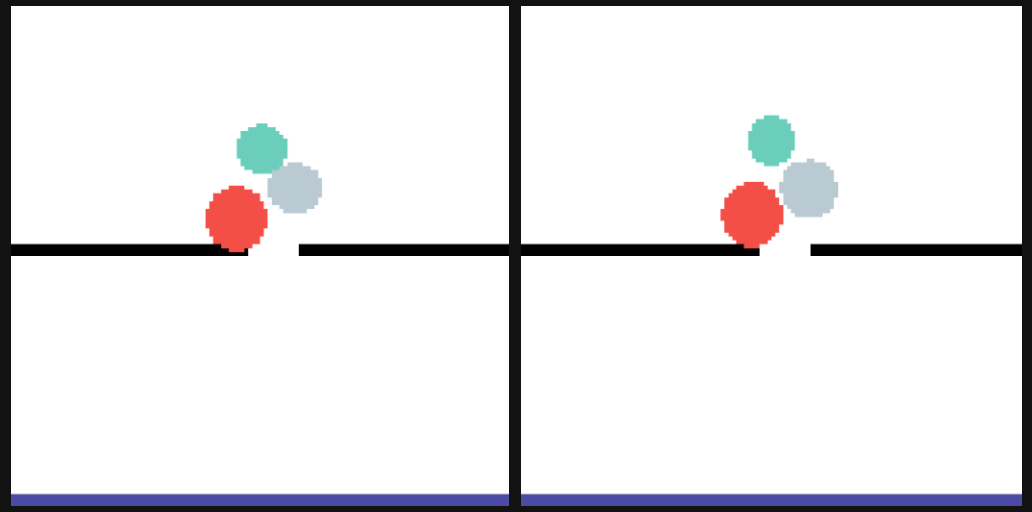
- Learn from object masks (represent shapes) and bounding boxes (represent locations)
- Use interaction network to model object relations



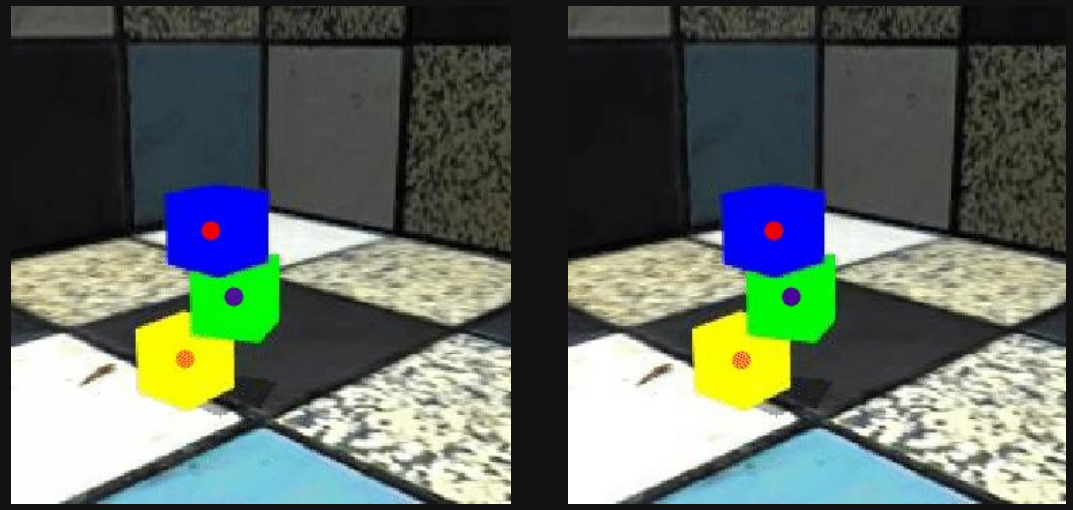
Vision-based prediction

Results of RPIN

Pred GT

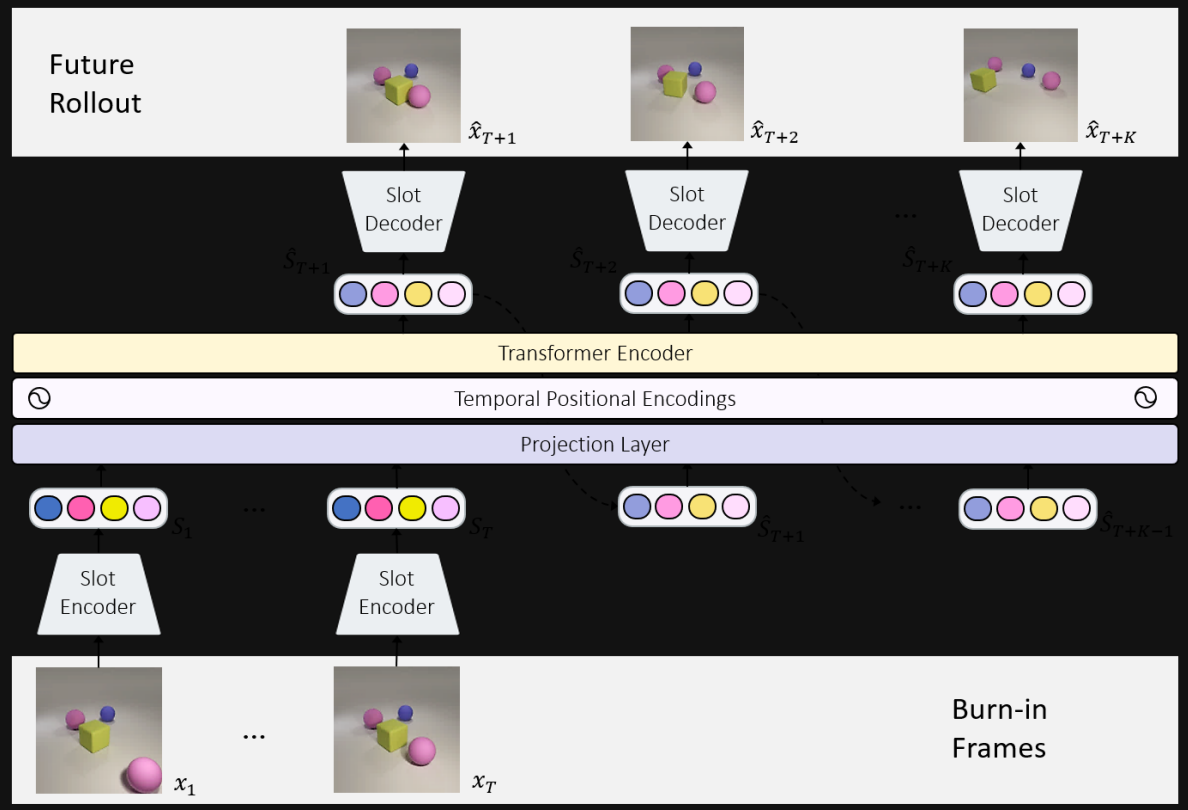


Pred GT



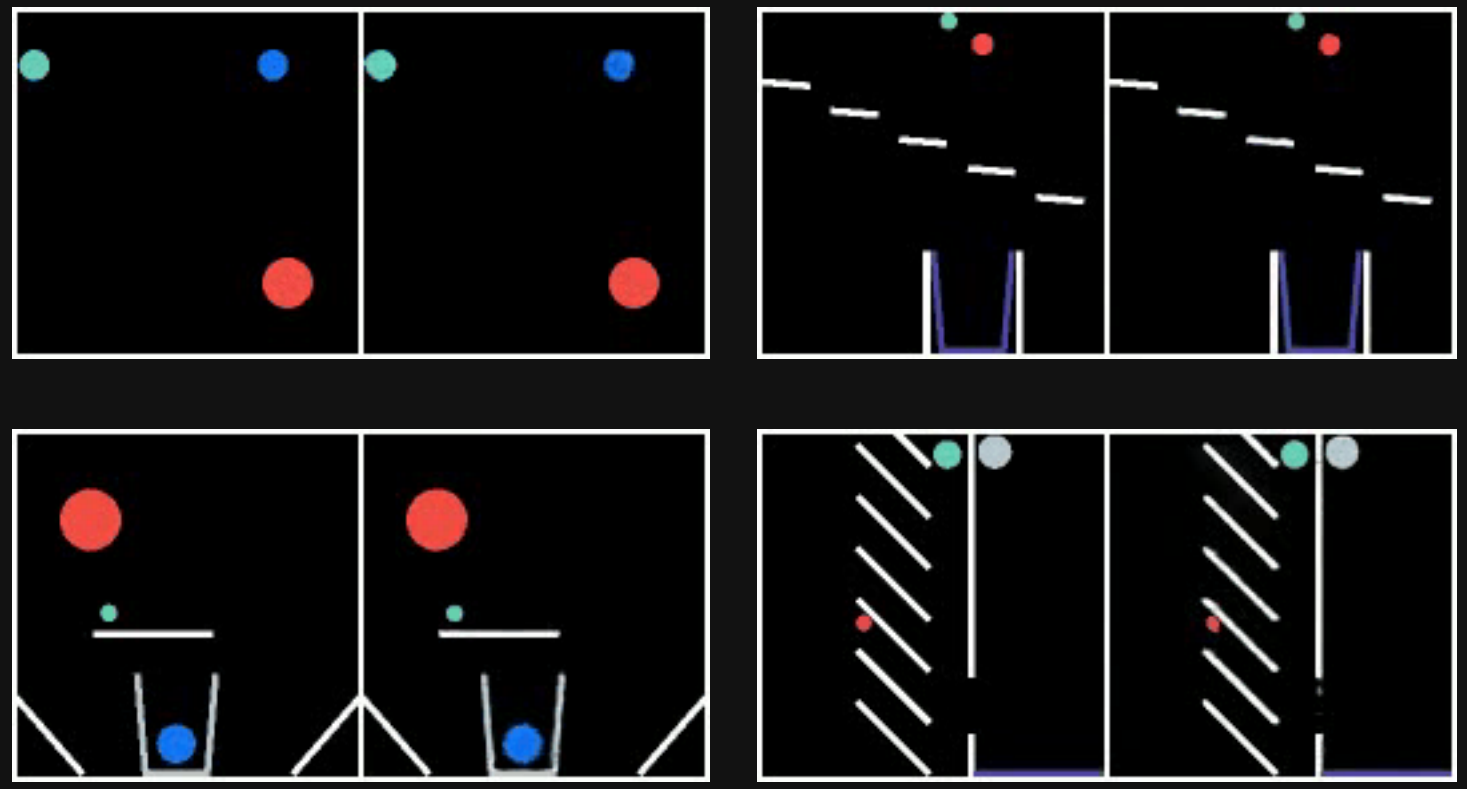
Unsupervised vision-based prediction

- SlotFormer. ICLR, 2023
  - Learn object-centric representations from videos but do not assume known object masks and bounding boxes (unsupervised encoding)



Unsupervised vision-based prediction

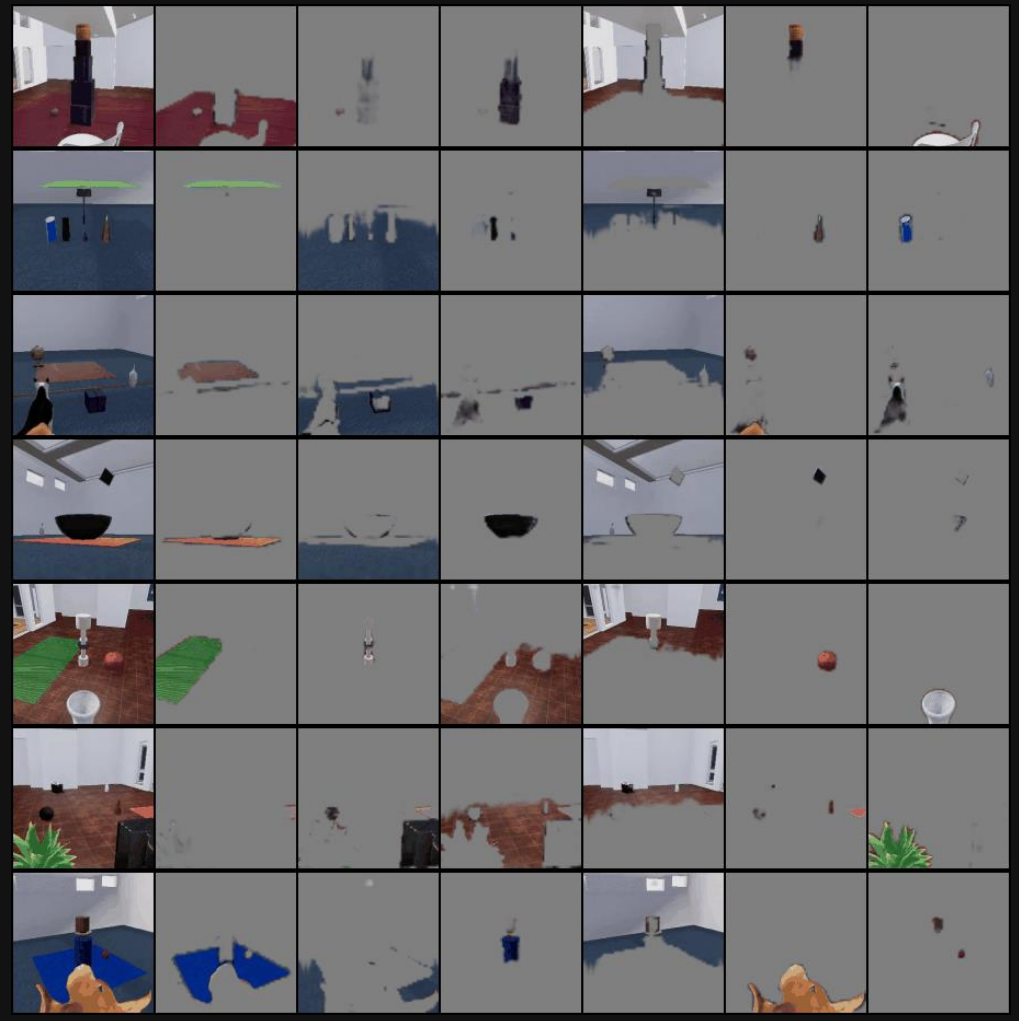
Results on PHYRE



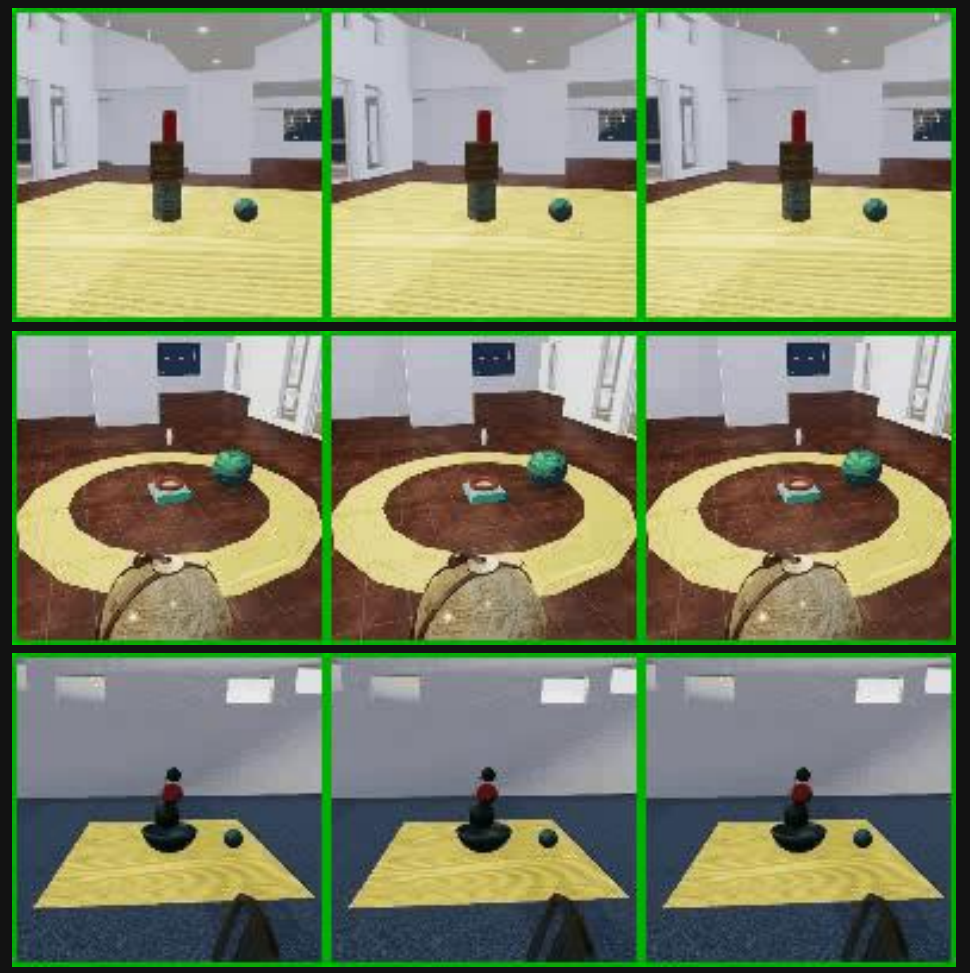
Unsupervised vision-based prediction

Results on Physion

Unsupervised learned per-slot masks



GT Reconstructed Prediction



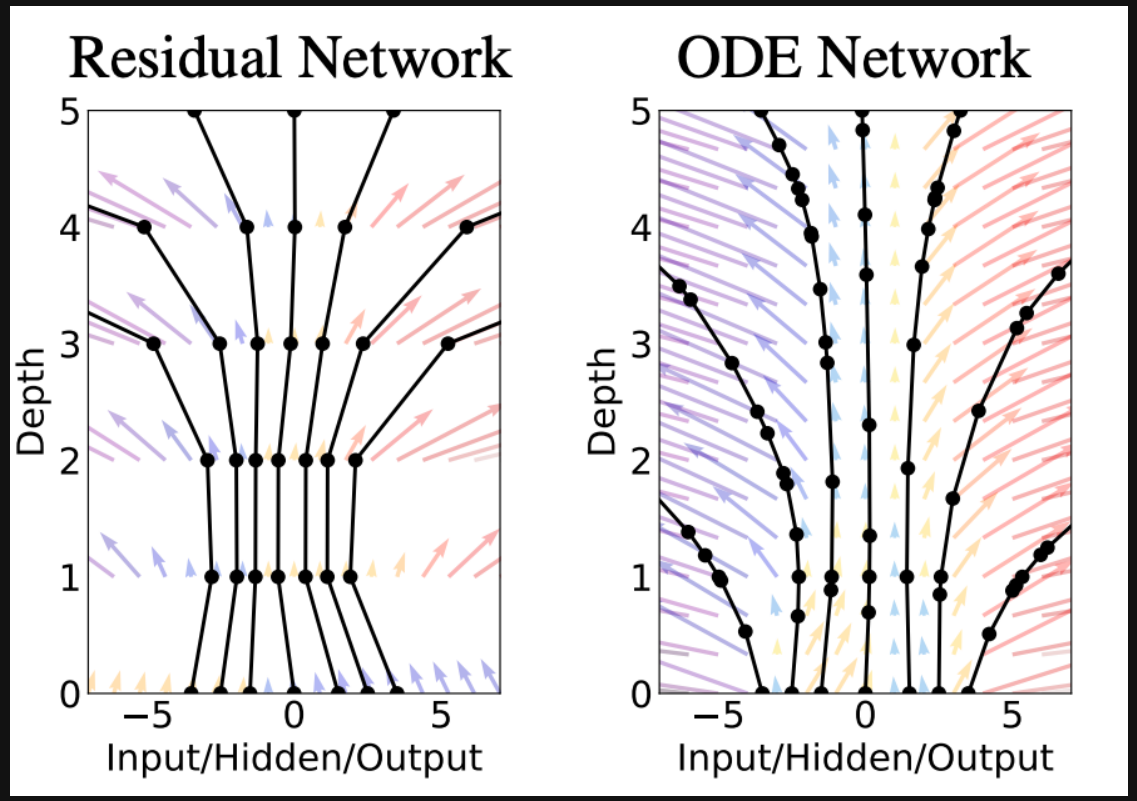
NeuralODE for continuous dynamics

- Traditional dynamic models (discrete)

$$\mathbf{h}_{t+1} = \mathbf{h}_t + f(\mathbf{h}_t, \theta_t)$$

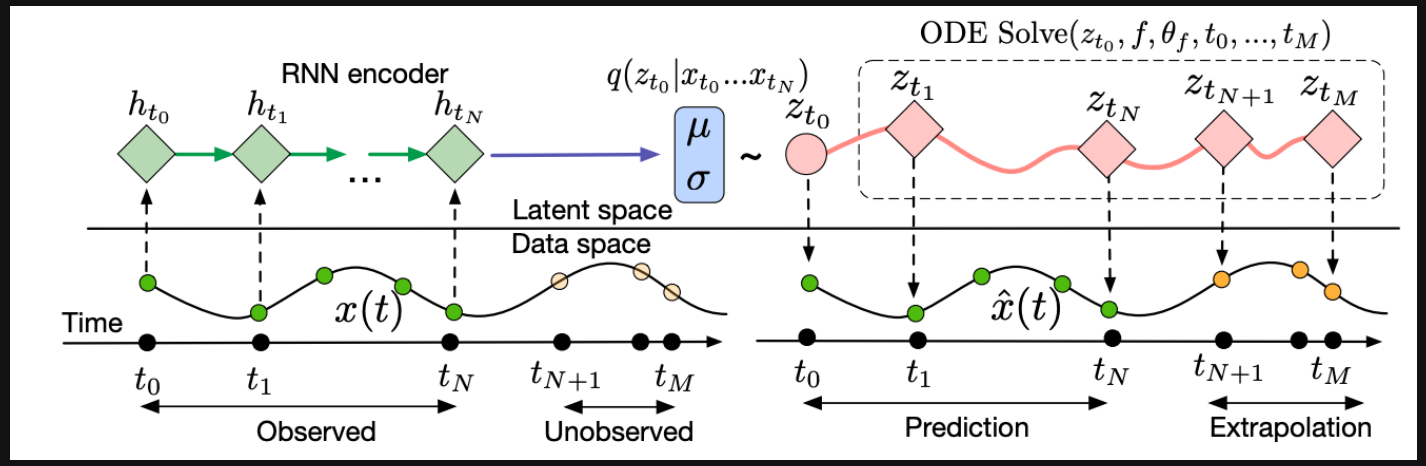
- Neural ODEs (continuous)

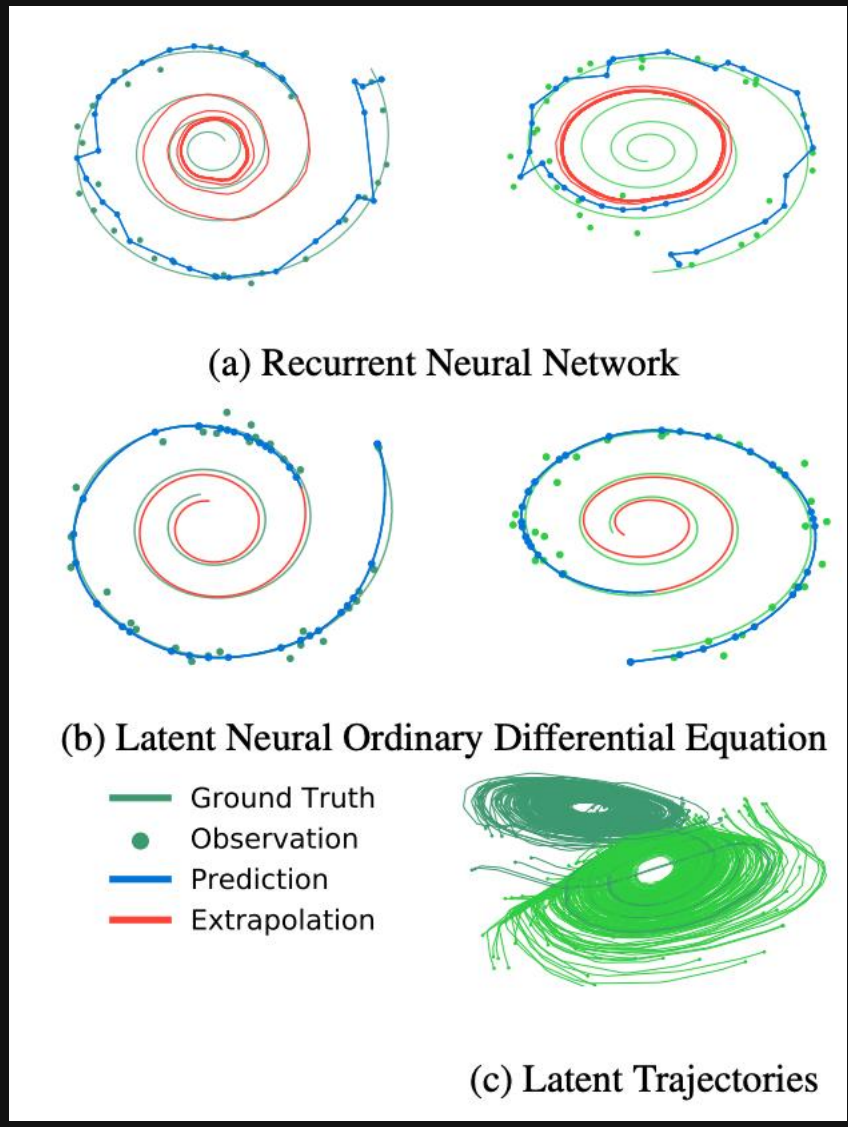
$$\frac{d\mathbf{h}(t)}{dt} = f(\mathbf{h}(t), t, \theta)$$



NeuralODE for continuous dynamics

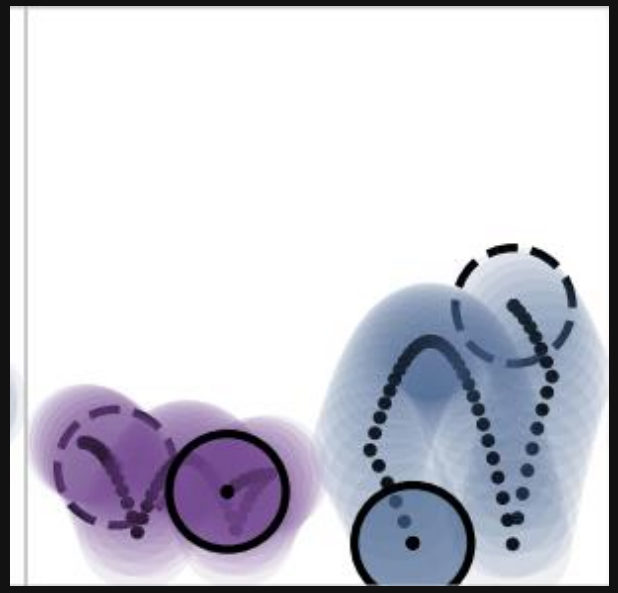
Neural ODEs can do better reconstruction and extrapolation than traditional RNN.



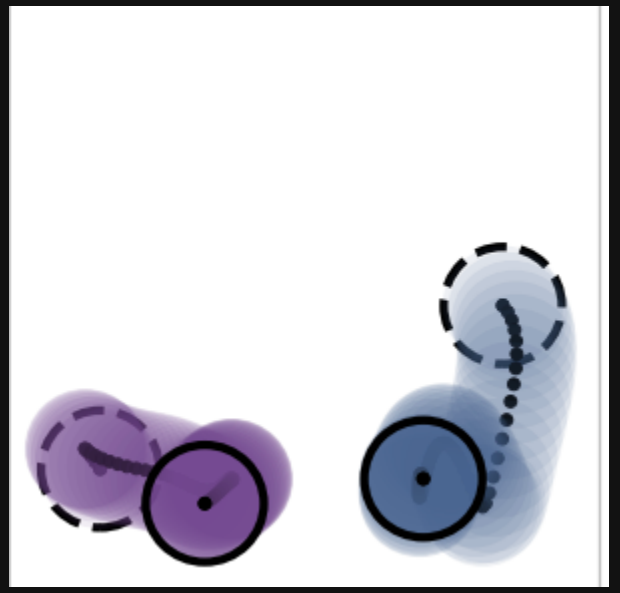


Discontinuities

- Neural ODEs cannot model discontinuities in the state.



Ground trith



Neural ODE

Neural Event ODE

- Neural Event ODEs: Introducing event functions and update functions to model discontinuities such as collisions.

Algorithm 1 The Neural Event ODE. In addition to an ODE, we also model a variable number of possible event locations and how each event affects the system.

Input: An initial condition (t_0, z_0), a final termination time T, a drift function f, an event function g, and an instantaneous update function h. *Parameters of f, h, g are notationally omitted.*

$i = 0$

while $t_i < T$ **do**

$t_{i+1}, z'_{i+1} = \texttt{ODESolveEvent}(z_i, f, g, t_i)$ ▷ Solve until the next event

$z_{i+1} = h(t_{i+1}, z'_{t+1})$ ▷ Instantaneously update the state

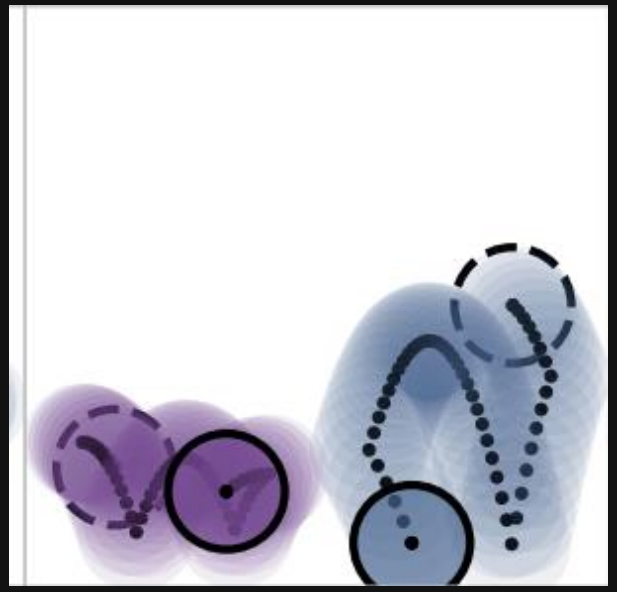
$i = i + 1$

end while

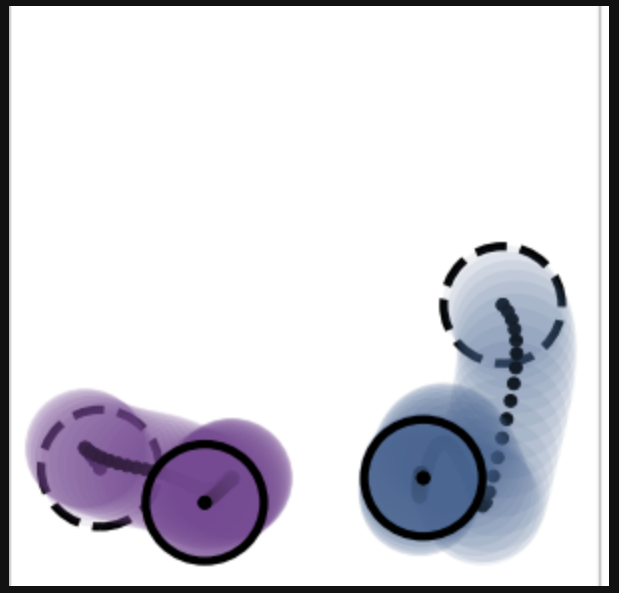
Return: event times $\{t_i\}$ and the piecewise continuous trajectory $\{z_i(t) \text{ for } t_i \leq t \leq t_{i+1}\}$

Neural Event ODE

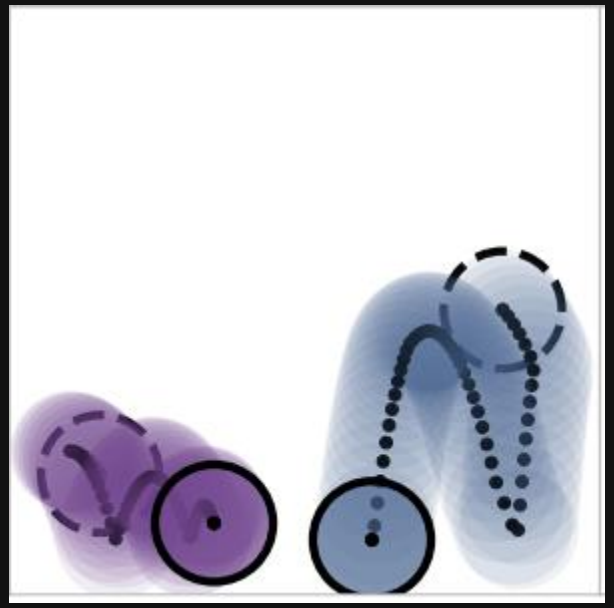
- Results



Ground trith



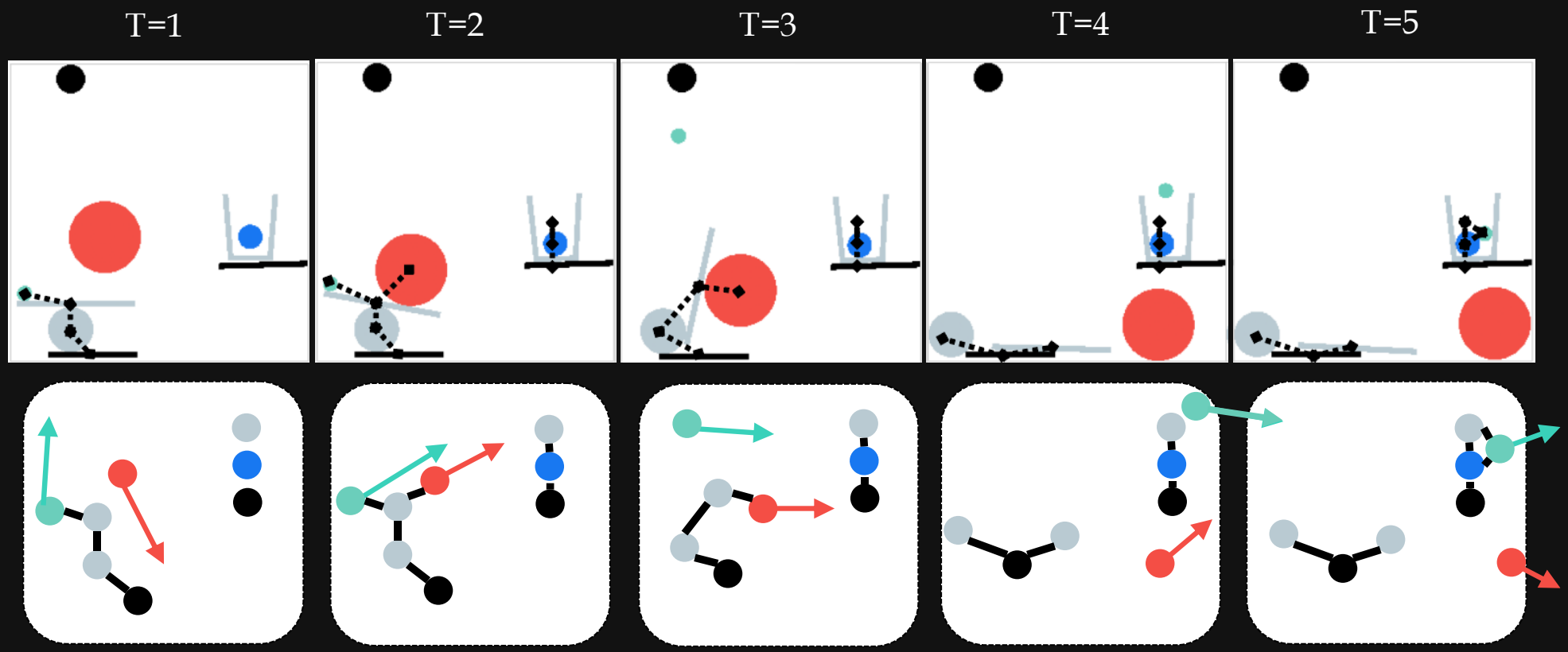
Neural ODE



Neural Event ODE

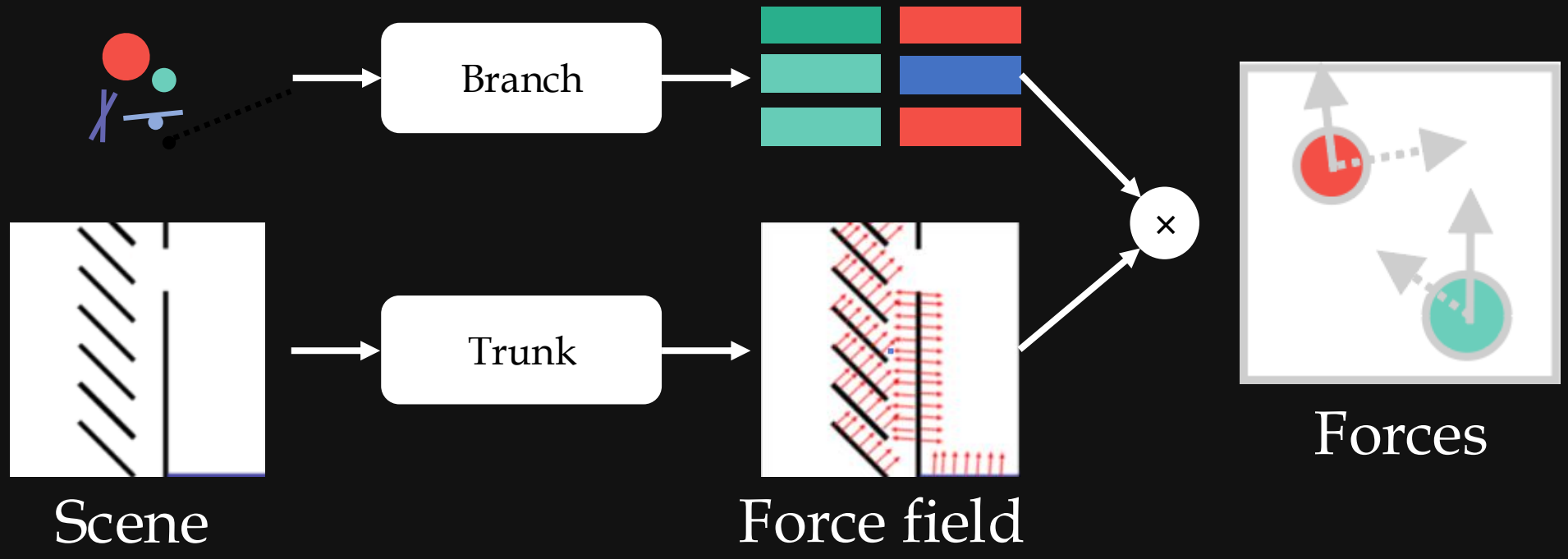
Neural Force Field: NeuralODE with Graphs for Interactions

Step 1: Build dynamic interaction graphs



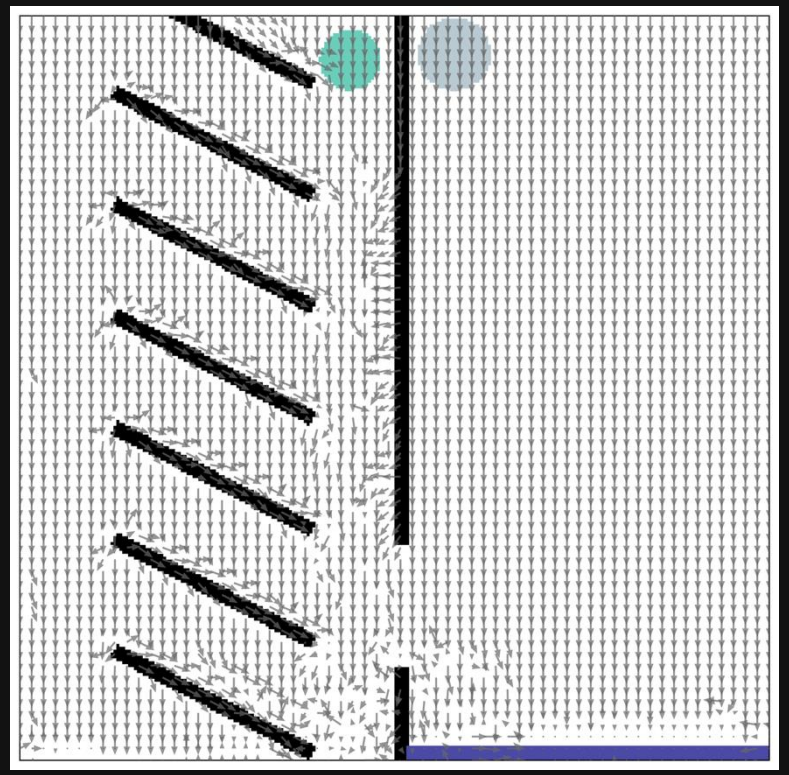
Neural Force Field

Step 2: Infer the force field through a DeepONet

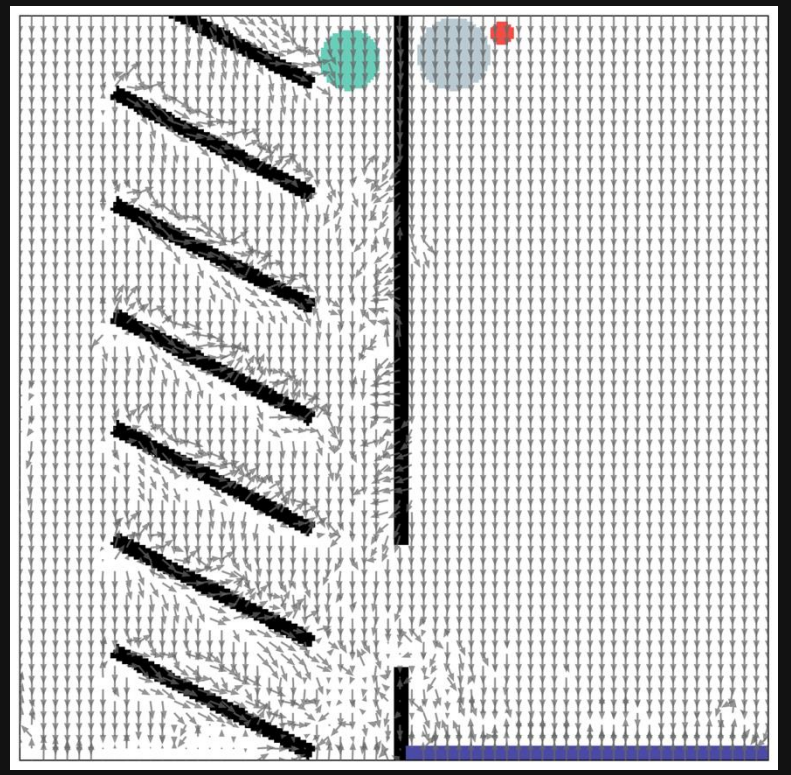


Neural Force Field

Reconstruct ODE-grounded force field from trajectories to model interpretable and generalizable physical interactions.



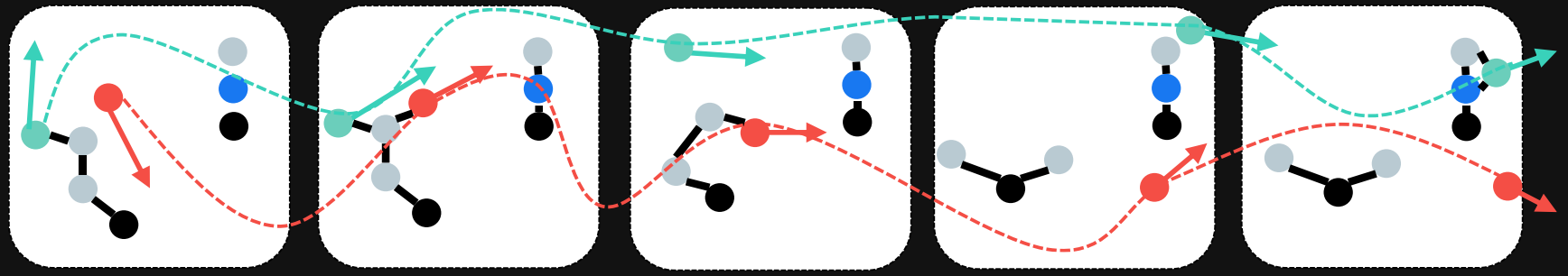
Ground truth force field

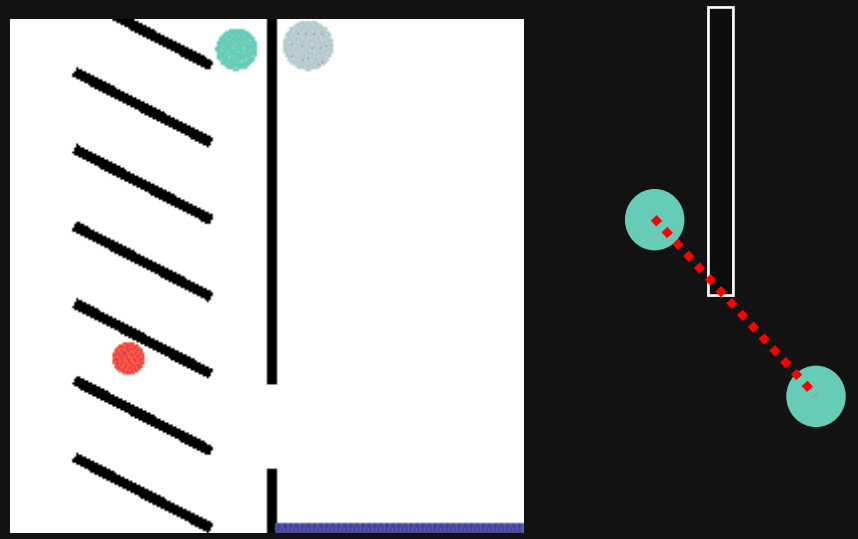


Inversion from a few trajectories
Reconstructed force field

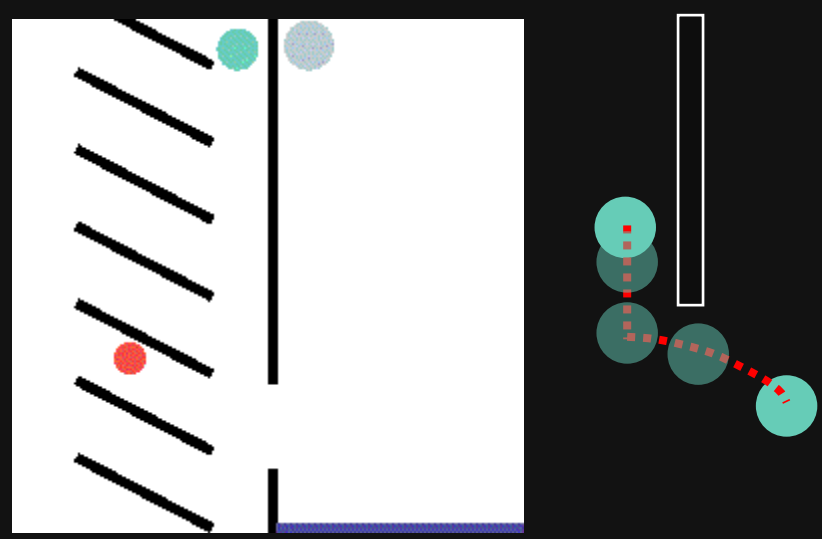
Neural Force Field

Step 3: Decoding trajectories by integration on the force field





Discrete decoding



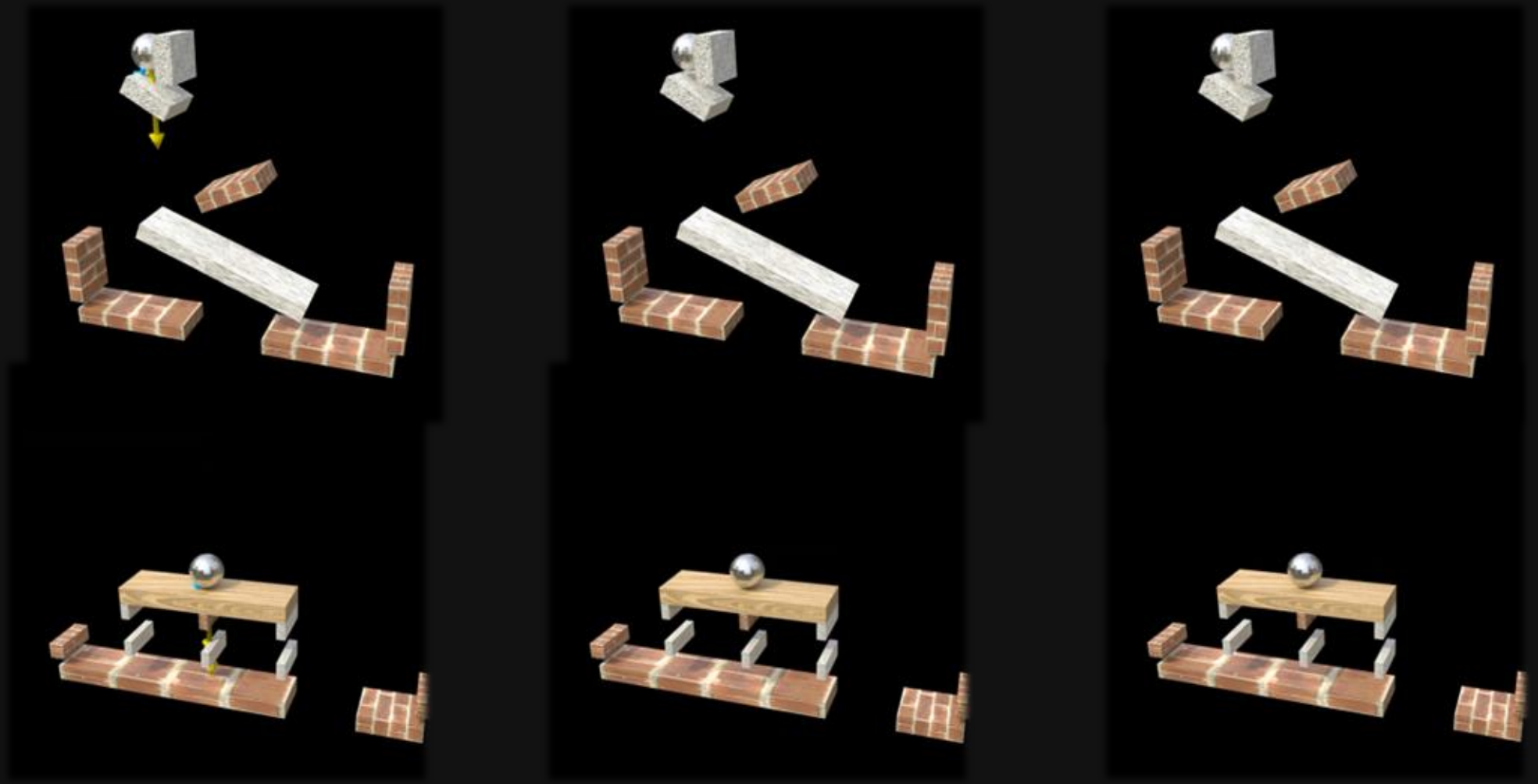
Continuous decoding

Neural Force Field

NFF

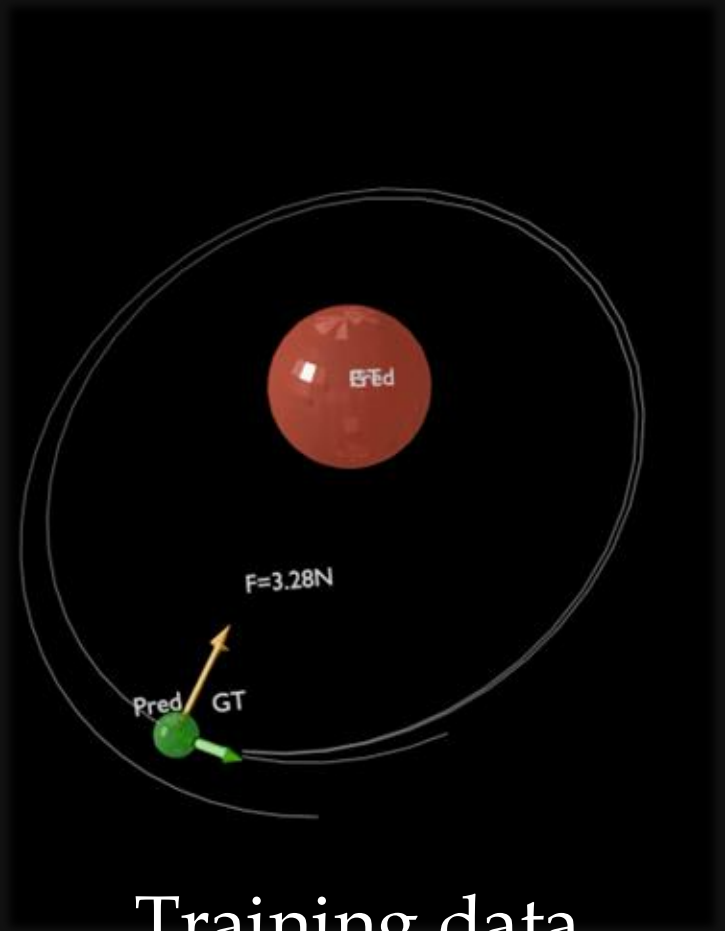
IN

SlotFormer



Cross-scenario generalization (trained on 100 trajectories)

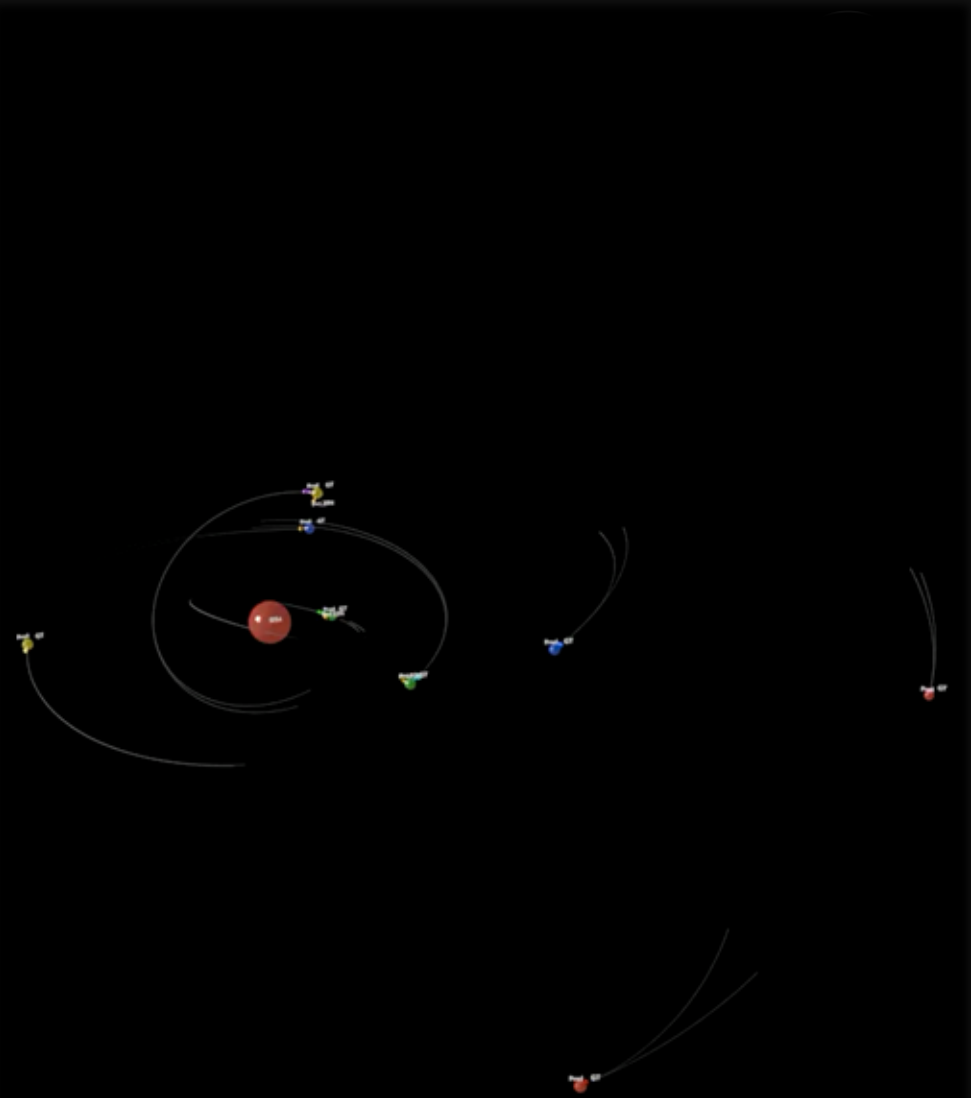
Neural Force Field



Training data



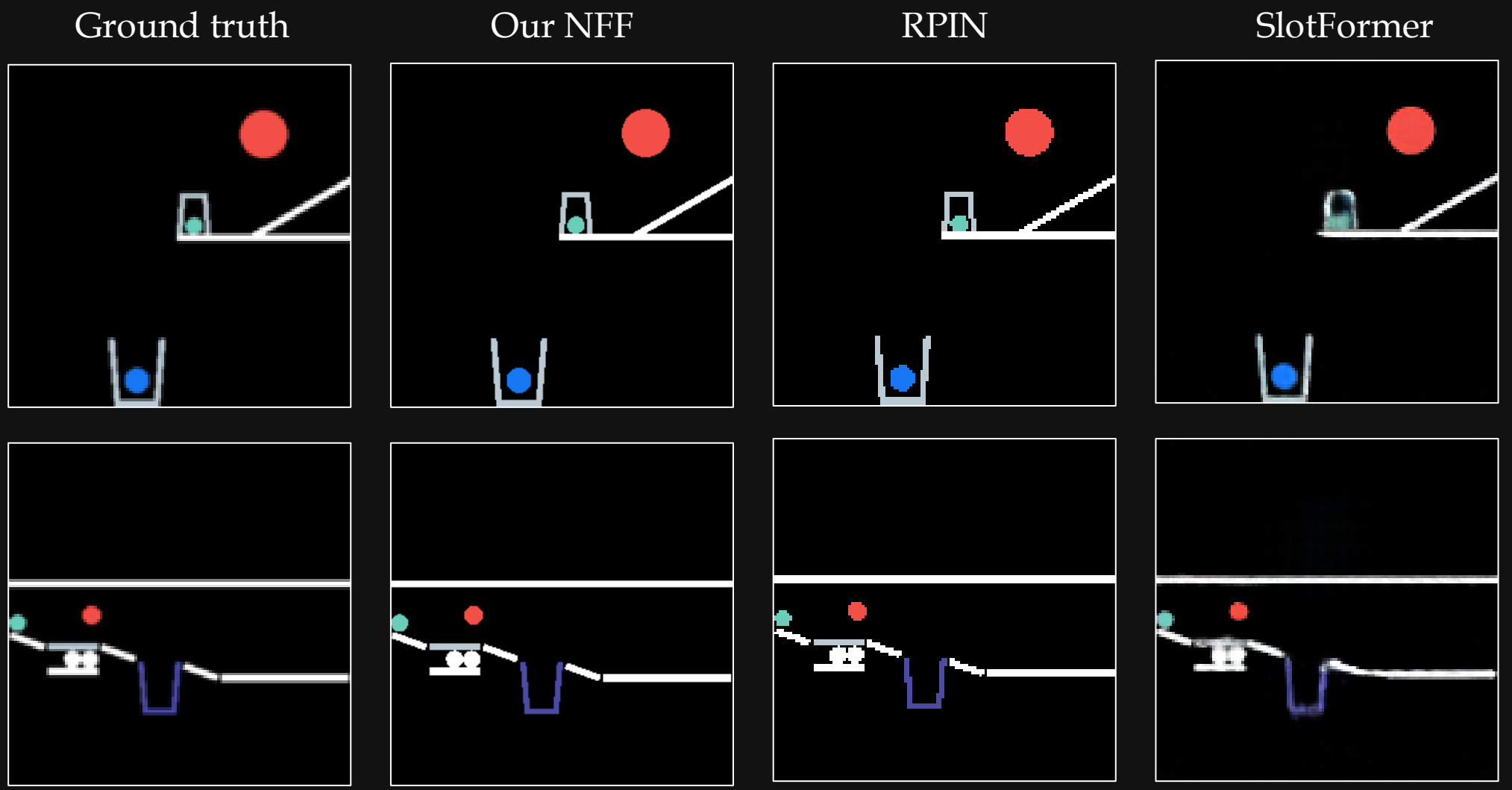
Within-scenario



Cross-scenario

Trained on 200 trajectories

Neural Force Field



Within-scenario generalization

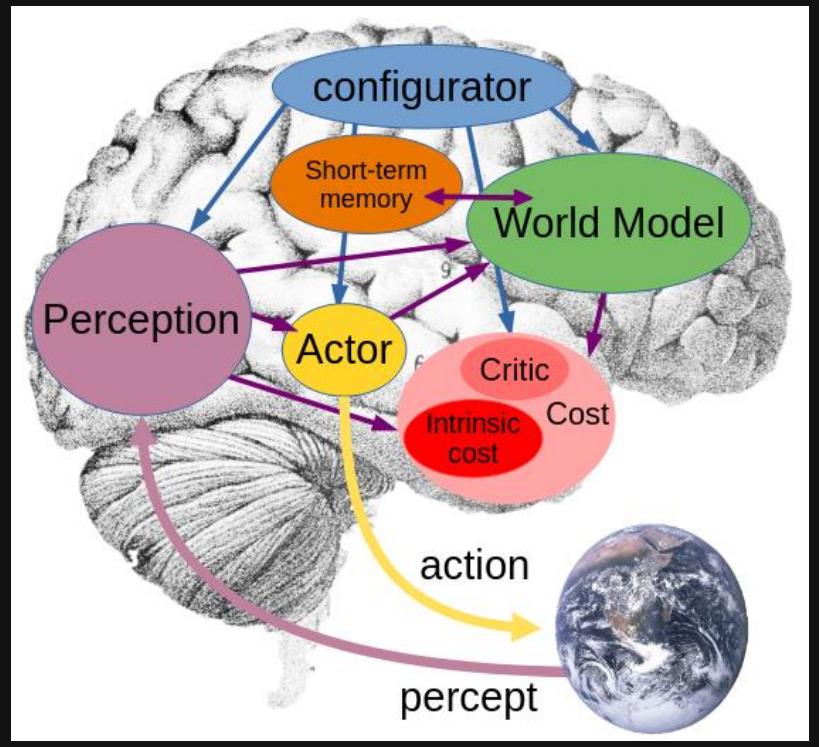
While others are trained on up to 3.2 million trajectories, NFF is trained on 12,000 trajectories (~267 times fewer)

How to achieve human-level intuitive physics in AI?

Key elements

Perception->prediction->interaction to reach a goal

o: observation, a: action, g: goal, s perceived state, z: predicted state



Goal-driven interaction

$$P(a \mid o, g) \propto P(a) \int P(g \mid z, a)\, P(z \mid o, a)\, dz$$

$$\propto P(a) \iint P(g \mid z, a)\, P(z \mid s, a)\, P(s \mid o)\, dz\, ds$$

$$\propto P(a) \iint \underbrace{P(g \mid z, a)}_{\text{Judgment}}\, \underbrace{P(z \mid s, a)}_{\text{Prediction}}\, \underbrace{P(o \mid s)\, P(s)}_{\text{Perception}}\, dz\, ds.$$

Decision making by maximize utility $P(g \mid z, a) \propto e^{\beta U(g,z,a)}$

$$a^\star = \arg\max_a \; \mathbb{E}_{z \sim P(z|o,a)} \left[U(g, z, a) \right]$$

$$= \arg\max_a \iint U(g, z, a)\, P(z \mid s, a)\, P(s \mid o)\, dz\, ds.$$

Imagination by generating sequences from predicted states

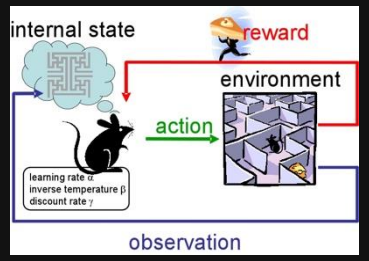
$$P(\tilde{o}_{t+1:T} \mid o_{1:t}, a_{t:T-1}) = \int \cdots \int \left[\prod_{\tau=t+1}^{T} P(\tilde{o}_\tau \mid s_\tau) \right] \left[\prod_{\tau=t+1}^{T} P(s_\tau \mid s_{\tau-1}, a_{\tau-1}) \right] P(s_t \mid o_{1:t})\, ds_{t:T}$$

Existing methods

o: observation, a: action, g: goal, s perceived state, z: predicted state

Goal-driven interaction

Model-free RL



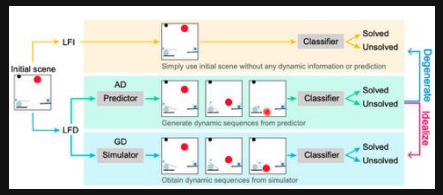
Limited generalization

$$P(a \mid o, g) \propto P(a) \int P(g \mid z, a)\, P(z \mid o, a)\, dz$$

$$\propto P(a) \iint P(g \mid z, a)\, P(z \mid s, a)\, P(s \mid o)\, dz\, ds$$

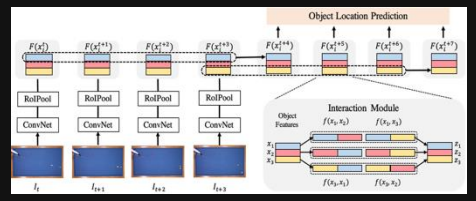
$$\propto P(a) \iint \underbrace{P(g \mid z, a)}_{\text{Judgment}} \underbrace{P(z \mid s, a)}_{\text{Prediction}} \underbrace{P(o \mid s)\, P(s)}_{\text{Perception}}\, dz\, ds.$$

Task-solution models



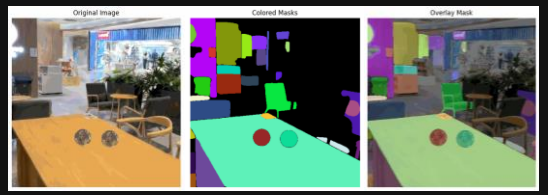
Task-specific

Dynamic prediction



Assume perfectly known states

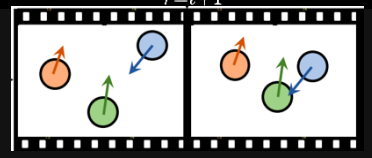
Reconstruction



No dynamic modeling

$$P(\tilde{o}_{t+1:T} \mid o_{1:t}, a_{t:T-1}) = \int \cdots \int \Big[\prod_{\tau=t+1}^{T} P(\tilde{o}_\tau \mid s_\tau)\Big] \Big[\prod_{\tau=t+1}^{T} P(s_\tau \mid s_{\tau-1}, a_{\tau-1})\Big] P(s_t \mid o_{1:t})\, ds_{t:T}$$

Imagination



Graphics engine rendering

How to integrate all functions?

A possible solution: Video generation models

o: observation, a: action, g: goal, s perceived state, z: predicted state

Goal-driven interaction

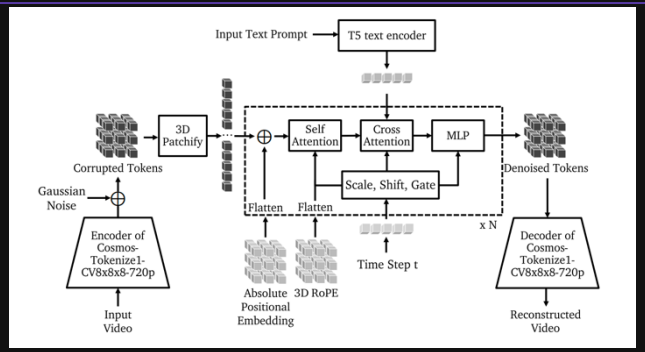
$$P(a \mid o, g) \propto P(a) \int P(g \mid z, a)\, P(z \mid o, a)\, dz$$

$$\propto P(a) \iint P(g \mid z, a)\, P(z \mid s, a)\, P(s \mid o)\, dz\, ds$$

$$\propto P(a) \iint P(g \mid z, a)\, P(z \mid s, a)\, P(o \mid s)\, P(s)\, dz\, ds.$$

Judgment Prediction Perception

VLMs: NVIDIA Cosmos, Google Veo, OpenAI Sora



Text prompts

Latent diffusion

No explicit perception

$$P(\tilde{o}_{t+1:T} \mid o_{1:t}, a_{t:T-1}) = \int \cdots \int \Big[\prod_{\tau=t+1}^{T} P(\tilde{o}_\tau \mid s_\tau) \Big] \Big[\prod_{\tau=t+1}^{T} P(s_\tau \mid s_{\tau-1}, a_{\tau-1}) \Big] P(s_t \mid o_{1:t})\, ds_{t:T}$$

Latent decoding

Video generation models

- Impressive vision understanding



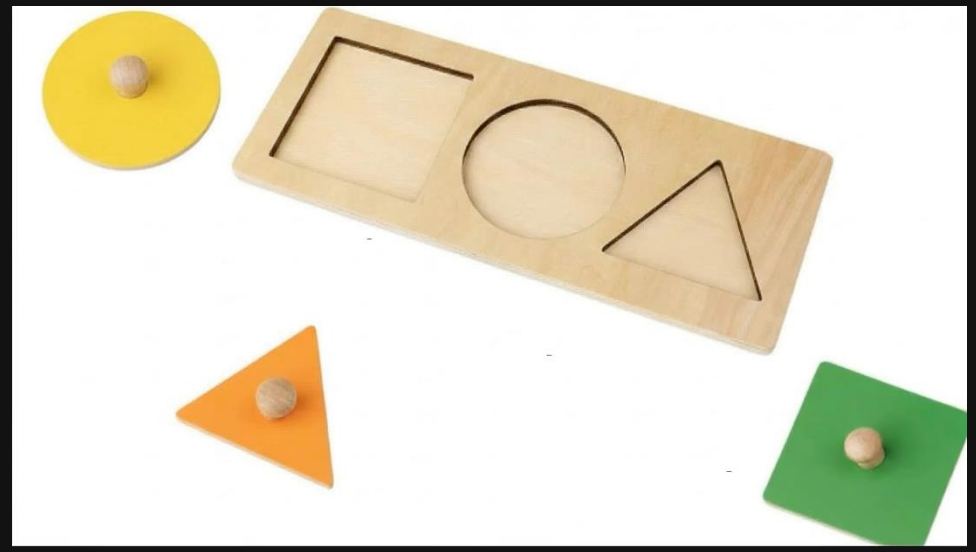
Categorizing objects

Video models are zero-shot learners and reasoners

Thaddäus Wiedemer*, Yuxuan Li, Paul Vicol, Shixiang Shane Gu, Nick Matarese, Kevin Swersky, Been Kim, Priyank Jaini*, Robert Geirhos*

Google DeepMind

* Joint leads



Shape fitting

Video generation models

- Reasoning

Video models are zero-shot learners and reasoners

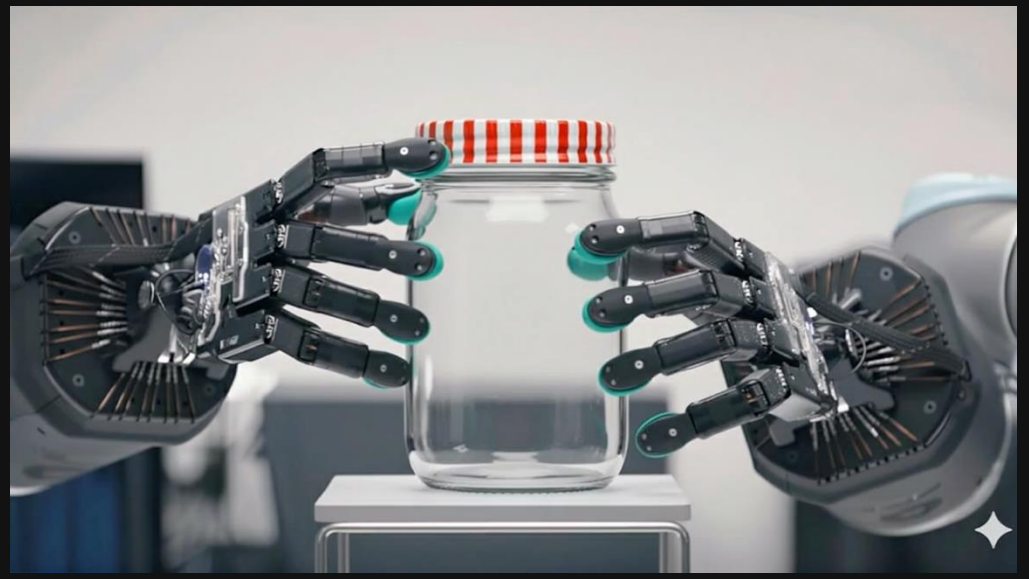
Thaddäus Wiedemer*, Yuxuan Li, Paul Vicol, Shixiang Shane Gu, Nick Matarese, Kevin Swersky, Been Kim, Priyank Jaini*, Robert Geirhos*

Google DeepMind

* Joint leads



Maze solving



Dexterous manipulation

Video generation models

- Still cannot consistently generate physically plausible videos



Falling onto the ground

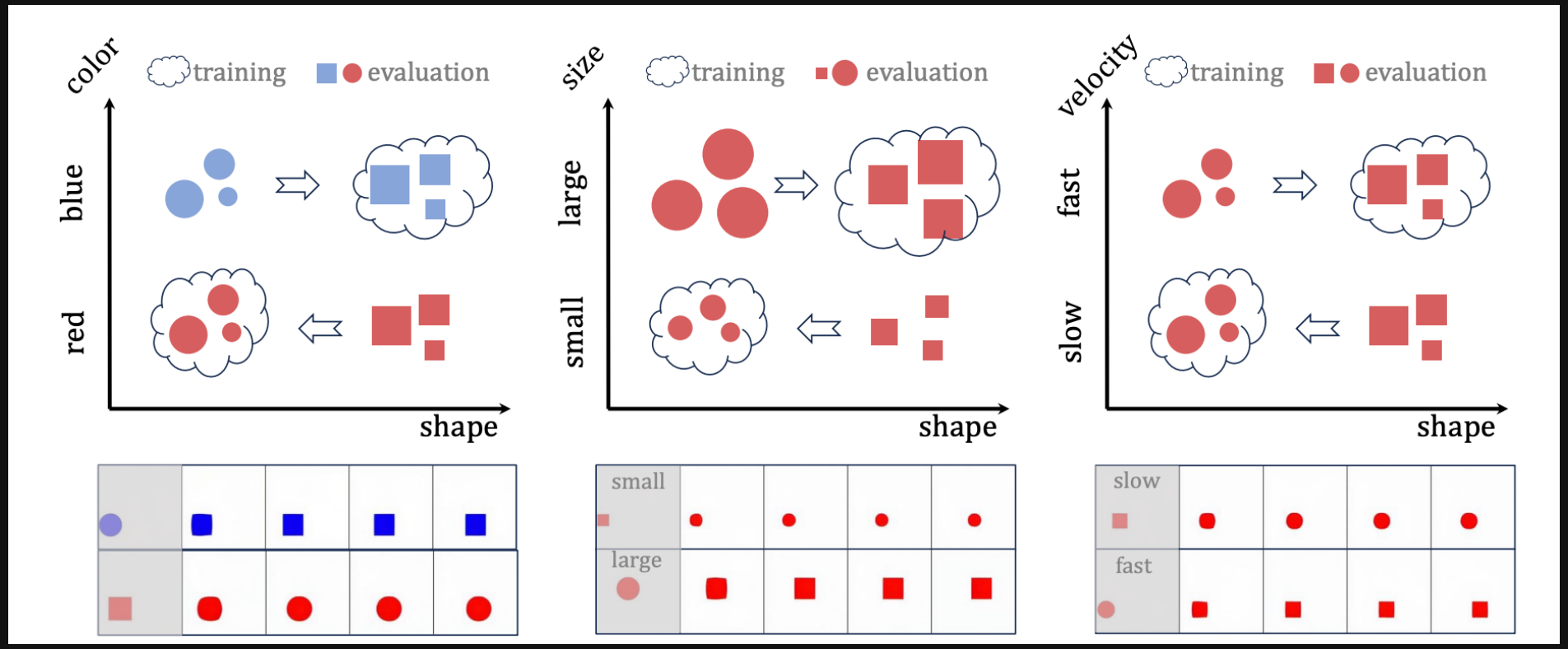


Real world

Why do current VLMs fail to understand physics?

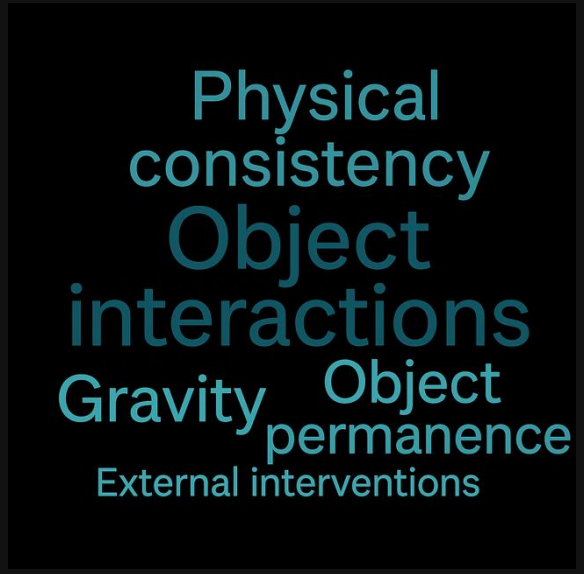
Video generation models

Overfitting: color > size > velocity > shape



Kang, B., Yue, Y., Lu, R., Lin, Z., Zhao, Y., Wang, K., ... & Feng, J. (2025). How far is video generation from world model: A physical law perspective. ICML.

How to build a physics-grounded world model?



Physics-grounded video prediction model

o: observation, a: action, g: goal, s perceived state, z: predicted state

Goal-driven interaction

$$P(a \mid o, g) \propto P(a) \int P(g \mid z, a)\, P(z \mid o, a)\, dz$$

$$\propto P(a) \iint P(g \mid z, a)\, P(z \mid s, a)\, P(s \mid o)\, dz\, ds$$

$$\propto P(a) \iint \underbrace{P(g \mid z, a)}_{\text{Judgment}} \underbrace{P(z \mid s, a)}_{\text{Prediction}} \underbrace{P(o \mid s)\, P(s)}_{\text{Perception}}\, dz\, ds.$$

Physics cues

Explicit integration

Explicit gaussian reconstruction

$$P(\tilde{o}_{t+1:T} \mid o_{1:t}, a_{t:T-1}) = \int \cdots \int \Big[\prod_{\tau=t+1}^{T} P(\tilde{o}_\tau \mid s_\tau) \Big] \Big[\prod_{\tau=t+1}^{T} P(s_\tau \mid s_{\tau-1}, a_{\tau-1}) \Big] P(s_t \mid o_{1:t})\, ds_{t:T}$$

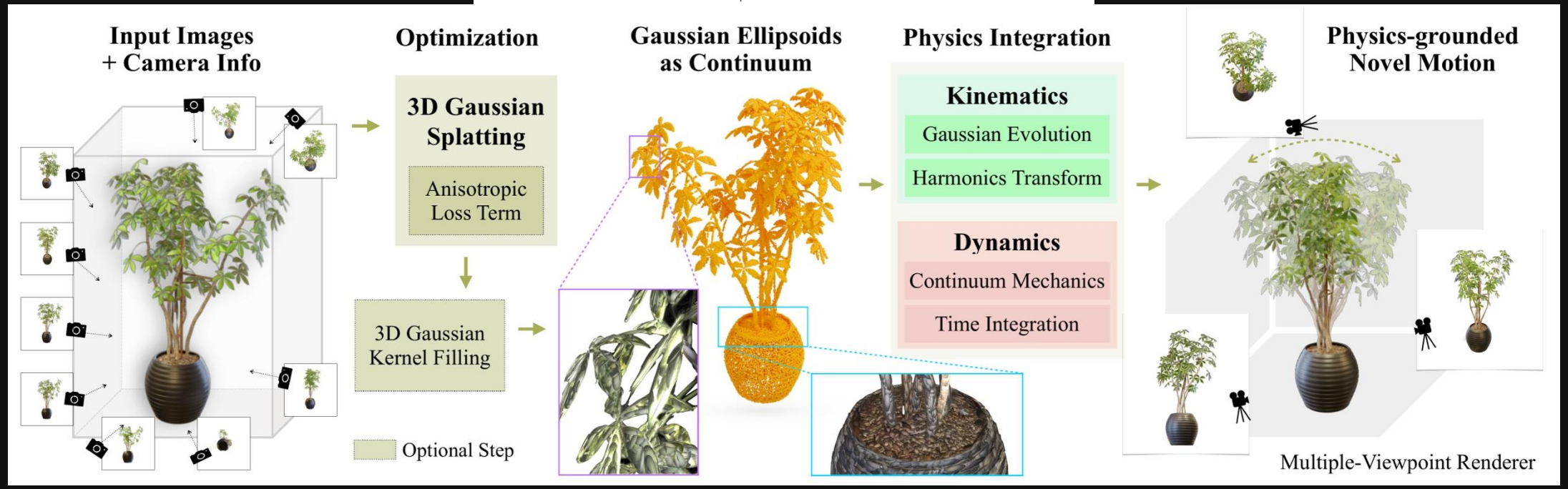
Gaussian splatting

Physics-grounded video prediction model

PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics

CVPR 2024

Tianyi Xie[1*], Zeshun Zong[1*], Yuxing Qiu[1*], Xuan Li[1*], Yutao Feng[2,3], Yin Yang[3], Chenfanfu Jiang[1]
[1]University of California, Los Angeles, [2]Zhejiang University, [3]University of Utah
[*]Equal Contributions



Physics-grounded video prediction model





Summary

1. Intuitive physics is essential for our human beings from infants to adults, supporting observation, prediction, and interacting with the physical world.
2. The computational mechanisms of intuitive physics includes VoE, prediction (heuristic and dynamic modeling), and interaction.
3. Current AI systems still fall short in human-like intuitive physics. Progress has been made in object-centric modeling, continuous modeling, and foundation world model training.

Thanks for your listening and sleeping